Energy band structure

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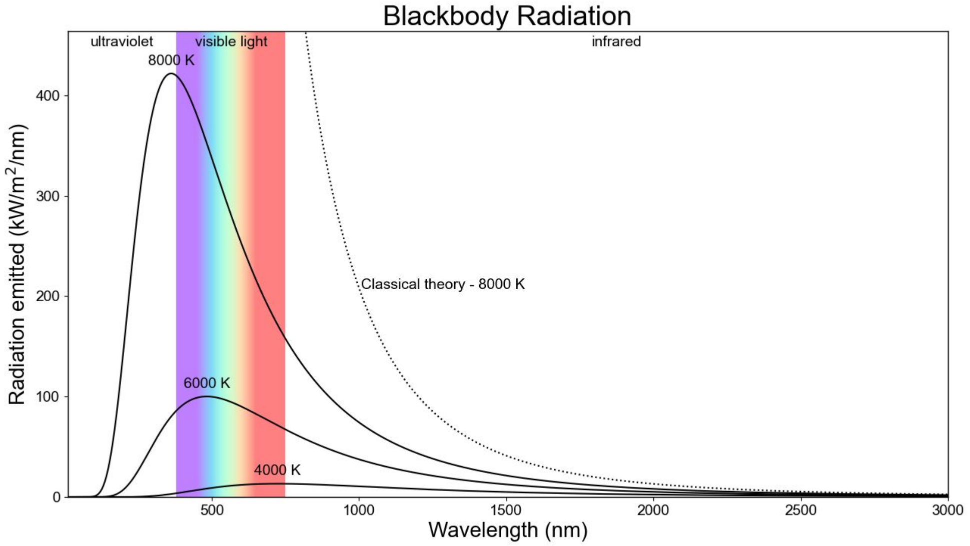
Why Newtonian physics is not an adequate theory for solids ?

- Solids - The last frontier of matter (Gas laws were discovered in 17th and 18th century)
- A proper theory of solids has to wait until the development of quantum mechanics and quantum statistical mechanics.
- Atoms are closely packed; one cannot neglect the interaction of electrons between the atoms
- No serious understanding of semiconductors and nanoscale devices is possible w/o knowledge in quantum mechanics

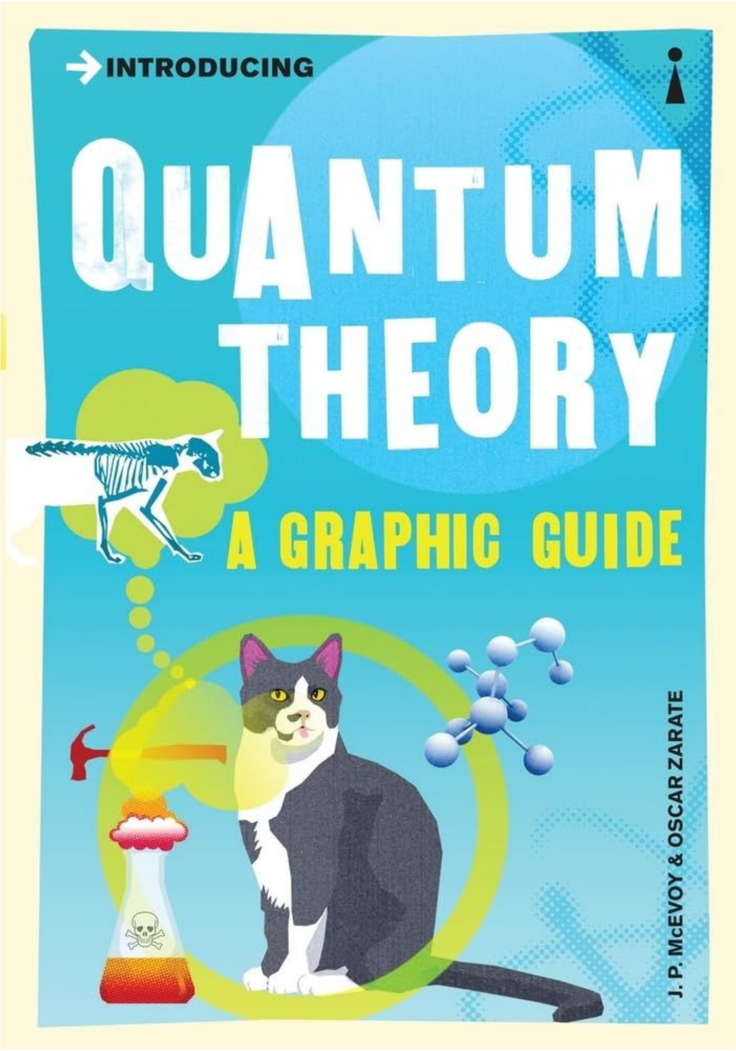
Failure of classical mechanics (1890-1930s)

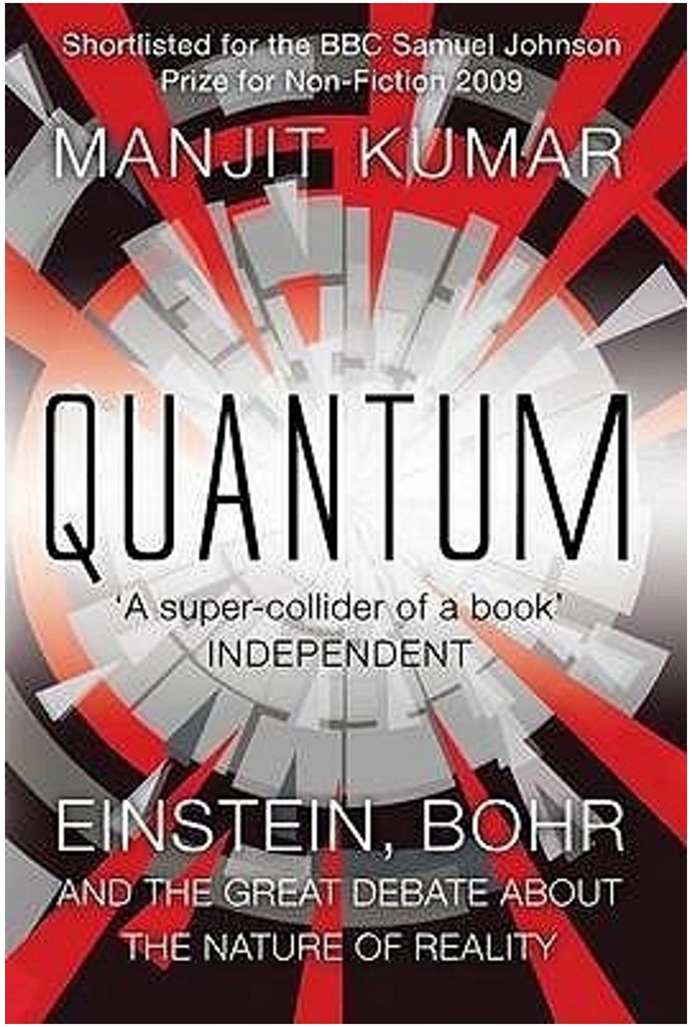
- UV catastrophe
- Photoelectric effect
- Specific heat capacity of solids at low temp
- Spectral lines of hydrogen
- Compton effect (X-ray scattering)
- Davisson and Thomson (Discovery of electron's wave nature)

Wave-particle duality of matter and EM waves



Two popular science books on the history of QM





1.13 DE BROGLIE'S HYPOTHESIS

The suggestion that matter may have wave-like properties was first put forward in 1924–25 by Louis de Broglie.[18] He argued that if light (which consists of waves according to the classical picture) can sometimes behave like particles, then it should

[17] N. Bohr, *Nature*, **121**, 580, 1923.

[18] L. de Broglie, *Phil. Mag.*, **47**, 446, 1924; *Annales de Physique*, **3**, 22, 1925.

be possible for matter (which consists of particles, classically) to exhibit wave-like behaviour under suitable circumstances. He made the hypothesis that the relation between the energy E of a particle and the frequency ν of the associated wave is exactly the same as that between the energy of a photon (the particle-like quantum of light) and the frequency of the light radiation:

$$E = h\nu = \hbar\omega \tag{1.28a}$$

where $\omega = 2\pi\nu$ is the angular frequency. He noted then that according to the theory of relativity, the energy E and the momentum $\mathbf{p}$ of a particle form the components of a single four-vector, and so do the angular frequency ω and the propagation vector $\mathbf{k}$ of wave. Since the components of both the four-vectors have to transform in an identical manner (according to the Lorentz transformation) for a given change of reference frame, any proportionality relation like (1.28a) between E and ω must necessarily hold also between $\mathbf{p}$ and $\mathbf{k}$. It was concluded, therefore, that

$$\mathbf{p} = \hbar\mathbf{k} \tag{1.28b}$$

In terms of the magnitudes of the vectors, this gives

$$p = \hbar k = \frac{h}{\lambda}, \quad \text{or} \quad \lambda = \frac{h}{p} \tag{1.28c}$$

nature. Confirmation that Nature does exhibit such an aesthetically pleasing symmetry between matter and radiation came from experiments by Davisson and Germer,[19] Kikuchi[20] and G. P. Thomson[21]. These experiments demonstrated the existence of 'de Broglie waves' associated with electrons in a very direct fashion. This was done

The Davisson-Germer experiment involved diffraction by a single crystal; Kikuchi reproduced the Laue type of diffraction pattern by transmission through very thin mica crystals, and Thomson obtained the analogue of the Debye-Scherrer rings by passage of electrons through thin (poly-crystalline) metal foils. In each case, the wavelength λ of the electron waves, as determined from the diffraction pattern, was found to be equal to (h/p) in agreement with de Broglie's theoretical prediction. Thus the dual nature of matter, like that of radiation, was firmly established.

Actually the theoretical exploitation of de Broglie's idea did not await its experimental verification. Immediately after the publication of the hypothesis, Erwin Schrödinger[23] proposed that the behaviour of matter waves associated with material particles (whether free or subject to forces) is governed by a certain differential equation for the *wave function* ψ (i.e., the function which represents the matter wave). He showed that this *wave equation*, together with physically-motivated conditions on the wave function, leads in a very natural way to stationary states characterized by discrete energy values. Thus Schrodinger was able to explain the basic fact of quantization as a consequence of the wave nature of matter, and to replace the *ad hoc* quantization rules of the Old Quantum Theory by mathematical conditions of a very general nature on the wave functions. The Schrödinger theory, called *wave mechanics*, is one of the alternative formulations of the general theory of quantum mechanics which, as far as we know at present, gives the correct fundamental theory of the physical world at the microscopic level. Much of this book will deal with the development of quantum mechanics following Schrödinger's approach. We defer the introduction of the wave equation to the next chapter, and devote the rest of this chapter to a discussion of the conceptual problems raised by the developments described in the preceding sections.

[23] E. Schrödinger, *Ann. d. Physik*, **79**, 361, 489, 1926; **80**, 437, 1926; **81**, 109, 1926.

Relevant Postulates of QM

Postulate 1

The state of a quantum mechanical system is completely specified by the function $\Psi(\mathbf{r},t)$ that depends on the coordinates of the particle, $\mathbf{r}$ and the time, t. This function is called the wavefunction or state function and has the property that $\Psi^*(\mathbf{r},t)\Psi(\mathbf{r},t)d\tau$ is the probability that the particle lies in the volume element $d\tau$ located at $\mathbf{r}$ and time t.

This is the *probabalistic* interpretation of the wavefunction. As a result the wavefunction must satisfy the condition that finding the particle *somewhere* in space is 1 and this gives us the normalisation condition,

$$\int_{-\infty}^{+\infty} \Psi^*(\mathbf{r},t)\Psi(\mathbf{r},t)d\tau = 1$$

The other conditions on the wavefunction that arise from the probabilistic interpretation are that it must be single-valued, continuous and finite. We normally write wavefunctions with a normalisation constant included.

The wave function $\psi(\mathbf{r}, t)$ is not physical, because it cannot be measured. The quantity that is measured, and hence physical, is $|\psi(\mathbf{r}, t)|^2$, or an expectation value of a quantum operator. Quantum operators are often associated with classical variables. Table 2.4 lists

Description	Classical theory	Quantum theory
Position	r	r
Potential	$V(\mathbf{r}, t)$	$V(\mathbf{r}, t)$
Momentum	p_x	$-i\hbar\frac{\partial}{\partial x}$
Energy	E	$i\hbar\frac{\partial}{\partial t}$

Energy $\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$

Postulate 2

To every observable in classical mechanics there corresponds a linear, Hermitian operator in quantum mechanics.

This postulate comes from the observation that the expectation value of an operator that corresponds to an observable must be real and therefore the operator must be Hermitian. Some examples of Hermitian operators are:

the operator Â is **Hermitian** by definition if

$$\int \psi^* \hat{A}\psi \, d\tau = \int (\hat{A}^* \psi^*)\psi \, d\tau$$

Postulate 3

In any measurement of the observable associated with operator $\hat{A}$, the only values that will ever be observed are the eigenvalues, a, that satisfy the eigenvalue equation,

$$\hat{A}\Psi = a\Psi$$

Postulate 4

If a system is in a state described by the normalised wavefunction, Ψ, then the average value of the observable corresponding to $\hat{A}$ is given by

$$< \hat{A} >= \int_{-\infty}^{+\infty} \Psi^* \hat{A} \Psi d\tau$$

An important consequence of the fourth postulate is that, after measurement of Ψ_i yields some eigenvalue the wave function immediately "collapses" into the corresponding eigenstate Ψ_i

Before a measurement, we do not know in advance with certainty in which eigenstate, among the various states $|\psi_n\rangle$, a system will be after the measurement; only a probabilistic outcome is possible. Postulate 4 states that the probability of finding the system in one particular nondegenerate eigenstate $|\psi_n\rangle$ is given by

$$P_n = \frac{|\langle \psi_n | \psi \rangle|^2}{\langle \psi | \psi \rangle}. \tag{3.31}$$

Postulate 5

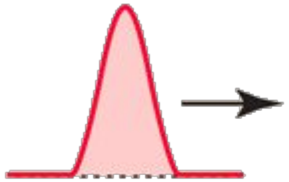
The wavefunction or state function of a system evolves in time according to the time-dependent Schrödinger equation

$$\mathcal{H}\Psi(\mathbf{r}, t) = i\hbar \frac{\partial \Psi}{\partial t}$$

Moving to other Fourier pair (position and wavenumber)

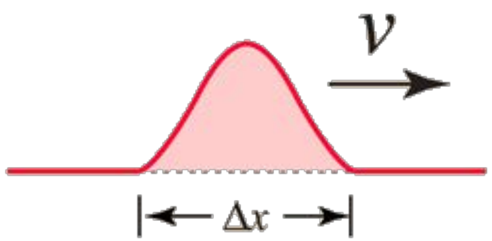
$$y(x,t) = \int A(k)\sin(kx - \omega t)\,dk$$

Producing a localized wave packet requires a continuous distribution of wavelengths.



Producing a wave packet that is more confined in space requires a wider distribution of wavelengths.

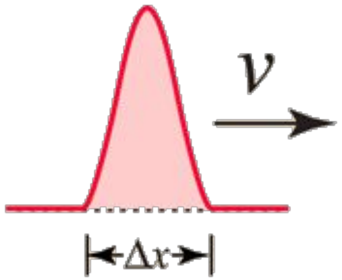
This kind of pulse requires a continuous frequency distribution.

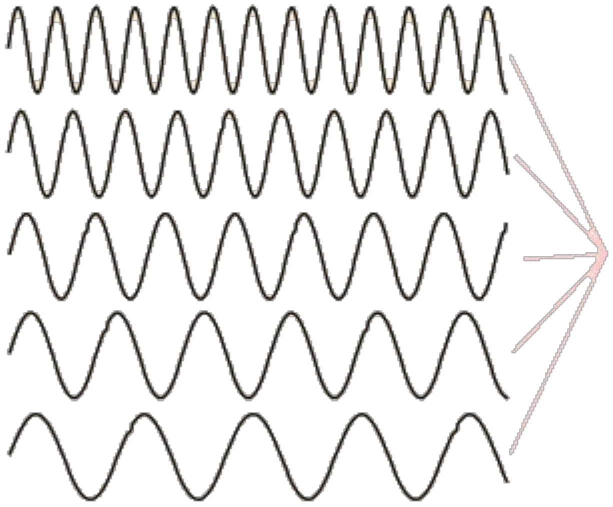


$$\Delta\omega\Delta t \approx 1$$

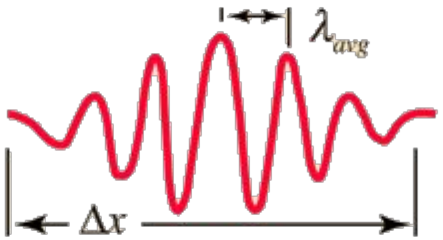
$$\Delta k\Delta x \approx 1$$

A shorter pulse requires a greater width in k.





Adding several waves of different wavelength together will produce an interference pattern which begins to localize the wave.



But that process spreads the wave number k values and makes it more uncertain. This is an inherent and inescapable increase in the uncertainty Δk when Δx is decreased. $\Delta k\Delta x \approx 1$

Moving to Heisenberg's uncertainty principle

$$\Delta k \Delta x \approx 1$$

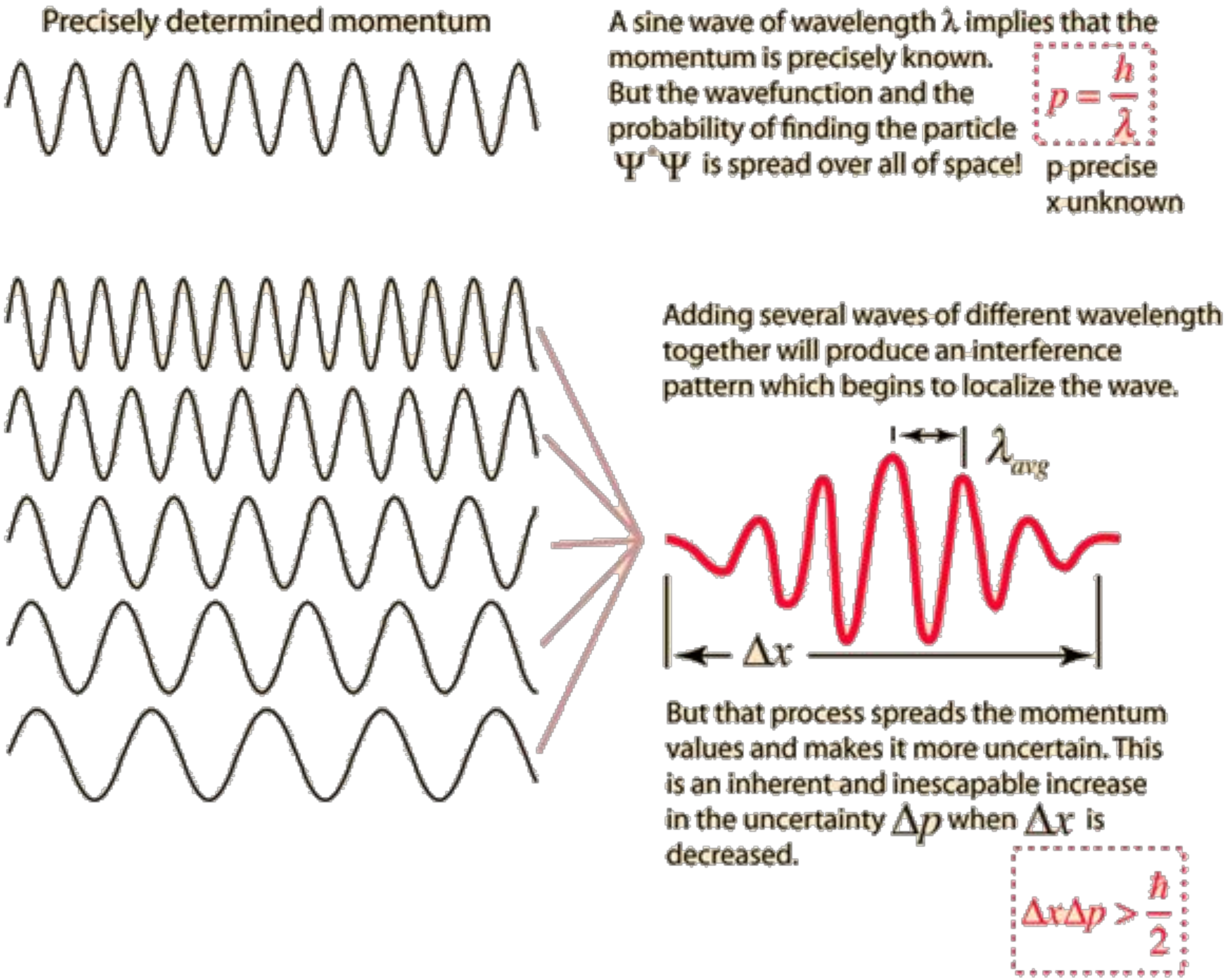
$$\hbar \Delta k \Delta x \approx \hbar$$

$$\hbar \Delta \left(\frac{2\pi}{\lambda} \right) \Delta x \approx \hbar$$

$$\Delta \left(\frac{h}{\lambda} \right) \Delta x \approx \hbar$$

Using de-Broglie formula relating momentum and wavelength

$$\Delta p \Delta x \approx \hbar$$



Schrodinger Wave Equation (SWE)

We will use total energy function or Hamiltonian of a particle mass m moving in potential V

$$\mathrm{H} \equiv KE + PE = \left[\frac{p^2}{2m} + V(r,t)\right]$$

Hamiltonian=total energy

Substituting the operator for p from previous page

$$\mathrm{H}\Psi(r,t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(r,t)\right]\Psi(r,t)$$

Replacing the Hamiltonian with energy operator from previous page

$$\mathrm{H}\Psi(r,t) = i\hbar\frac{\partial}{\partial t}\Psi(r,t)$$

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(r,t)\right]\Psi(r,t) = i\hbar\frac{\partial}{\partial t}\Psi(r,t)$$

3.4 Linearity of quantum mechanics: linear superposition

The time-dependent Schrödinger equation is also linear in the wavefunction Ψ, just as the time-independent Schrödinger equation was. Again, no higher powers of Ψ appear anywhere in the equation. In the time-independent equation, this allowed us to say that, if ψ was a solution, then so also was $A\psi$, where A is any constant, and the same kind of behavior holds here for the solutions Ψ of the time-dependent Schrödinger equation. Another consequence of linearity is the possibility of linear superposition of solutions, which we can state as follows.

$$\begin{gathered}\text{If } \Psi_a(\mathbf{r},t) \text{ and } \Psi_b(\mathbf{r},t) \text{ are solutions,} \\ \text{then so also is } \Psi_{a+b}(\mathbf{r},t) = \Psi_a(\mathbf{r},t) + \Psi_b(\mathbf{r},t).\end{gathered} \tag{3.13}$$

This is easily verified by substitution into the time-dependent Schrödinger equation[4].

We can also multiply the individual solutions by arbitrary constants and still have a solution to the equation, i.e.,

$$\Psi_c(\mathbf{r},t) = c_a\Psi_a(\mathbf{r},t) + c_b\Psi_b(\mathbf{r},t) \tag{3.14}$$

where c_a and c_b are (complex) constants, is also a solution.

The concept of linear superposition solutions is, at first sight, a very strange one from a classical point of view. There is no classical precedent for saying that a particle or a system is in a linear superposition of two or more possible states. In classical mechanics, a particle simply has a "state" that is defined by its position and momentum, for example. Here we are saying that a particle may exist in a superposition of states each of which may have different energies (or possibly positions or momenta). Actually, however, it will turn out that, to recover the kind of behavior we expect from particles in the classical picture of the world, we need to use linear superpositions, as we will illustrate below.

Time independent Schrodinger Equation

We now consider the special case of a closed system in which energy is conserved and potential energy is time independent in such a way that $V = V(\mathbf{r})$. Separation of the wave function into orthogonal spatial components allows us to solve one-dimensional problems. In quantum mechanics the Hamiltonian is separable if the potential is separable. This is the case if, for example, $V(\mathbf{r}) = V(x)V(y)V(z)$.

We now return to the one-dimensional system in which $V = V(x)$ and show how the time and spatial parts of the wave function may be separated out using the method of

separation of variables. If we assume the wave function can be expressed as a product $\psi(x,t) = \psi(x)\phi(t)$, then substitution into the one-dimensional Schrödinger equation gives

$$\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x)\phi(t) + V(x)\psi(x)\phi(t) = i\hbar\frac{\partial}{\partial t}\psi(x)\phi(t) \tag{2.40}$$

We then divide both sides by $\psi(x)\phi(t)$, so that the left-hand side is a function of x only and the right-hand side is a function of t only. This is true if both sides are equal to a constant E. It follows that

$$\boxed{E\phi(t) = i\hbar\frac{\partial}{\partial t}\phi(t)} \tag{2.41}$$

is the *time-dependent Schrödinger equation*,

$$\boxed{\left(\frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\psi(x) = E\psi(x)} \tag{2.42}$$

is the *time-independent Schrödinger equation* in one dimension, and the constant E is just the *energy eigenvalue* of the particle described by the wave function.

It is important to remember that these two equations apply when the potential may be considered independent of time. Of course, the potential is not truly time-independent in the sense that it must have been created at some time in the past. However, we assume that the transients associated with this creation have negligible effect on the calculated results, so that the use of a static potential is an excellent approximation.

The solution to the time-dependent Schrödinger equation is of simple harmonic form

$$\phi(t) = e^{-i\omega t} \tag{2.43}$$

The solution to the time-independent Schrödinger equation is $\psi(x)$, which is independent of time. The wave function $\psi(x, t) = \psi(x)\phi(t)$ is called a *stationary state* because the probability density $|\psi(x, t)|^2$ is independent of time. A particle in such a time-independent state will remain in that state until acted upon by some external entity that forces it out of that state.

Because the probability of finding a particle with wave function $\psi_n(\mathbf{r}, t)$ somewhere in space is unity, it makes sense to require wave functions that are solutions to the time-independent Schrödinger equation be normalized such that

$$\int_{-\infty}^{\infty} \psi_n^*(\mathbf{r})\psi_n(\mathbf{r})\,d^3r = 1 \tag{2.44}$$

Wave function description of an electron in free space

Free particle: V(r)=0

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(r)\right]\ \Psi(r) = E\Psi(r) \qquad\qquad \Psi(r) = Ae^{ik.r}$$

Wavefunction: $$\Psi(r,t) = \Psi(r)\ e^{-iEt/\hbar}$$

$$\omega = \frac{E}{\hbar}$$

Plane wave

$$\Psi(r,t) = Ae^{i(k.r-\omega t)}$$

In the absence of any confining potential, the solution is a plane wave – free particle

$$\psi_n(x,t) = Ae^{ik_x x}e^{-i\omega_n t} \tag{2.52}$$

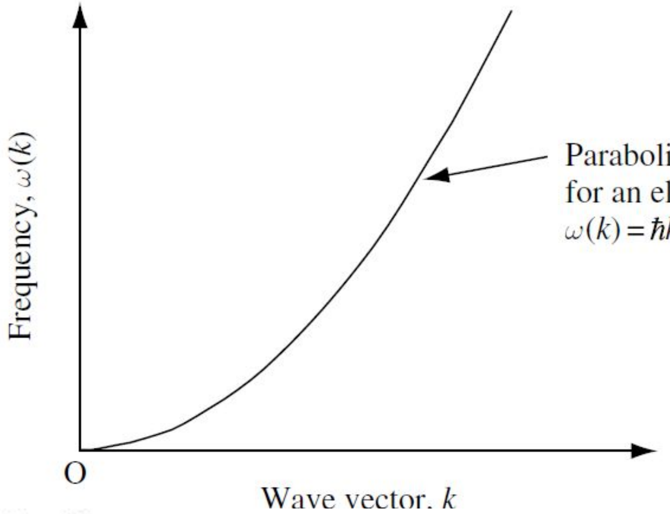
To find the momentum associated with this wave function, we apply the momentum operator to the wave function:

$$\hat{p}_x\psi_n(x,t) = -i\hbar\frac{\partial}{\partial x}\psi_n(x,t) = \hbar k_x Ae^{ik_x x}e^{-i\omega_n t} = \hbar k_x\psi_n(x,t) \tag{2.53}$$

Since, in quantum mechanics, the real eigenvalue of an operator can, at least in principle, be measured, we may safely assume that the x-directed momentum of the particle is just $p_x = \hbar k_x$.

From these results, we conclude that an electron with mass m_0 and energy E in free space has momentum with magnitude of $p = \hbar k = \sqrt{2m_0 E}$ and a nonlinear dispersion relation that gives $E_k = \hbar\omega = \hbar^2 k^2/2m_0$, or

$$\boxed{\omega(k) = \frac{\hbar k^2}{2m_0}}$$



problem we need to confront first: *This wave function is not normalizable*:

$$\int_{-\infty}^{+\infty} \Psi_k^* \Psi_k \, dx = |A|^2 \int_{-\infty}^{+\infty} dx = |A|^2 (\infty). \qquad (2.100)$$

In the case of the free particle, then, the separable solutions do not represent physically realizable states. A free particle cannot exist in a stationary state; or, to put it another way, *there is no such thing as a free particle with a definite energy*.

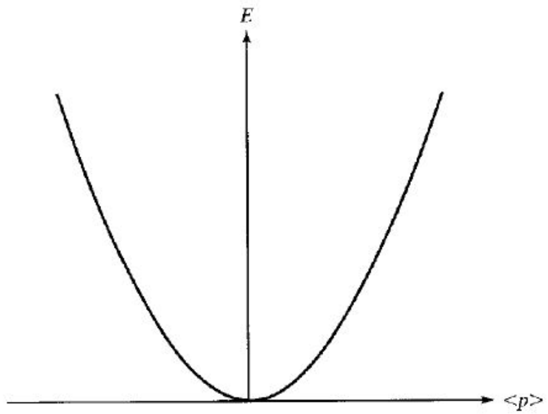
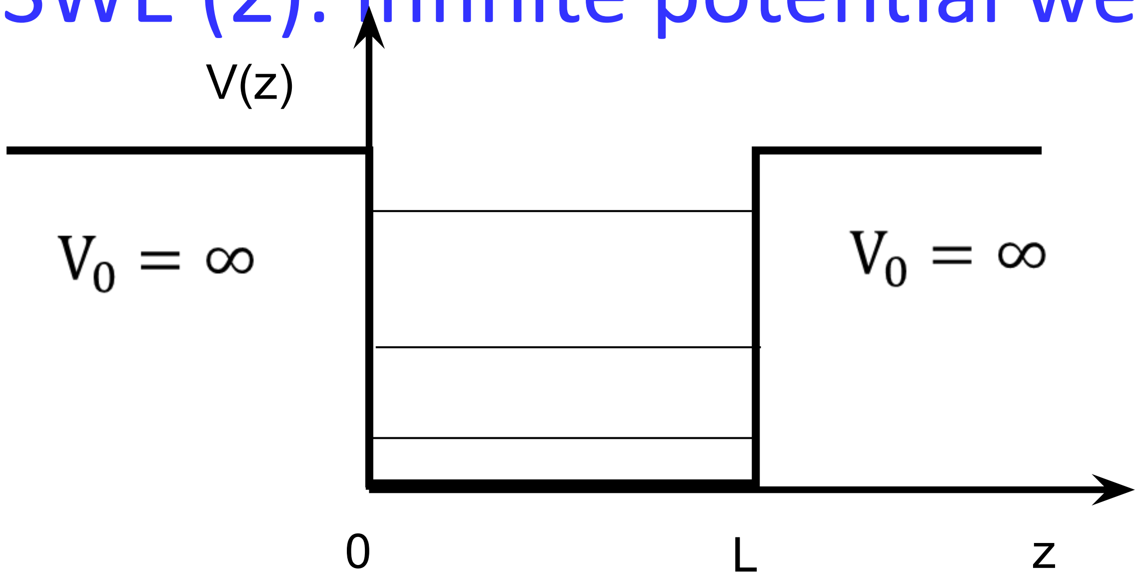


Figure 2.3 Energy–momentum relationship for a free particle.

Solution to time independent SWE (2): Infinite potential well

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dz^2}+V(z)\right]\ \Psi(z)=E\Psi(z)$$



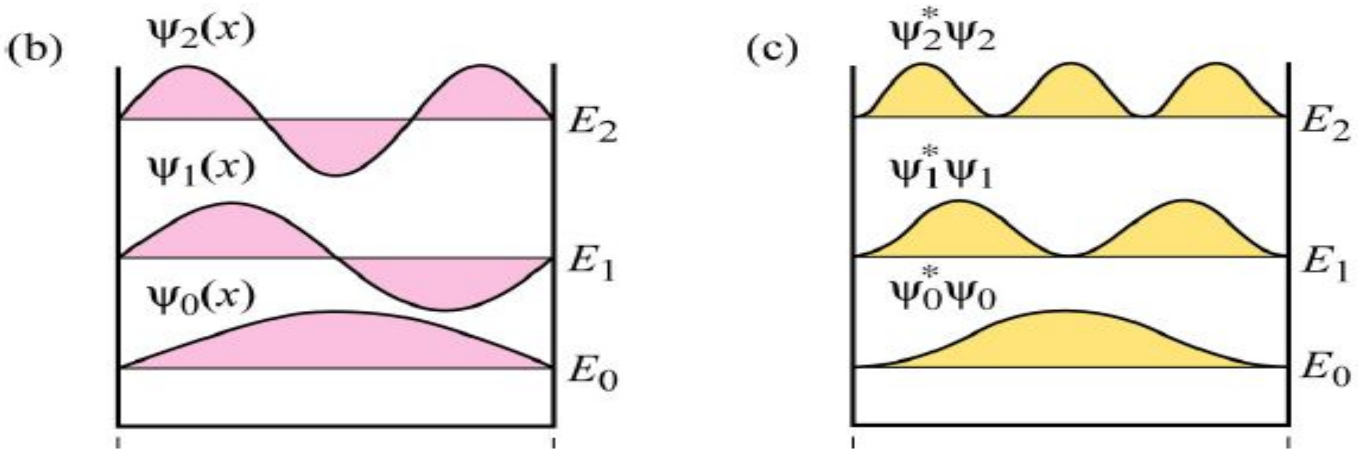
Solution: For z<0 and z>L, $\Psi(z) = 0$

For 0<z<L, $\Psi(z) = A\sin\left(\frac{n\pi}{L}z\right)$

$$\int|\Psi(z)|^2 = 1 \Rightarrow A = \sqrt{\frac{2}{L}}$$

$$\Psi(z) = \sqrt{\frac{2}{L}}\,sin\left(\frac{n\pi}{L}z\right)$$

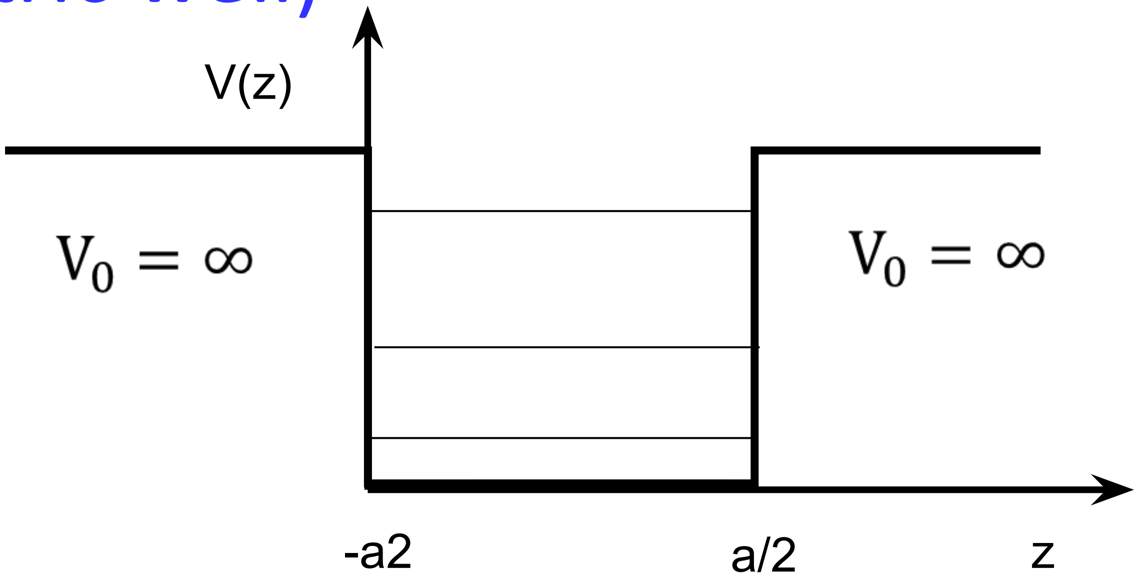
$$E_n = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$$



Infinite potential well (Symmetric well)

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dz^2}+V(z)\right]\Psi(z)=E\Psi(z)$$

For z<-a/2 and z>a/2, $\Psi(z)=0$

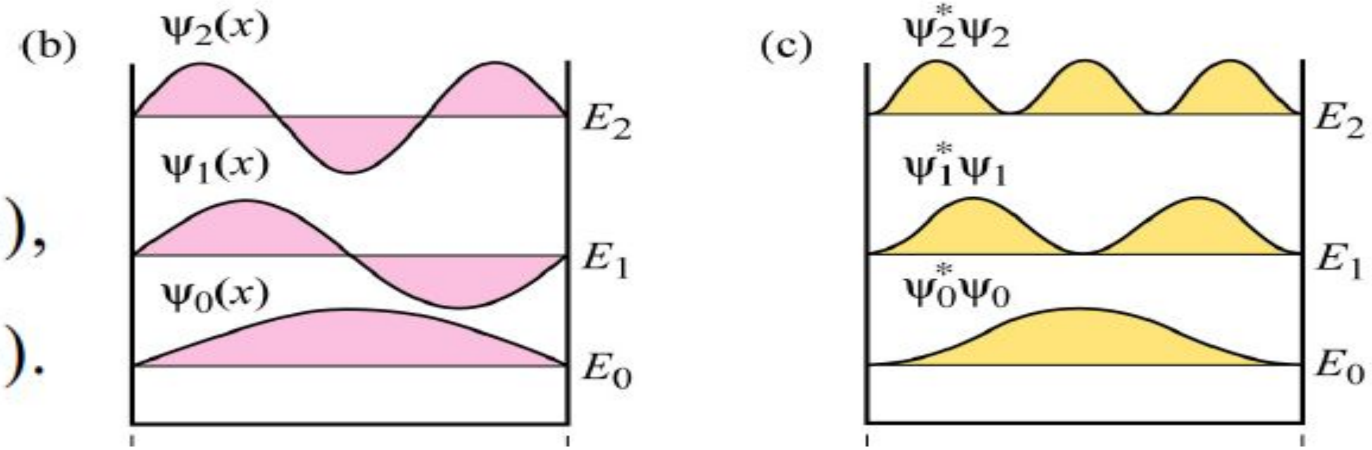


For -L/2<z<L/2,

$$\psi_n(x)=\sqrt{\frac{2}{a}}\sin\left[\frac{n\pi}{a}\left(x+\frac{a}{2}\right)\right]$$

$$E_n=\frac{\hbar^2}{2m}k_n^2=\frac{\hbar^2\pi^2}{2ma^2}n^2 \qquad (n=1,2,3,\cdots).$$

$$\sqrt{\frac{2}{a}}\cos\left(\frac{n\pi}{a}x\right) \qquad (n=1,3,5,7,\cdots),$$
$$\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right) \qquad (n=2,4,6,8,\cdots).$$



Dirac delta function well

Let's consider a potential of the form

$$V(x) = -\alpha\delta(x), \tag{2.117}$$

where α is some positive constant.[45] This is an artificial potential, to be sure (so was the infinite square well), but it's delightfully simple to work with, and illuminates the basic theory with a minimum of analytical clutter. The Schrödinger equation for the delta-function well reads

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} - \alpha\delta(x)\,\psi = E\psi; \tag{2.118}$$

it yields both bound states ($E < 0$) and scattering states ($E > 0$).

We'll look first at the bound states. In the region $x < 0$, $V(x) = 0$, so

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi = \kappa^2\psi, \tag{2.119}$$

where

$$\kappa \equiv \frac{\sqrt{-2mE}}{\hbar}. \tag{2.120}$$

(E is negative, by assumption, so κ is real and positive.) The general solution to Equation 2.119 is

$$\psi(x) = Ae^{-\kappa x} + Be^{\kappa x}, \tag{2.121}$$

but the first term blows up as $x \to -\infty$, so we must choose $A = 0$:

$$\psi(x) = Be^{\kappa x}, \quad (x < 0). \tag{2.122}$$

In the region $x > 0$, $V(x)$ is again zero, and the general solution is of the form $F\exp(-\kappa x) + G\exp(\kappa x)$; this time it's the second term that blows up (as $x \to +\infty$), so

$$\psi(x) = Fe^{-\kappa x}, \quad (x > 0). \tag{2.123}$$

It remains only to stitch these two functions together, using the appropriate boundary conditions at $x = 0$. I quoted earlier the standard boundary conditions for ψ:

$$\boxed{\begin{cases} 1.\ \psi & \text{is always continuous;} \\ 2.\ d\psi/dx & \text{is continuous except at points where the potential is infinite.} \end{cases}} \tag{2.124}$$

In this case the first boundary condition tells us that $F = B$, so

$$\psi(x) = \begin{cases} Be^{\kappa x}, & (x \le 0), \\ Be^{-\kappa x}, & (x \ge 0); \end{cases} \tag{2.125}$$

$\psi(x)$ is plotted in Figure 2.13. The second boundary condition tells us nothing; this is (like the walls of the infinite square well) the exceptional case where V is infinite at the join, and it's clear from the graph that this function has a kink at $x = 0$. Moreover, up to this point the delta function has not come into the story at all. It turns out that the delta function determines the *discontinuity in the derivative* of ψ, at $x = 0$. I'll show you now how this works, and as a byproduct we'll see why $d\psi/dx$ is ordinarily continuous.

The idea is to *integrate* the Schrödinger equation, from $-\epsilon$ to $+\epsilon$, and then take the limit as $\epsilon \to 0$:

$$-\frac{\hbar^2}{2m}\int_{-\epsilon}^{+\epsilon}\frac{d^2\psi}{dx^2}\,dx + \int_{-\epsilon}^{+\epsilon}V(x)\,\psi(x)\,dx = E\int_{-\epsilon}^{+\epsilon}\psi(x)\,dx \tag{2.126}$$

The first integral is nothing but $d\psi/dx$, evaluated at the two end points; the last integral is *zero*, in the limit $\epsilon \to 0$, since it's the area of a sliver with vanishing width and finite height. Thus

$$\Delta\left(\frac{d\psi}{dx}\right) \equiv \lim_{\epsilon\to 0}\left(\left.\frac{\partial\psi}{\partial x}\right|_{+\epsilon} - \left.\frac{\partial\psi}{\partial x}\right|_{-\epsilon}\right) = \frac{2m}{\hbar^2}\lim_{\epsilon\to 0}\int_{-\epsilon}^{+\epsilon}V(x)\,\psi(x)\,dx. \tag{2.127}$$

Ordinarily, the limit on the right is again zero, and that's why $d\psi/dx$ is ordinarily continuous. But when $V(x)$ is *infinite* at the boundary, this argument fails. In particular, if $V(x) = -\alpha\delta(x)$, Equation 2.116 yields

$$\Delta\left(\frac{d\psi}{dx}\right) = -\frac{2m\alpha}{\hbar^2}\psi(0). \tag{2.128}$$

For the case at hand (Equation 2.125),

$$\begin{cases} d\psi/dx = -B\kappa e^{-\kappa x}, \text{ for } (x > 0), & \text{so } d\psi/dx|_{+} = -B\kappa, \\ d\psi/dx = +B\kappa e^{+\kappa x}, \text{ for } (x < 0), & \text{so } d\psi/dx|_{-} = +B\kappa, \end{cases}$$

and hence $\Delta(d\psi/dx) = -2B\kappa$. And $\psi(0) = B$. So Equation 2.128 says

$$\kappa = \frac{m\alpha}{\hbar^2}, \tag{2.129}$$

and the allowed energy (Equation 2.120) is

$$E = -\frac{\hbar^2\kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2}. \tag{2.130}$$

Finally, we normalize ψ:

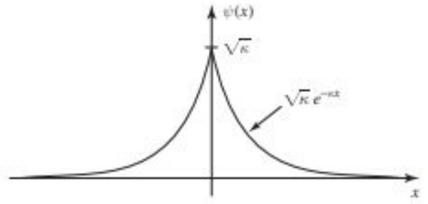
$$\int_{-\infty}^{+\infty}|\psi(x)|^2\,dx = 2|B|^2\int_0^{\infty}e^{-2\kappa x}\,dx = \frac{|B|^2}{\kappa} = 1,$$

so (choosing the positive real root):

$$B = \sqrt{\kappa} = \frac{\sqrt{m\alpha}}{\hbar}. \tag{2.131}$$

Evidently the delta function well, regardless of its "strength" α, has *exactly one* bound state:

$$\boxed{\psi(x) = \frac{\sqrt{m\alpha}}{\hbar}e^{-m\alpha|x|/\hbar^2}; \quad E = -\frac{m\alpha^2}{2\hbar^2}.} \tag{2.132}$$



2.13: Bound state wave function for the delta-function potential

What about *scattering* states, with $E > 0$? For $x < 0$ the Schrödinger equation reads

$$\frac{d^2\psi}{dx^2} = -\frac{2mE}{\hbar^2}\psi = -k^2\psi,$$

where

$$k \equiv \frac{\sqrt{2mE}}{\hbar} \tag{2.133}$$

is real and positive. The general solution is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \tag{2.134}$$

and this time we cannot rule out either term, since neither of them blows up. Similarly, for $x > 0$,

$$\psi(x) = Fe^{ikx} + Ge^{-ikx}. \tag{2.135}$$

The continuity of $\psi(x)$ at $x = 0$ requires that

$$F + G = A + B. \tag{2.136}$$

The derivatives are

$$\begin{cases} d\psi/dx = ik\left(Fe^{ikx} - Ge^{-ikx}\right), \text{ for } (x > 0), & \text{so } d\psi/dx\big|_+ = ik(F - G), \\ d\psi/dx = ik\left(Ae^{ikx} - Be^{-ikx}\right), \text{ for } (x < 0), & \text{so } d\psi/dx\big|_- = ik(A - B), \end{cases}$$

and hence $\Delta(d\psi/dx) = ik(F - G - A + B)$. Meanwhile, $\psi(0) = (A + B)$, so the second boundary condition (Equation 2.128) says

$$ik(F - G - A + B) = -\frac{2m\alpha}{\hbar^2}(A + B), \tag{2.137}$$

or, more compactly,

$$F - G = A(1 + 2i\beta) - B(1 - 2i\beta), \quad \text{where } \beta \equiv \frac{m\alpha}{\hbar^2 k}. \tag{2.138}$$

Having imposed both boundary conditions, we are left with two equations (Equations 2.136 and 2.138) in four unknowns (A, B, F, and G)—*five*, if you count k. Normalization won't help—this isn't a normalizable state. Perhaps we'd better pause, then, and examine the physical significance of these various constants. Recall that $\exp(ikx)$ gives rise (when coupled with the wiggle factor $\exp(-iEt/\hbar)$) to a wave function propagating to the *right*, and $\exp(-ikx)$ leads to a wave propagating to the *left*. It follows that A (in Equation 2.134) is the amplitude of a wave coming in from the left, B is the amplitude of a wave returning to the left; F (Equation 2.135) is the amplitude of a wave traveling off to the right, and G is the amplitude of a wave coming in from the right (see Figure 2.14). In a typical scattering experiment particles are fired in from one direction—let's say, from the left. In that case the amplitude of the wave coming in from the *right* will be *zero*:

$$G = 0 \quad \text{(for scattering from the left)}; \tag{2.139}$$

A is the amplitude of the **incident wave**, B is the amplitude of the **reflected wave**, and F is the amplitude of the **transmitted wave**. Solving Equations 2.136 and 2.138 for B and F, we find

$$B = \frac{i\beta}{1 - i\beta}A, \quad F = \frac{1}{1 - i\beta}A. \tag{2.140}$$

(If you want to study scattering from the *right*, set $A = 0$; then G is the incident amplitude, F is the reflected amplitude, and B is the transmitted amplitude.)

Now, the probability of finding the particle at a specified location is given by $|\Psi|^2$, so the *relative*[46] probability that an incident particle will be reflected back is

$$R \equiv \frac{|B|^2}{|A|^2} = \frac{\beta^2}{1 + \beta^2}. \tag{2.141}$$

R is called the **reflection coefficient**. (If you have a *beam* of particles, it tells you the *fraction* of the incoming number that will bounce back.) Meanwhile, the probability that a particle will continue right on through is the **transmission coefficient**[47]

$$T \equiv \frac{|F|^2}{|A|^2} = \frac{1}{1 + \beta^2}. \tag{2.142}$$

Of course, the *sum* of these probabilities should be 1—and it *is*:

$$R + T = 1. \tag{2.143}$$

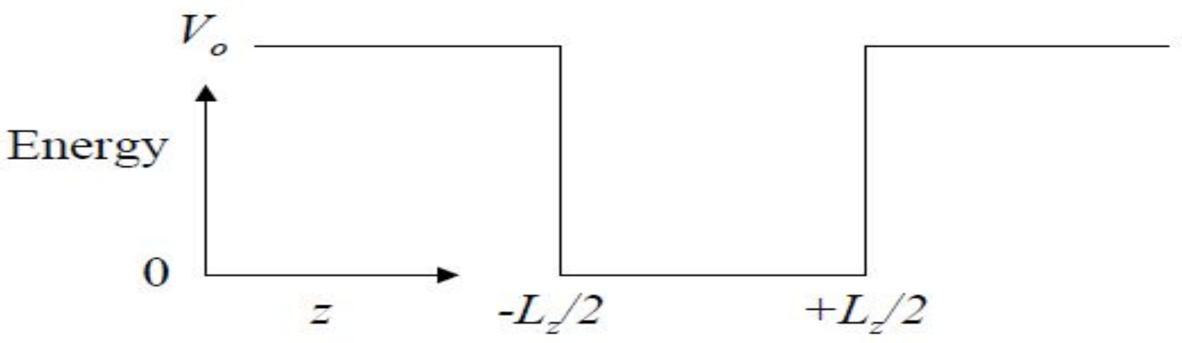
Notice that R and T are functions of β, and hence (Equations 2.133 and 2.138) of E:

$$\boxed{R = \frac{1}{1 + \left(2\hbar^2 E/m\alpha^2\right)}, \quad T = \frac{1}{1 + \left(m\alpha^2/2\hbar^2 E\right)}.} \tag{2.144}$$

The higher the energy, the greater the probability of transmission (which makes sense).

This is all very tidy, but there is a sticky matter of principle that we cannot altogether ignore: These scattering wave functions are not normalizable, so they don't actually represent possible particle states. We know the resolution to this problem: form normalizable linear combinations

Finite potential well



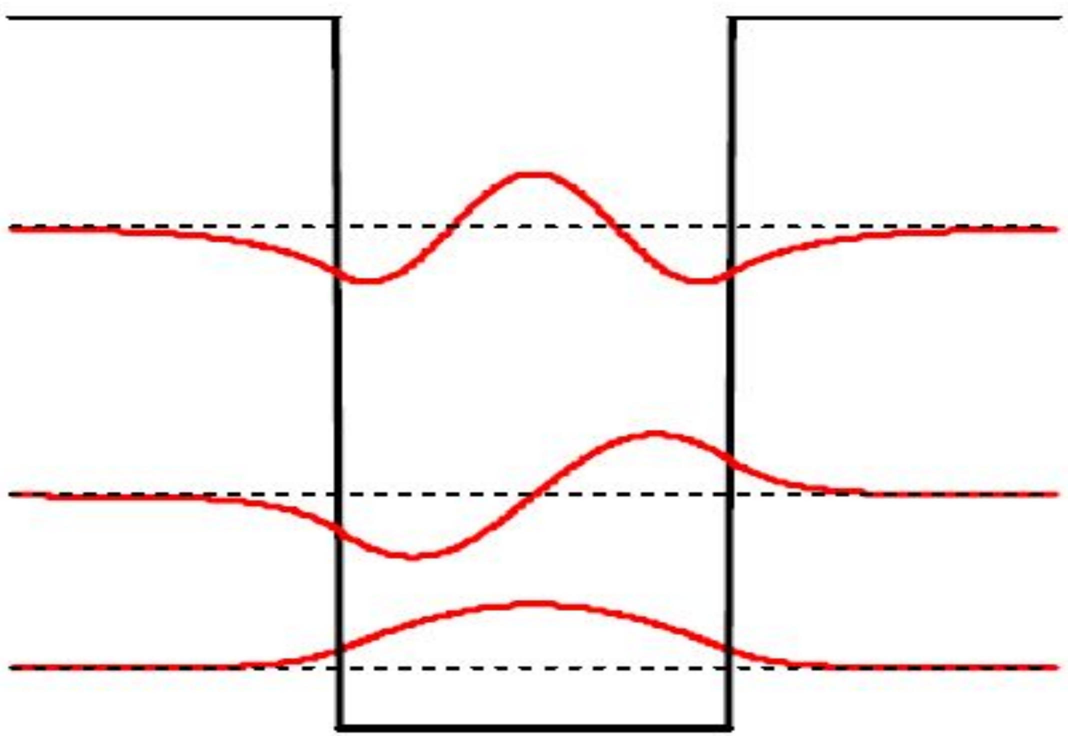
$$\psi(z) = G\exp(\kappa z),$$
$$\psi(z) = A\sin kz + B\cos kz\ , \text{-}$$
$$\psi(z) = F\exp(-\kappa z)$$

$$\kappa = \sqrt{2m(V_o - E)/\hbar^2}$$

$$(\psi \rightarrow 0 \text{ as } x \rightarrow \pm\infty)$$

Solve for G, A, B and F by applying continuity of Ψ and $d\Psi/dz$ at $z = \mp L_z/2$

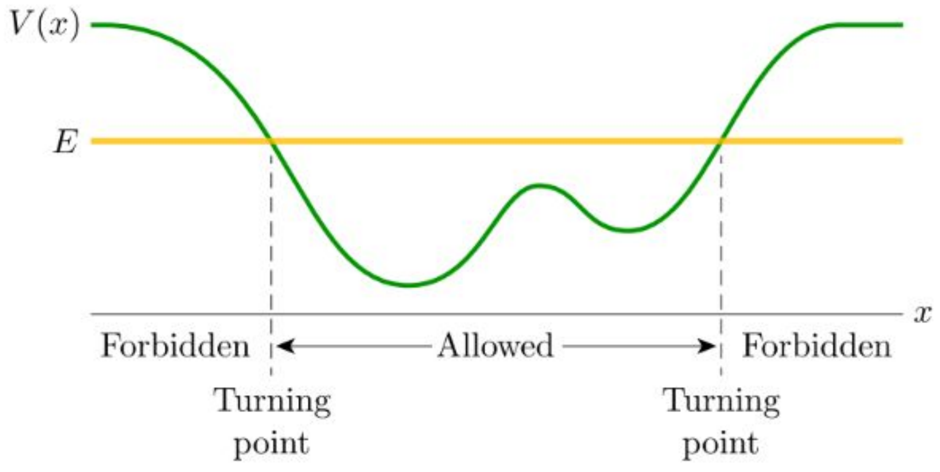
Only numerical or graphical solution possible

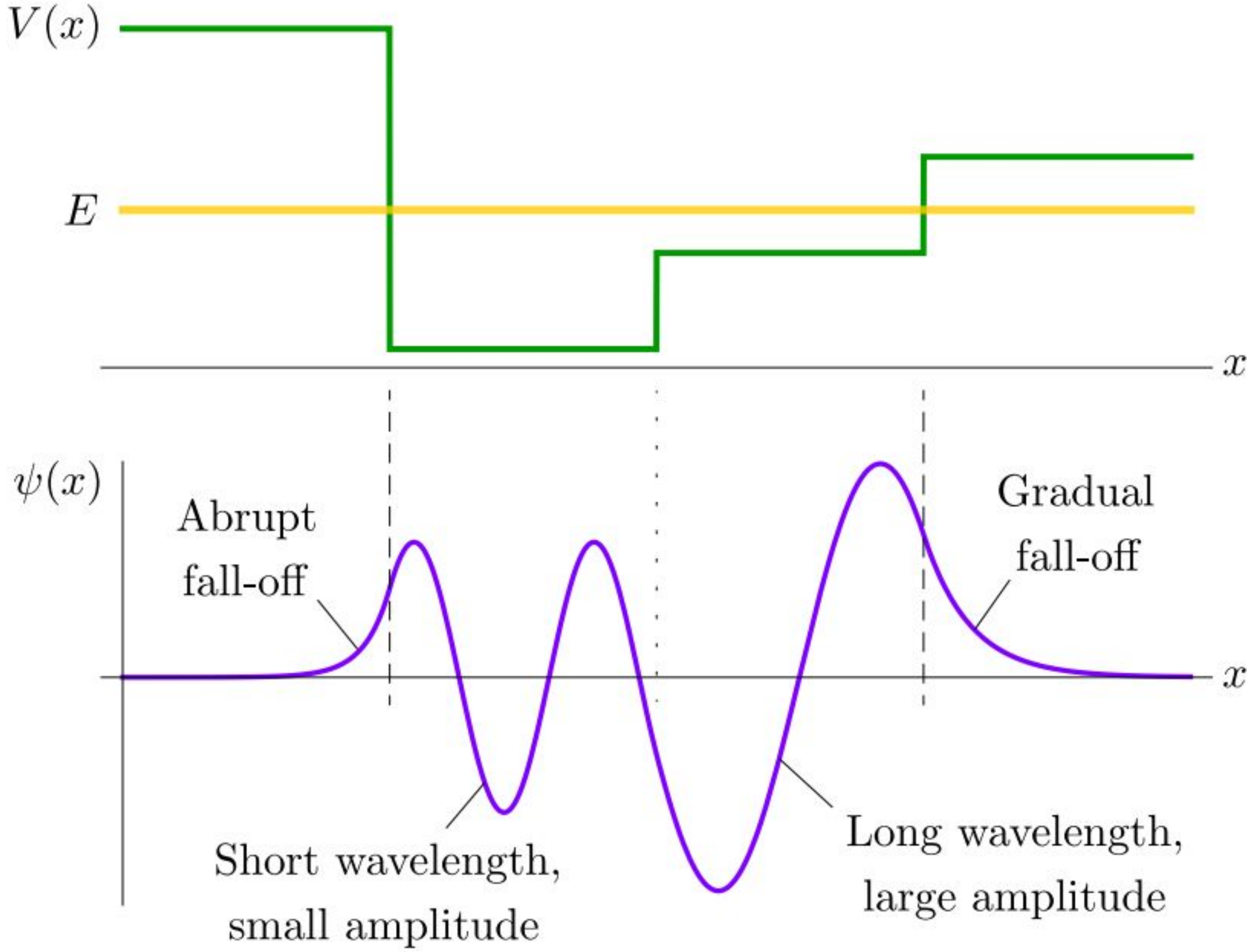


Is the first eigenvalue lower or higher in energy compared to that of an infinite potential well ?

The energies in the finite well are lower than those in the infinite well because the finite well allows the wavefunctions to “spill out” into the classically forbidden regions, resulting in longer wavelengths.

Asymmetric Finite potential well

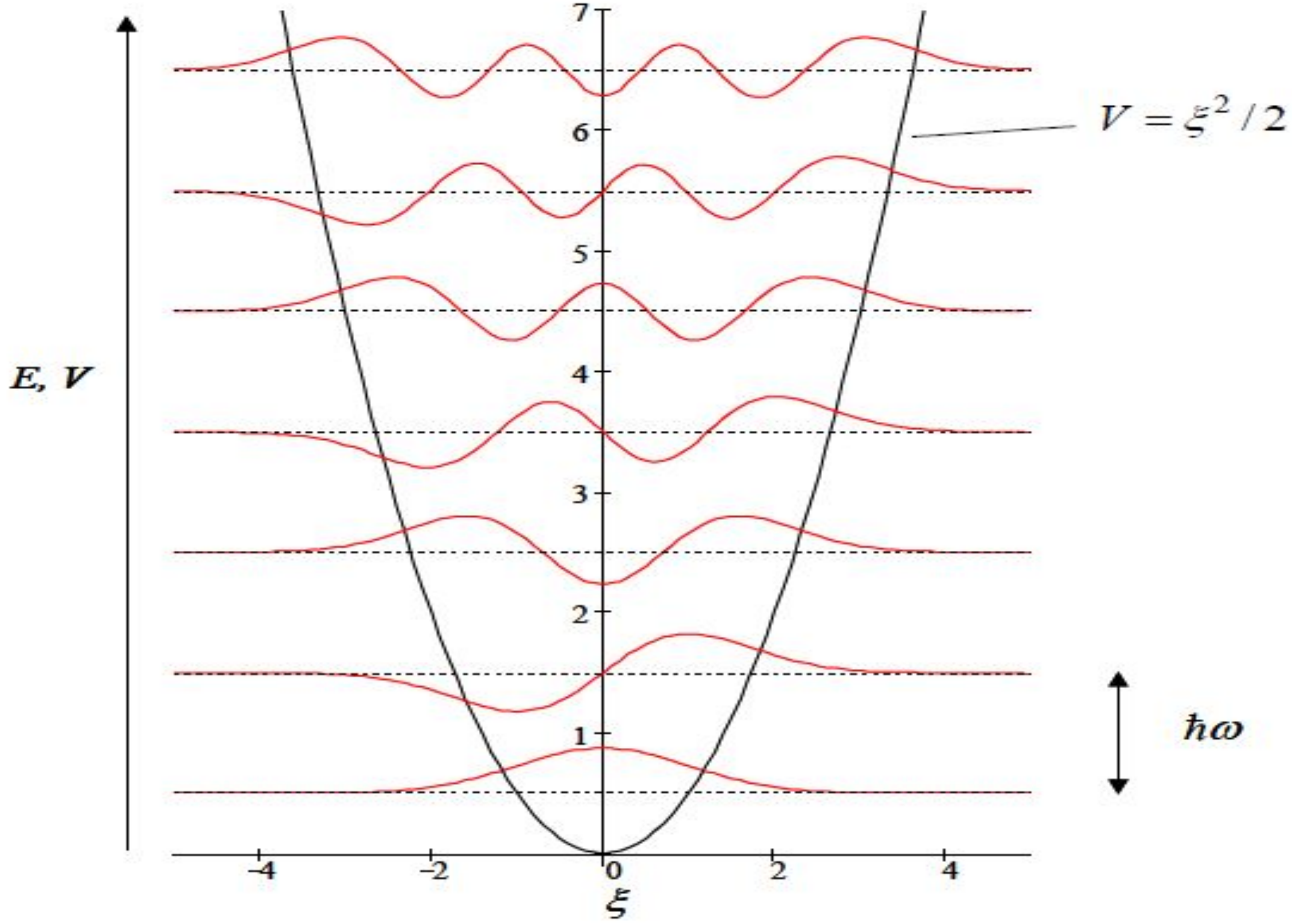


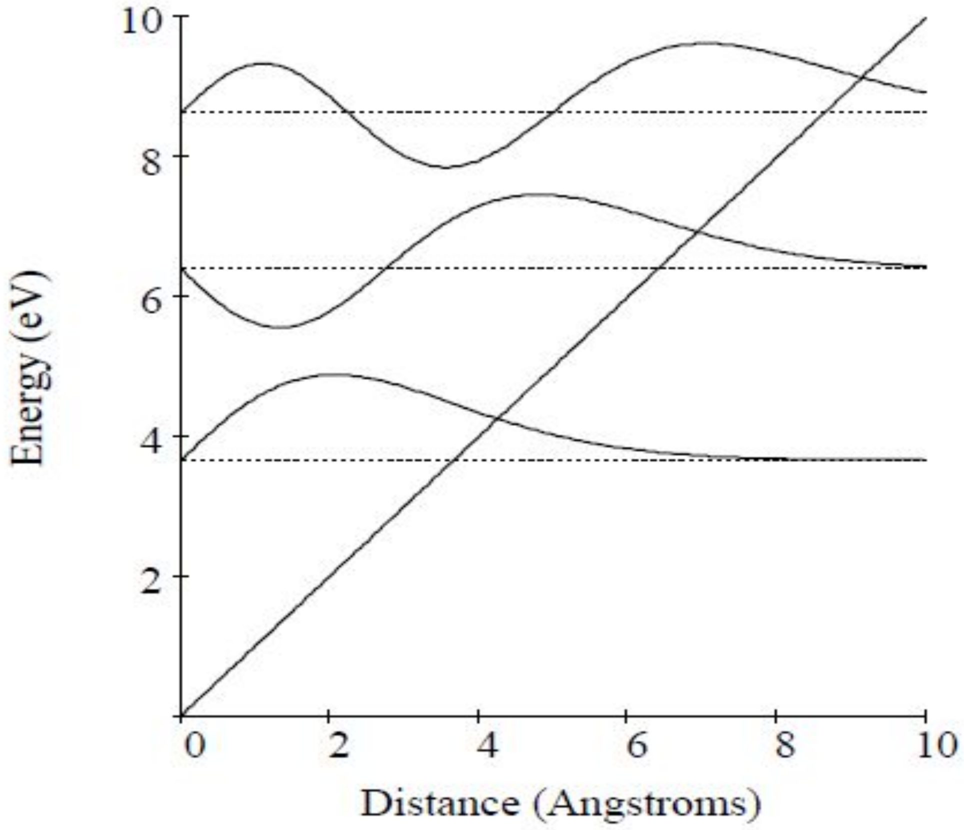


• In a classically allowed region, the solutions oscillate. Both the amplitude and the wavelength of the oscillations will be smaller where E − V (x) is large.

• In a classically forbidden region, the solutions have exponential-like behavior, dying out as the distance from a classically allowed region increases. The wavefunction dies out more abruptly where V (x) − E is large.

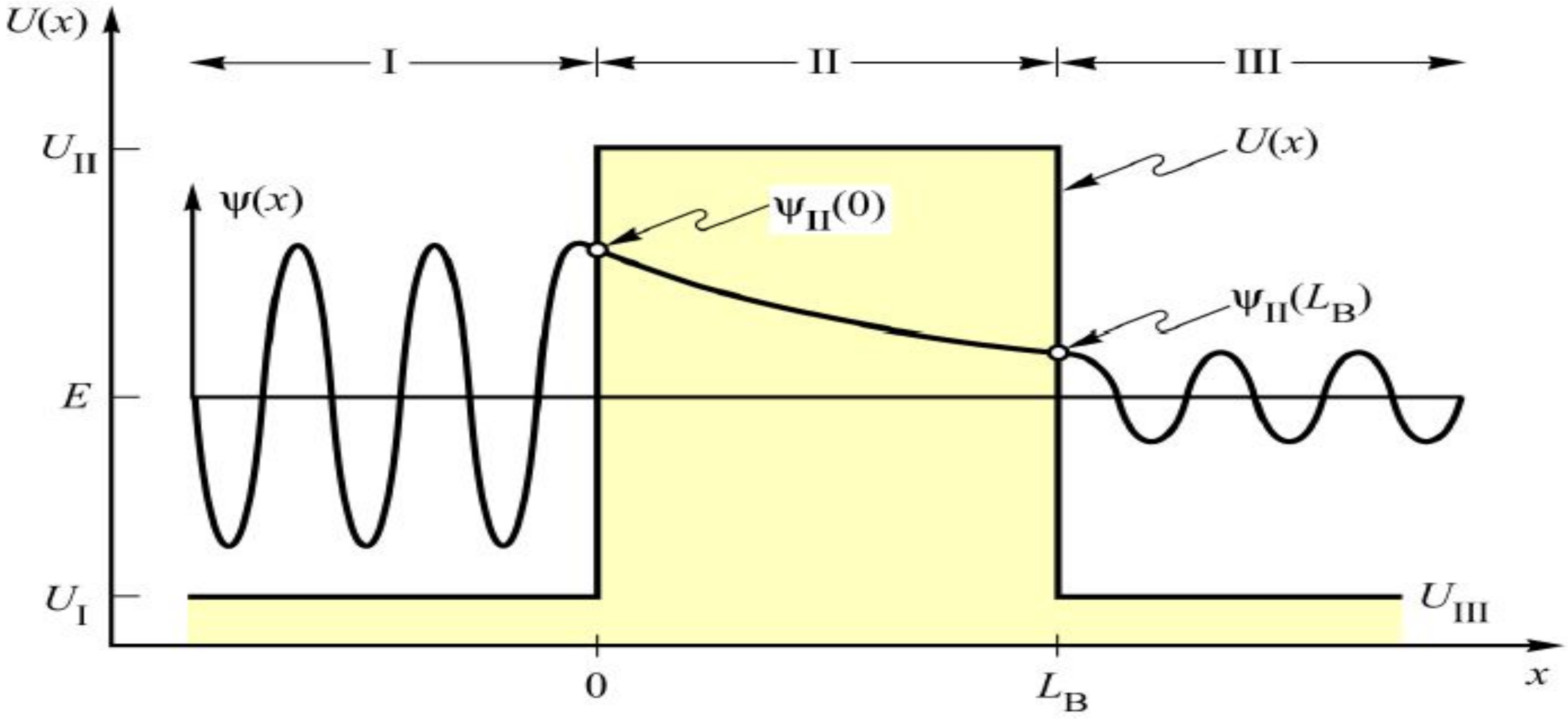
Parabolic and linearly varying potential wells





Eigenvalues are spaced uniformly for a parabolic potential well. Wave functions are asymmetric for an asymmetric triangular potential well

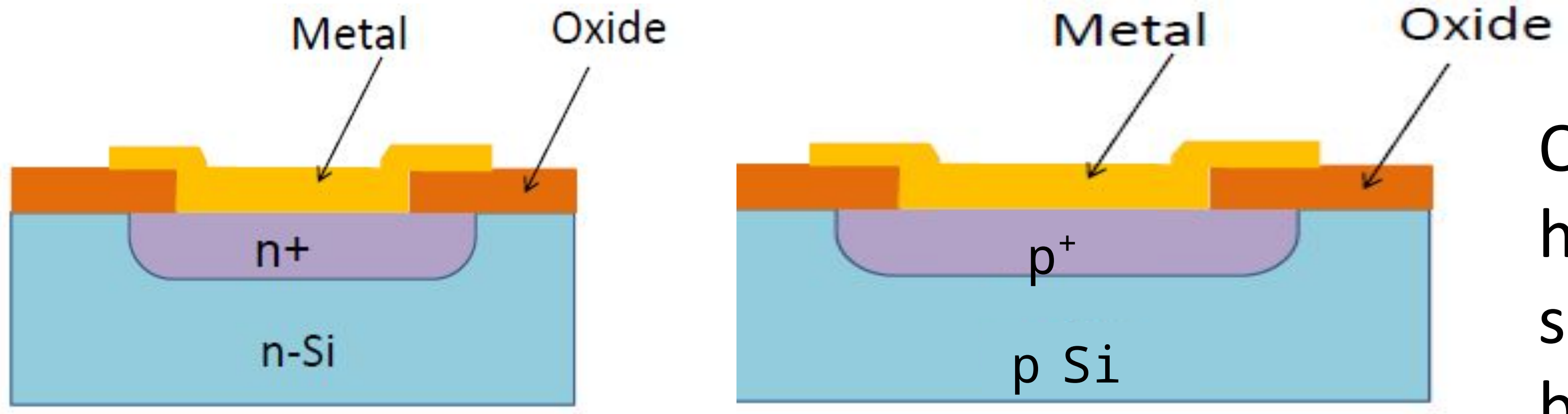
Tunneling through a potential barrier



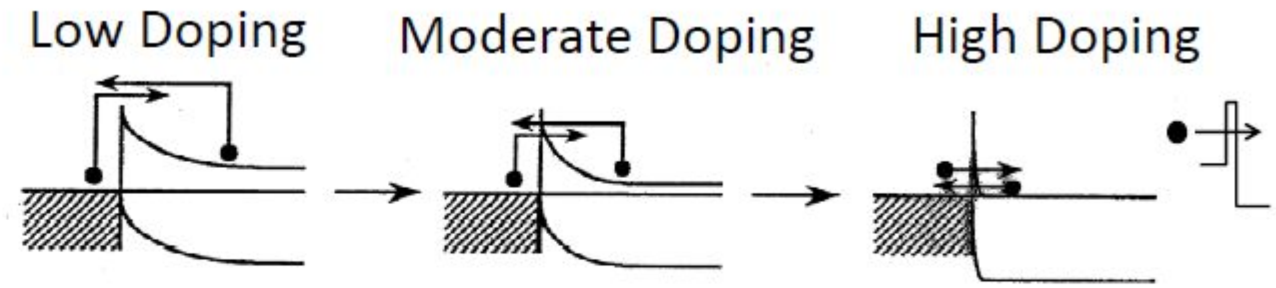
$$T = \frac{\psi_{II}^{*}(L_B)\,\psi_{II}(L_B)}{\psi_{II}^{*}(0)\,\psi_{II}(0)}$$

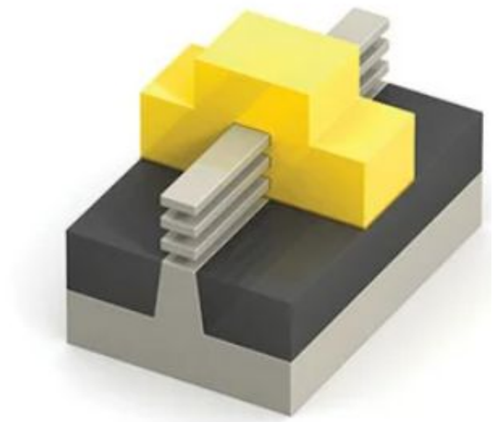
Finite tunneling probability even if energy of particle is less than potential barrier

How Ohmic contacts are made in ICs

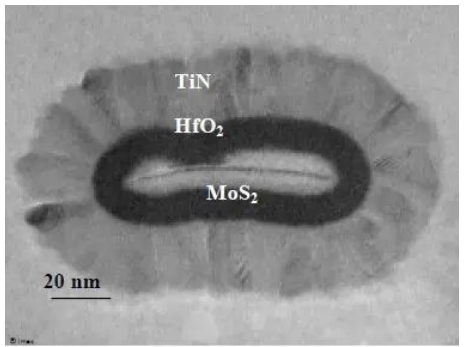


Ohmic contacts are made by heavily doping (~10^{19}/cm^3) the surface region immediately beneath the contact. The resulting barrier width is very small to enable tunneling





Stacked nanosheet FET

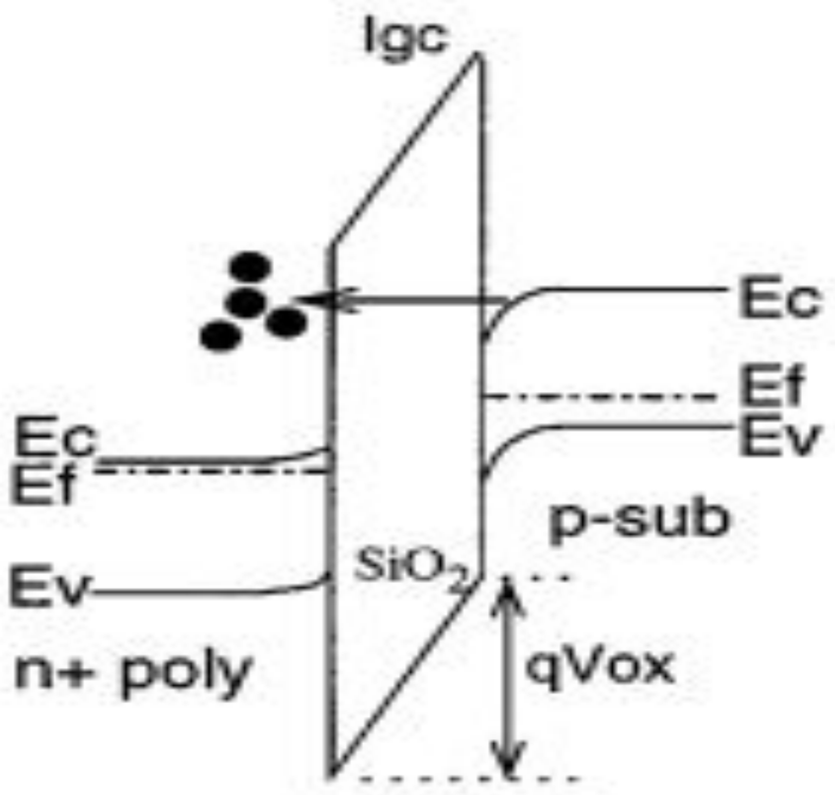


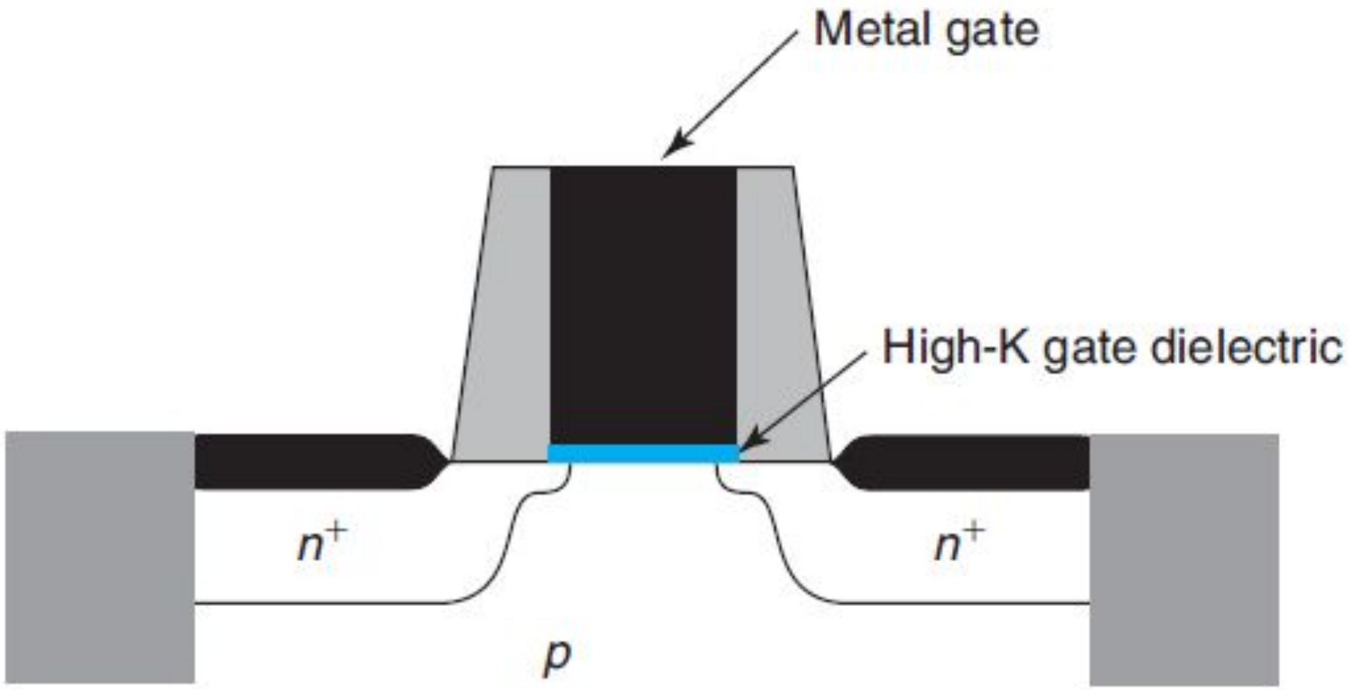
Lack of good Ohmic contacts is one of the biggest challenges for realizing MOSFETs with 2D materials as channel for post-Si (circa 2037)

https://www.imec-int.com/en/articles/introducing-2d-material-based-devices-logic-scaling-roadmap
https://spectrum.ieee.org/the-nanosheet-transistor-is-the-next-and-maybe-last-step-in-moores-law

Excessive gate leakage current through ~ 2 nm thin barrier

forced MOS Field Effect Transistor to undergo a heart replacement



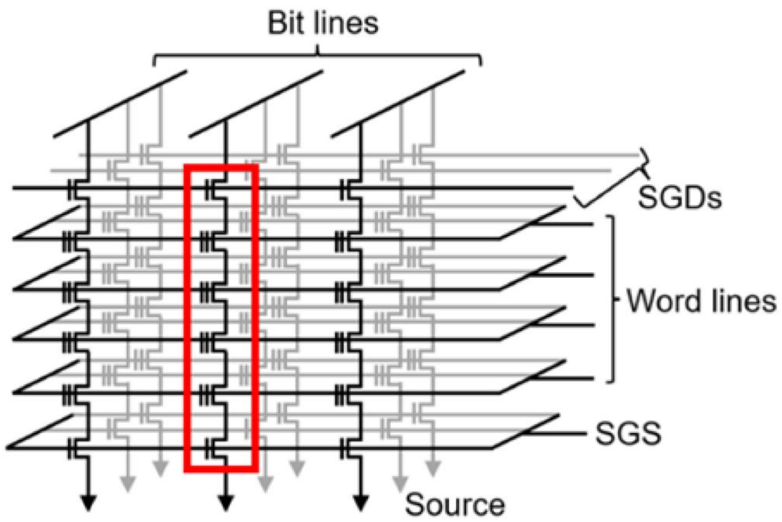


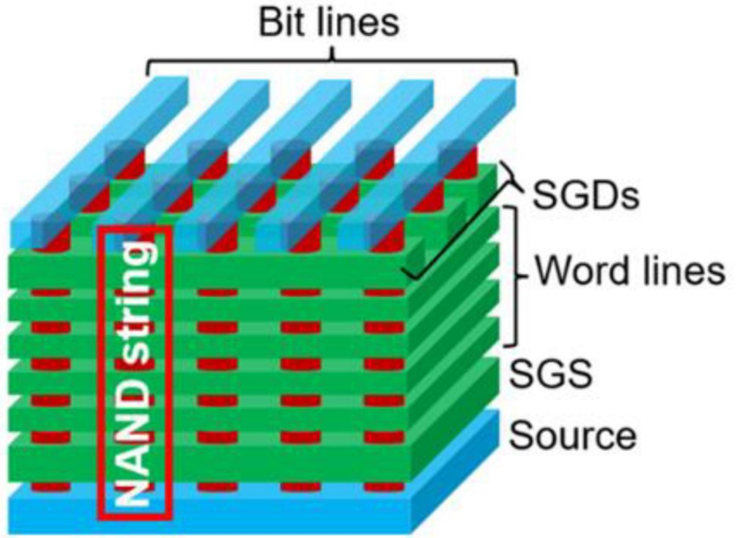
Hf based higher dielectric constant material as gate insulator and metal in place of polysilicon gate (since 45/32 nm in 2007)

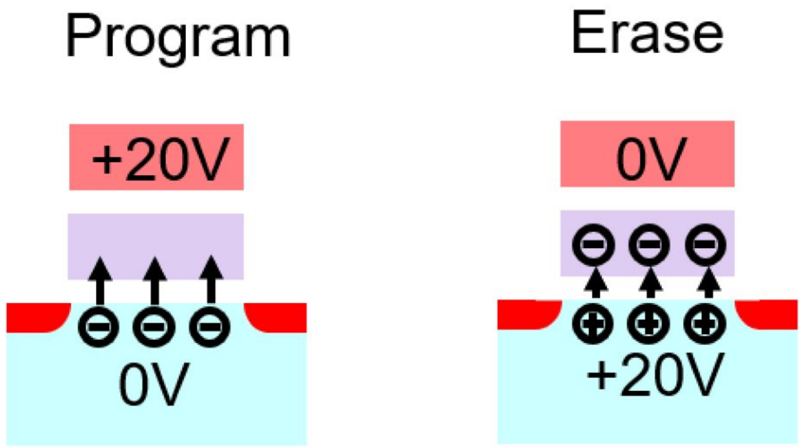
https://spectrum.ieee.org/transistor-history-2670915657

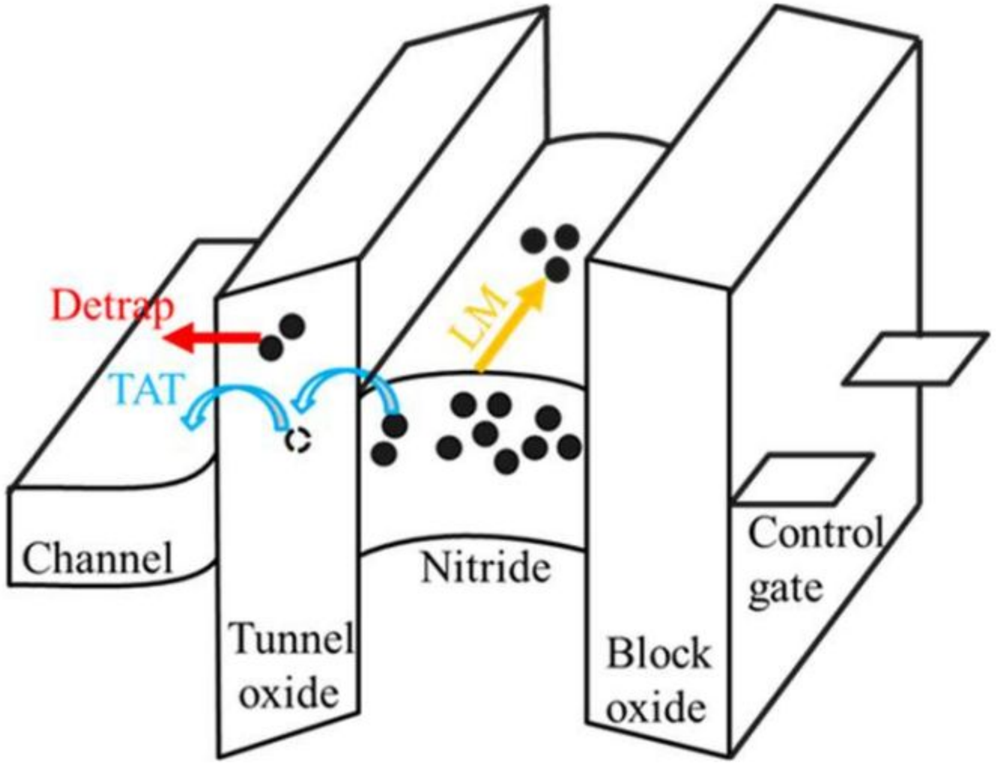
3D NAND Flash memory

The lowest cost-per-bit memory behind iPod, Solid State Drive..



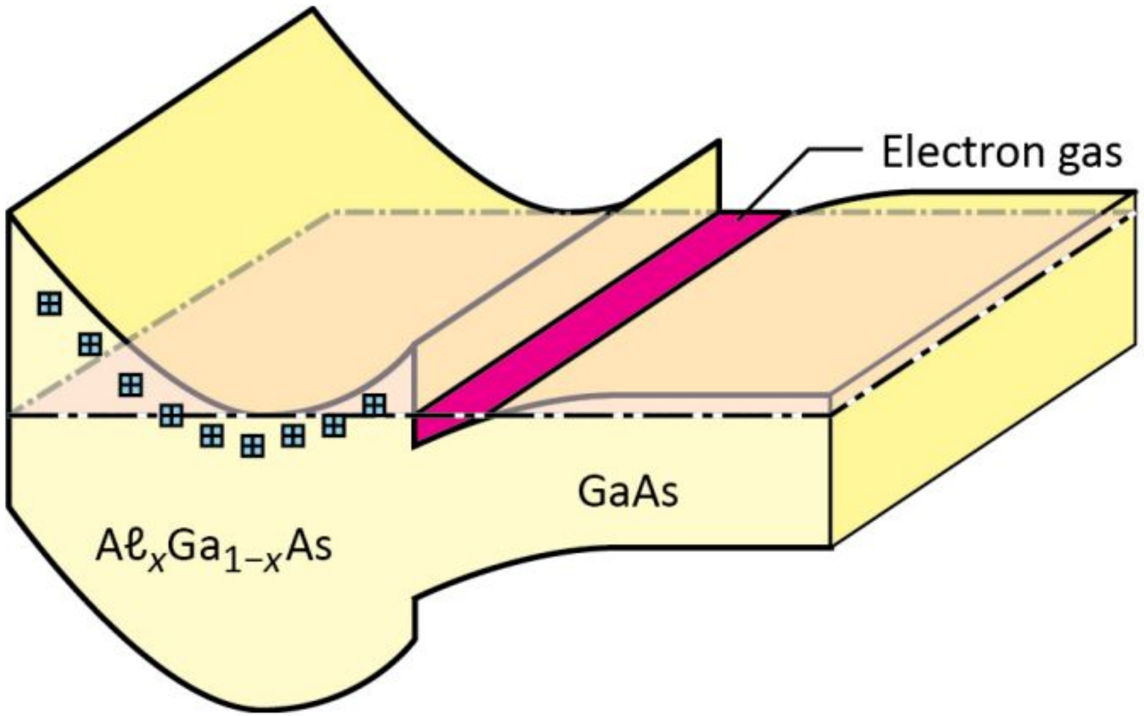


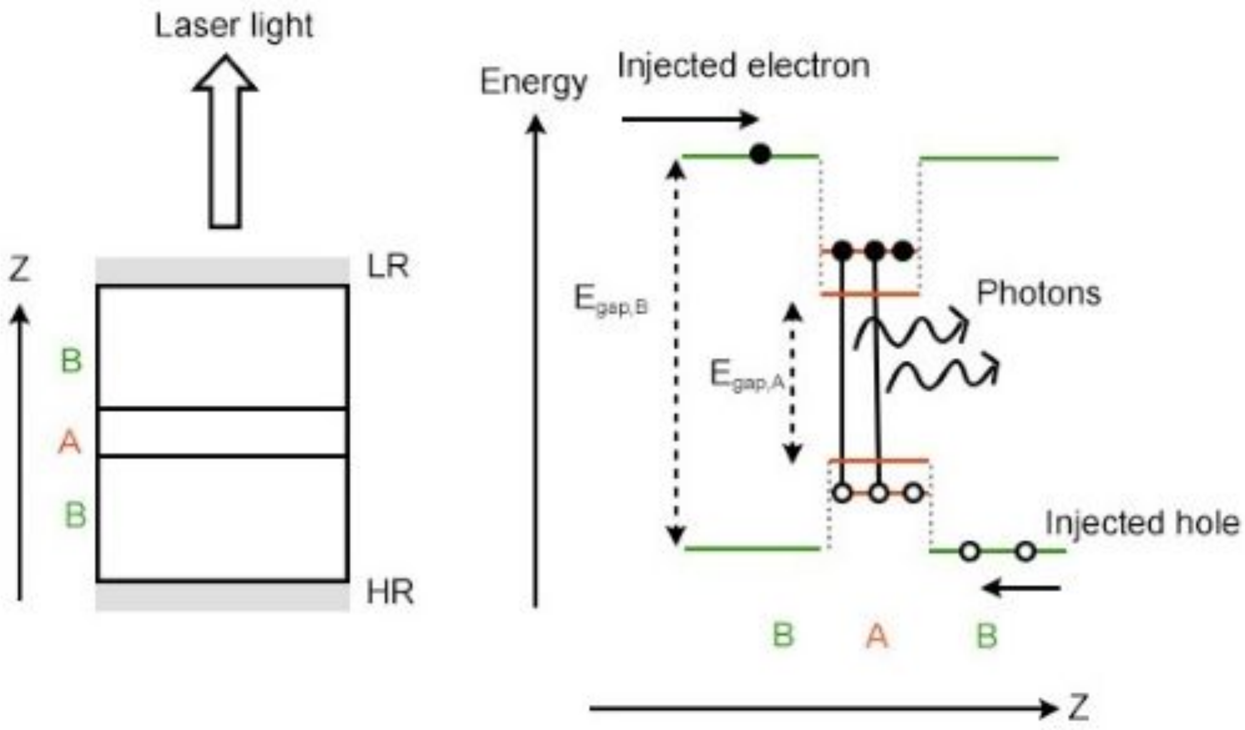




Program and erase done by tunneling carriers into/out of the quantum well

Other devices with quantum wells



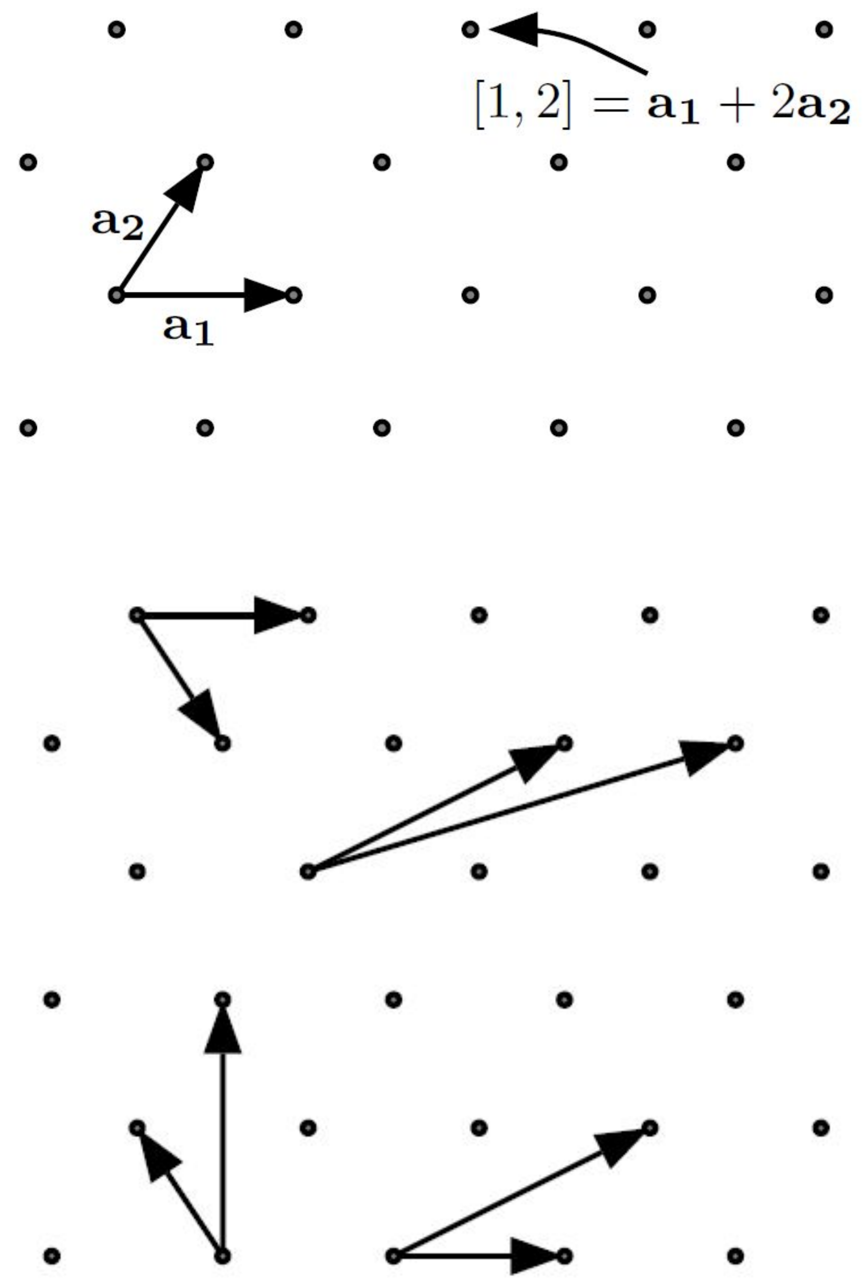


GaAs/GaN High electron mobility transistors

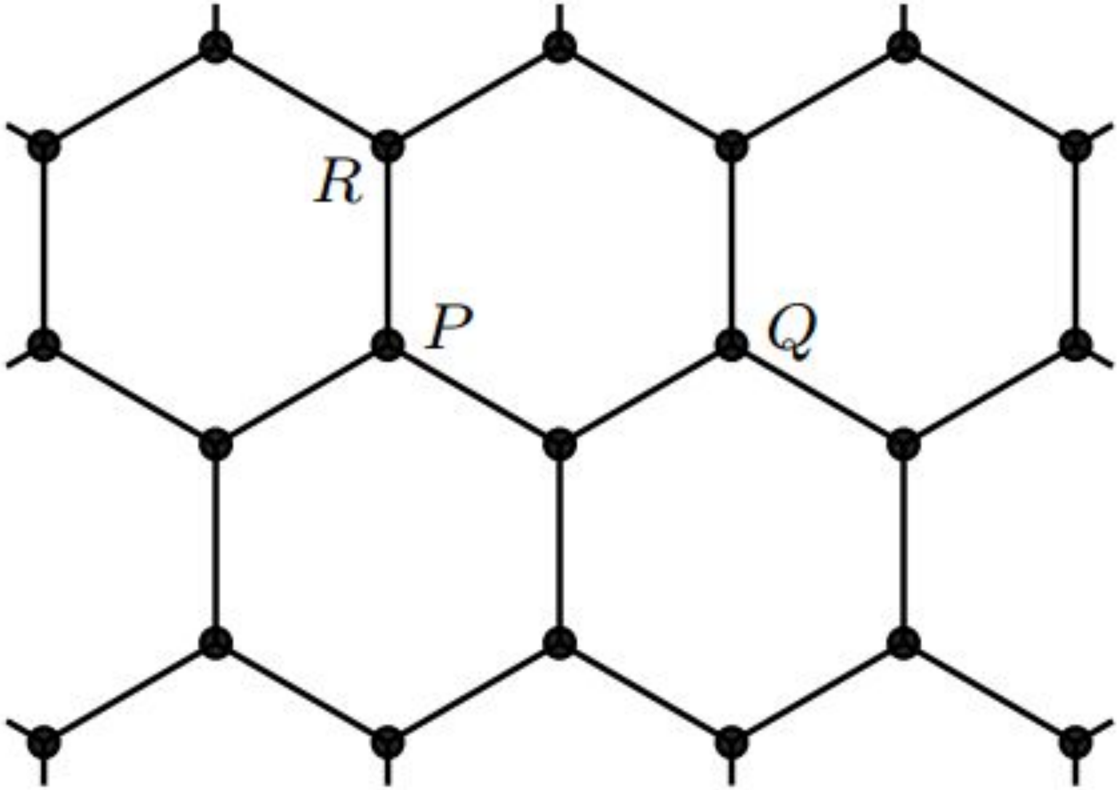
Heterojunction quantum wells are the heart of LED and LASER

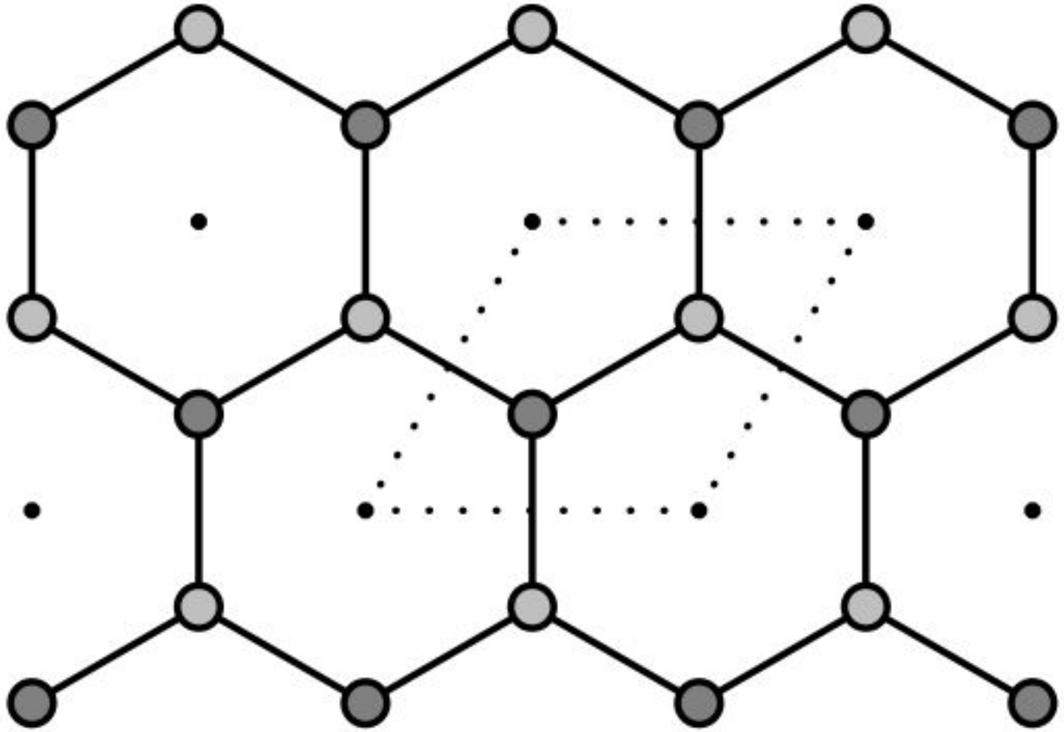
What is a Bravais Lattice?

- Lattice is an infinite set of points defined by integer sums of a set of linearly independent primitive lattice vectors (needn't be position of atoms)
- Lattice is a set of points where the environment of any given point is equivalent to the environment of any other given point
- Choice of primitive lattice vectors for a lattice is not unique



Vertices of Hexagonal comb (Can it be considered as a Bravais lattice?)

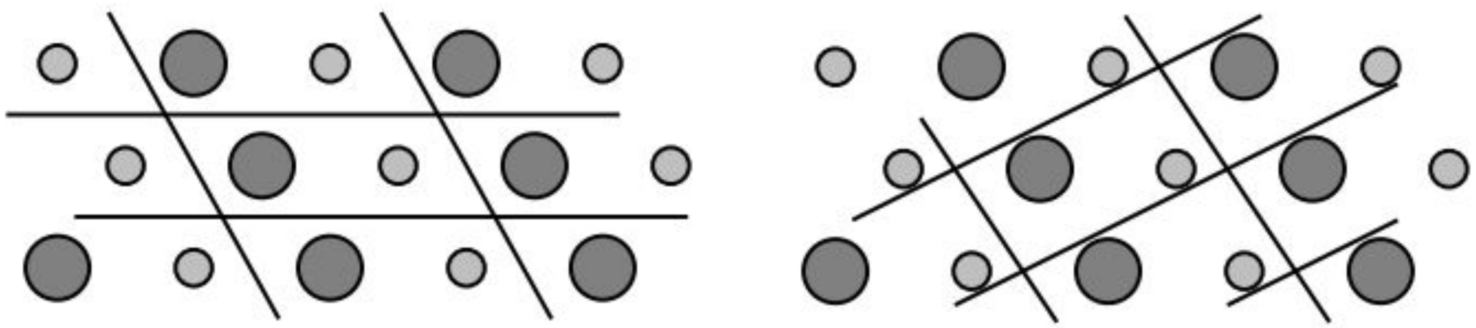




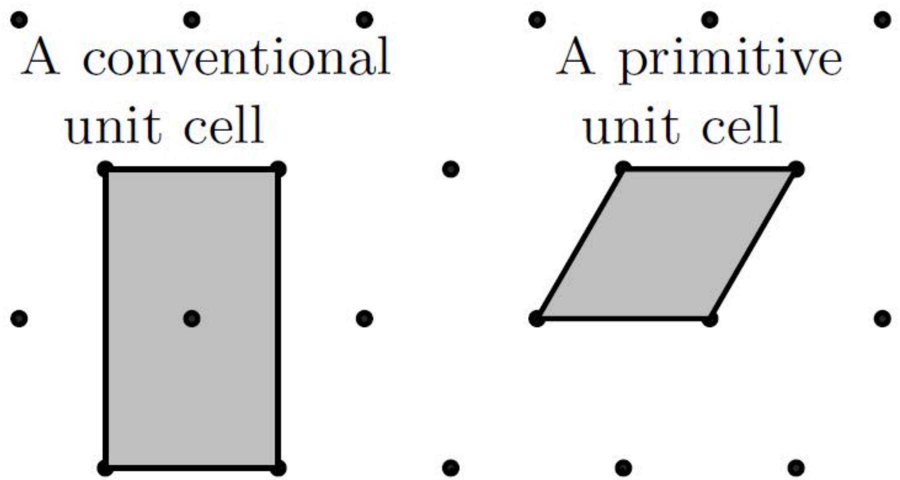
- Not all periodic arrangements of points are lattices
- For complicated arrangements of points, understand **unit cell**

Unit cell and primitive unit cell

- **Unit cell** is a region of space such that when many identical units are stacked together it tiles (completely fills) all of space and reconstructs the full structure.

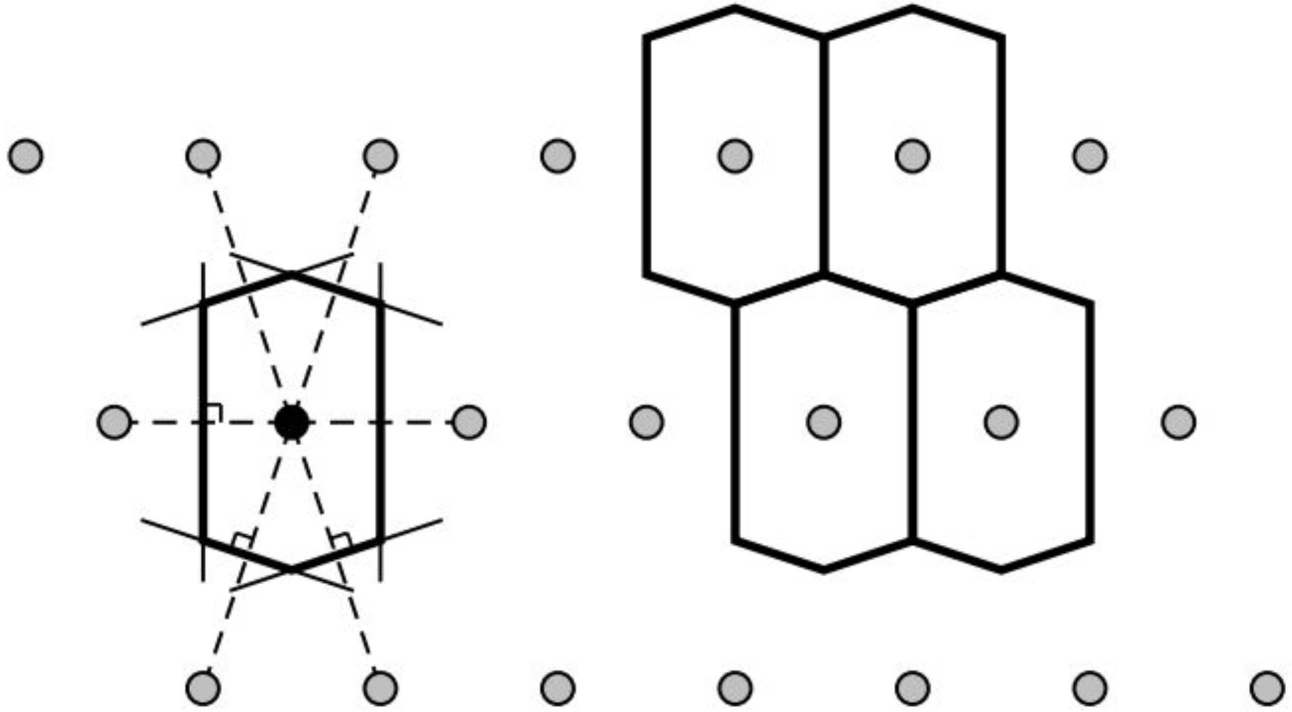


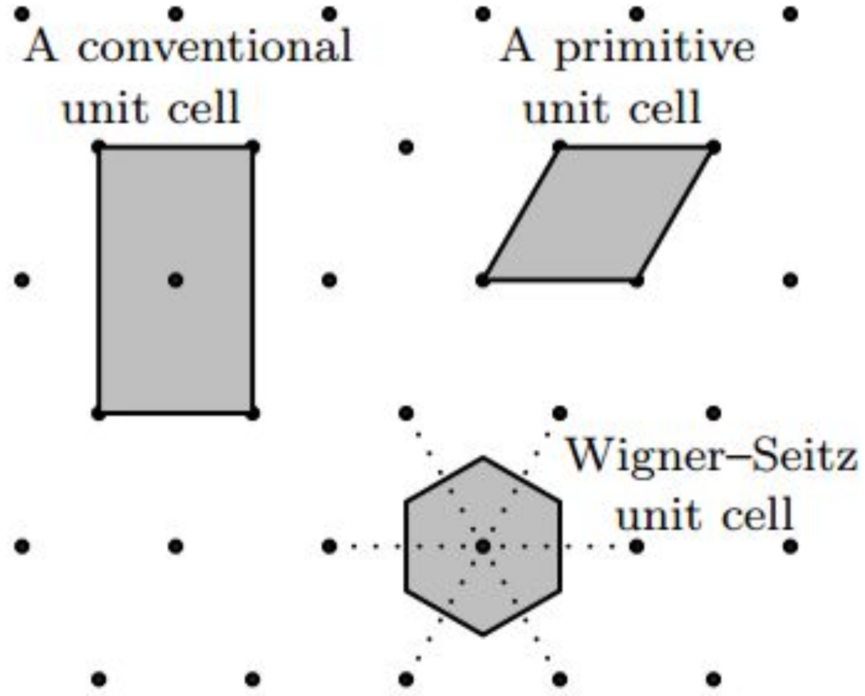
- **Primitive unit cell** for a periodic crystal is a unit cell containing exactly one lattice point.
- The choice of a unit cell or a primitive unit cell isn’t unique



Wigner-Seitz primitive cell

Given a lattice point, the set of all points in space which are closer to that given lattice point than to any other lattice point constitute the **Wigner–Seitz cell** of the given lattice point.





Volume or area of all primitive unit cells are the same (density of lattice points x volume of cell=1 for all such cells)

3D Bravais Lattices (2)

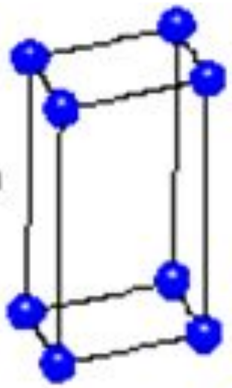
HEXAGONAL

$a = b \neq c$

$\alpha = \beta = 90°$

$\gamma = 120°$

P

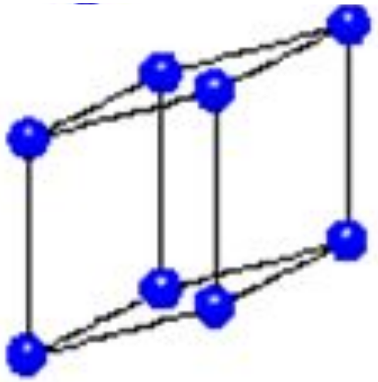


TRIGONAL

$a = b = c$

$\alpha = \beta = \gamma \neq 90°$

P



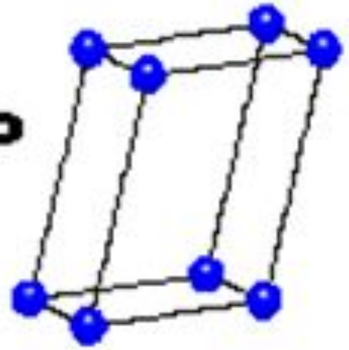
MONOCLINIC

$a \neq b \neq c$

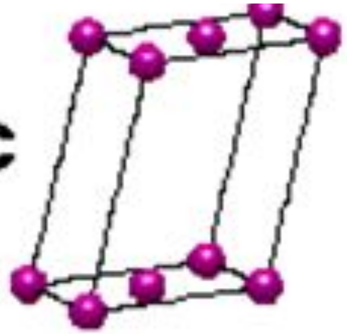
$\alpha = \gamma = 90°$

$\beta \neq 120°$

P



C

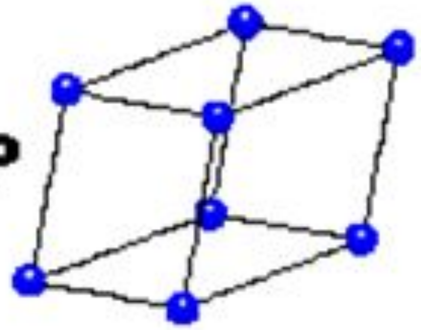


TRICLINIC

$a \neq b \neq c$

$\alpha \neq \beta \neq \gamma \neq 90°$

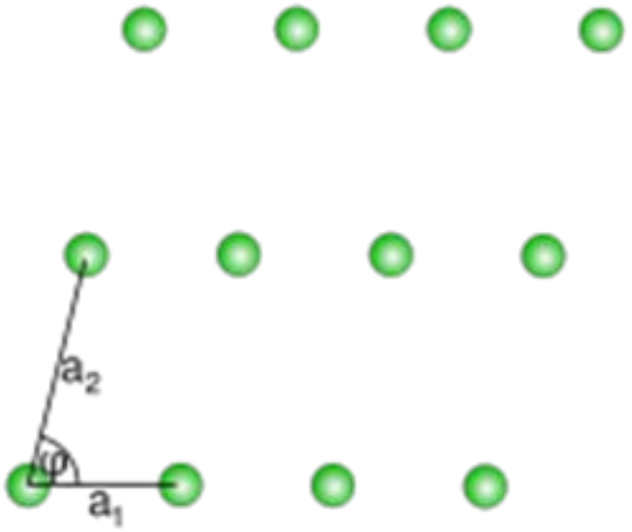
P

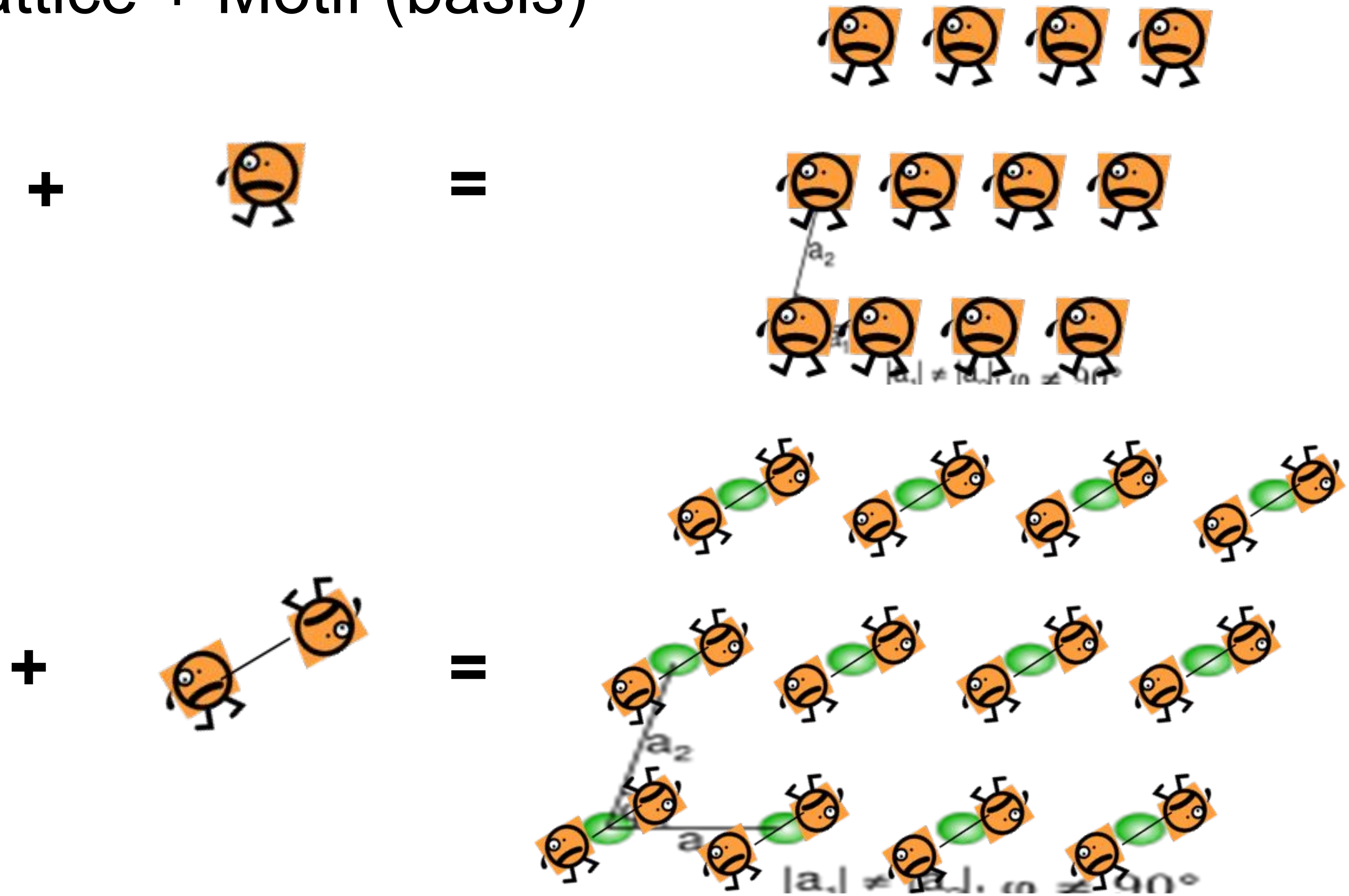


4 Types of Unit Cell
P = Primitive
I = Body-Centred
F = Face-Centred
C = Side-Centred
+
7 Crystal Classes
→ 14 Bravais Lattices

Lattice and Crystal Structure

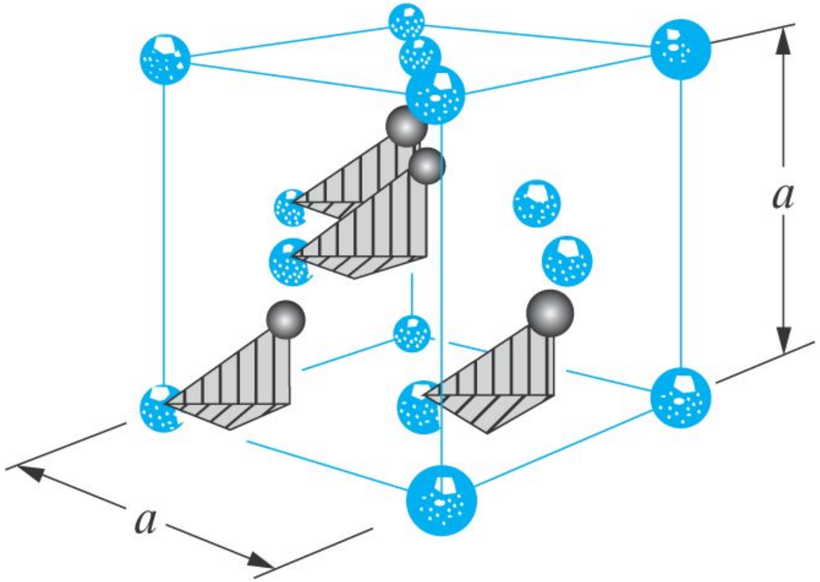
Crystal Structure = Lattice + Motif (basis)

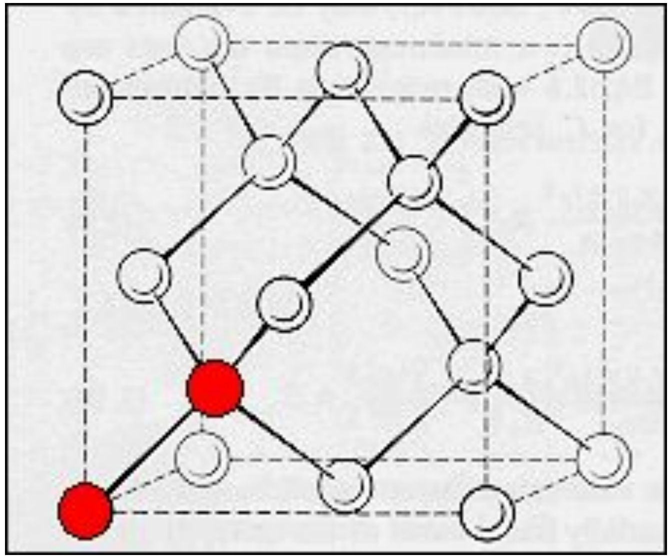




Basis could be atom, atoms, molecules or ions

Diamond lattice of Silicon





Diamond cubic structure is NOT a Bravais lattice

a=5.43 A^o

Silicon crystallizes in a diamond cubic structure, which is a face-centered cubic (FCC) Bravais lattice with a two-atom basis. The structure consists of two interpenetrating FCC sublattices, with one shifted by a vector (¼,¼,¼) from other. The two basis atoms are located at positions and (0,0,0) and (¼,¼,¼) within the unit cell. Each silicon atom is tetrahedrally bonded to four nearest neighbors. There are four such tetrahedrons in each cube.

https://skfb.ly/oLWv8

Crystal planes and Miller indices

- How to find Miller Indices (h, k, l) for a plane?

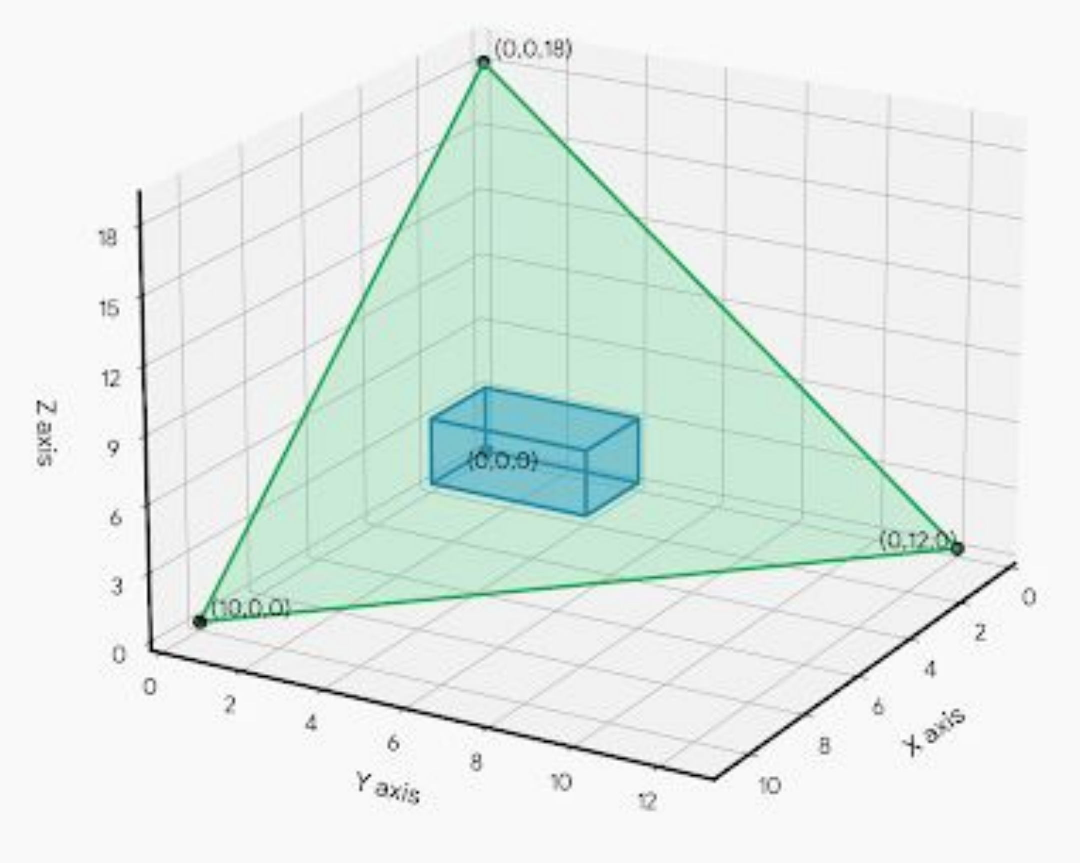
  ❖ Origin shouldn't be on the plane

  ❖ Determine intercepts x, y, z (10, 12, 18)

  ❖ Divide by unit cell parameters a, b, c (2, 4, 3)
  x/a, y/b, z/c (5, 3, 6)

  ❖ Take the inverse (to get rid of ∞)
  a/x, b/y, c/z (1/5, 1/3, 1/6)

  ❖ Find smallest set of integers
  h, k, l such that h:k:l::a/x:b/y:c/z (6, 10, 5)

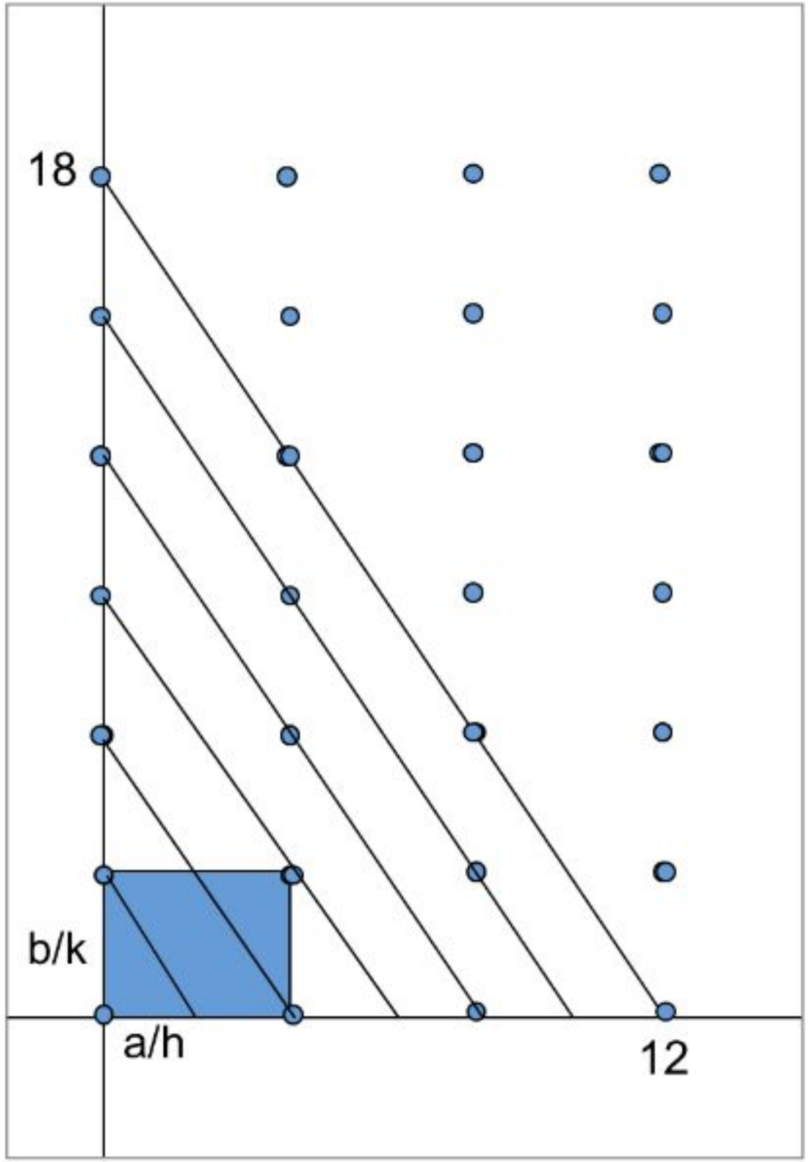


❑ All parallel planes have the same (hkl) indices

❑ {hkl} to denote crystallographic equivalent (hkl) planes

$$\{100\} = (100), (010), (001), (\bar{1}00), (0\bar{1}0), (00\bar{1})$$

Miller Indices example



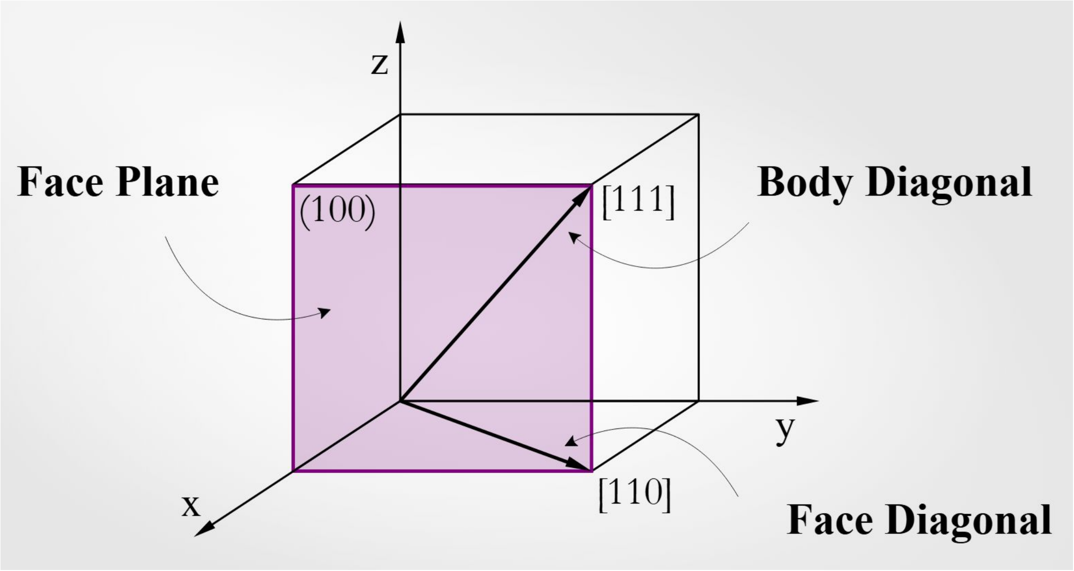
- Plane intercepts x, y (12, 18)
- Unit cell parameters a, b (4, 3)
  x/a, y/b (3, 6)
- Take the inverse
  a/x, b/y (1/3, 1/6)
- h:k::a/x:b/y (2, 1)

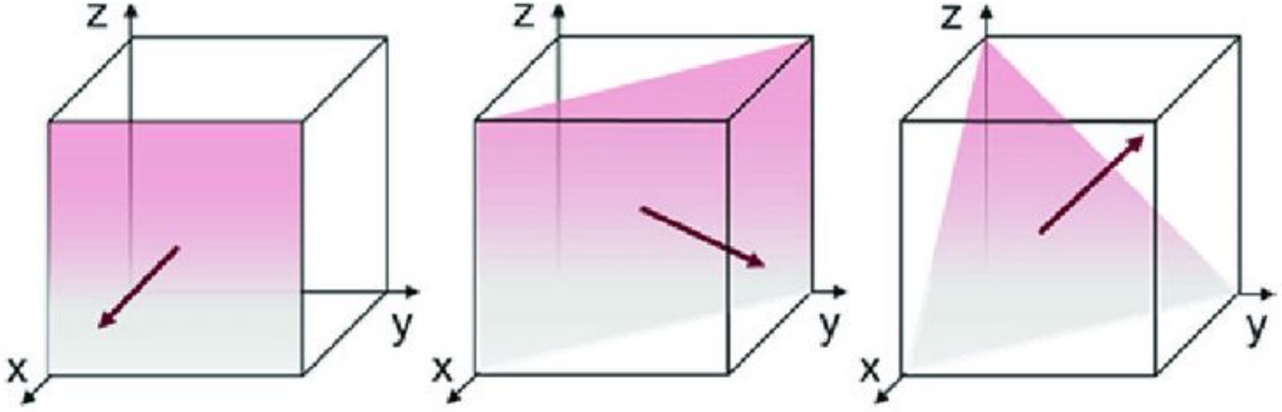
❖ (h, k) = (2, 1)

❖ a/h, b/k = 2, 3 (Intercepts of plane closest to origin)

Crystal directions [h,k,l]

- Determined by coordinates of point through which the direction vector passes and represented by the smallest set of integers.
- In cubic crystals (SC, FCC, BCC), [hkl] is perpendicular to the (hkl) plane

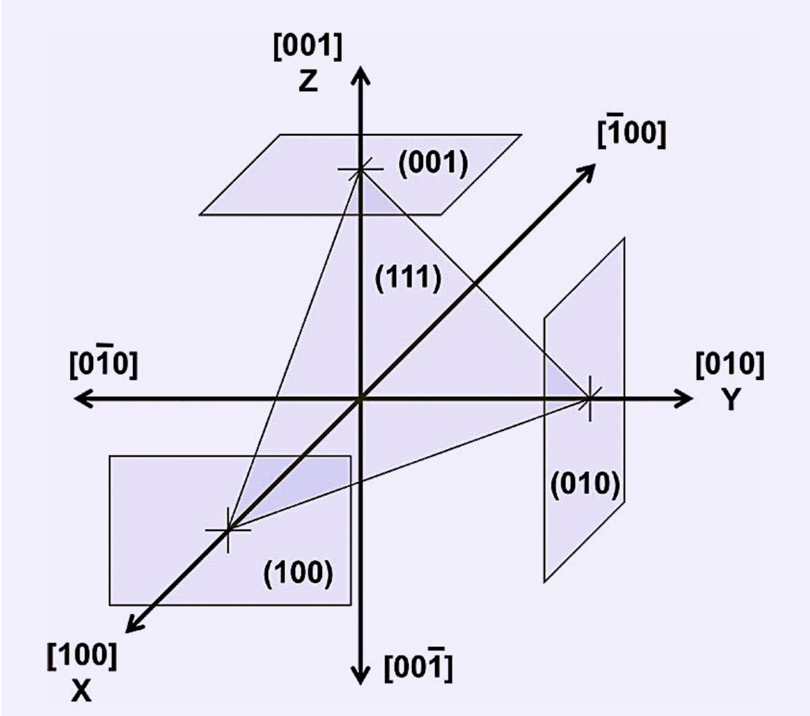




{100}-plane with <100>-direction

{110}-plane with <110>-direction

{111}-plane with <111>-direction



{hkl} to denote family of equivalent (hkl) planes

<hkl> to denote family of equivalent [hkl] directions

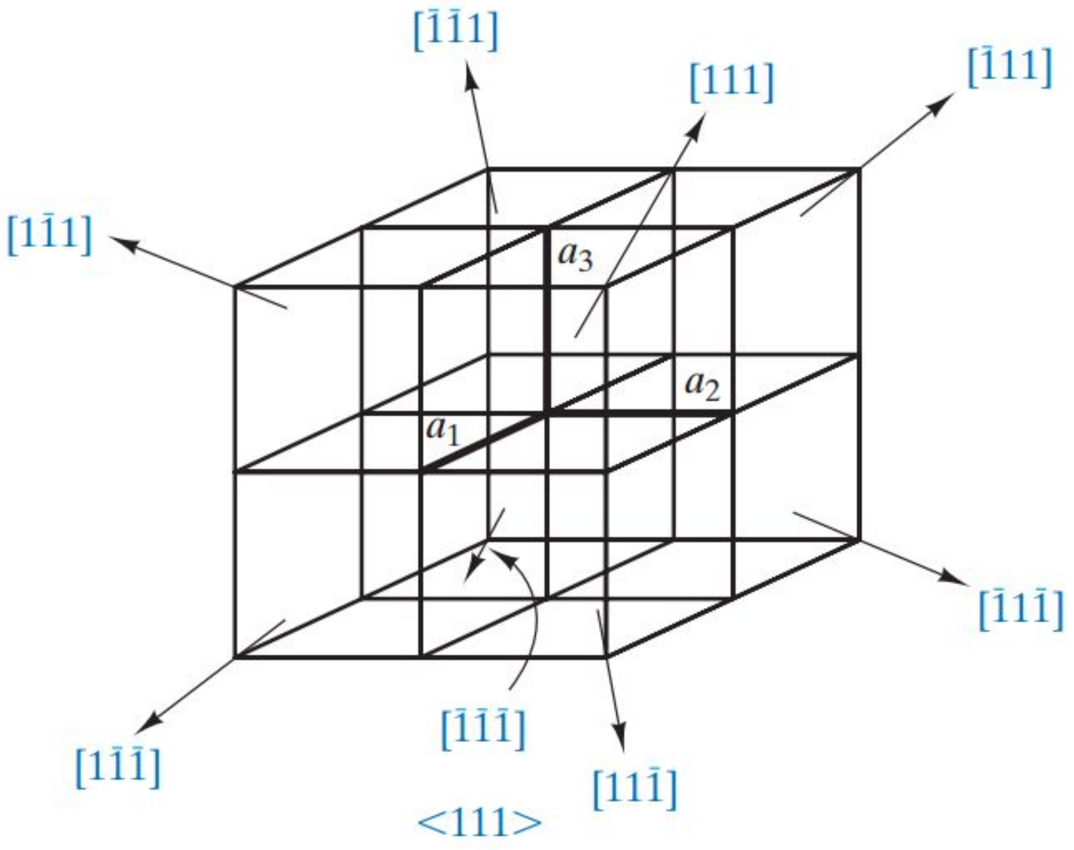
Crystal directions [h,k,l]

- <hkl> to denote crystallographic equivalent [hkl] directions

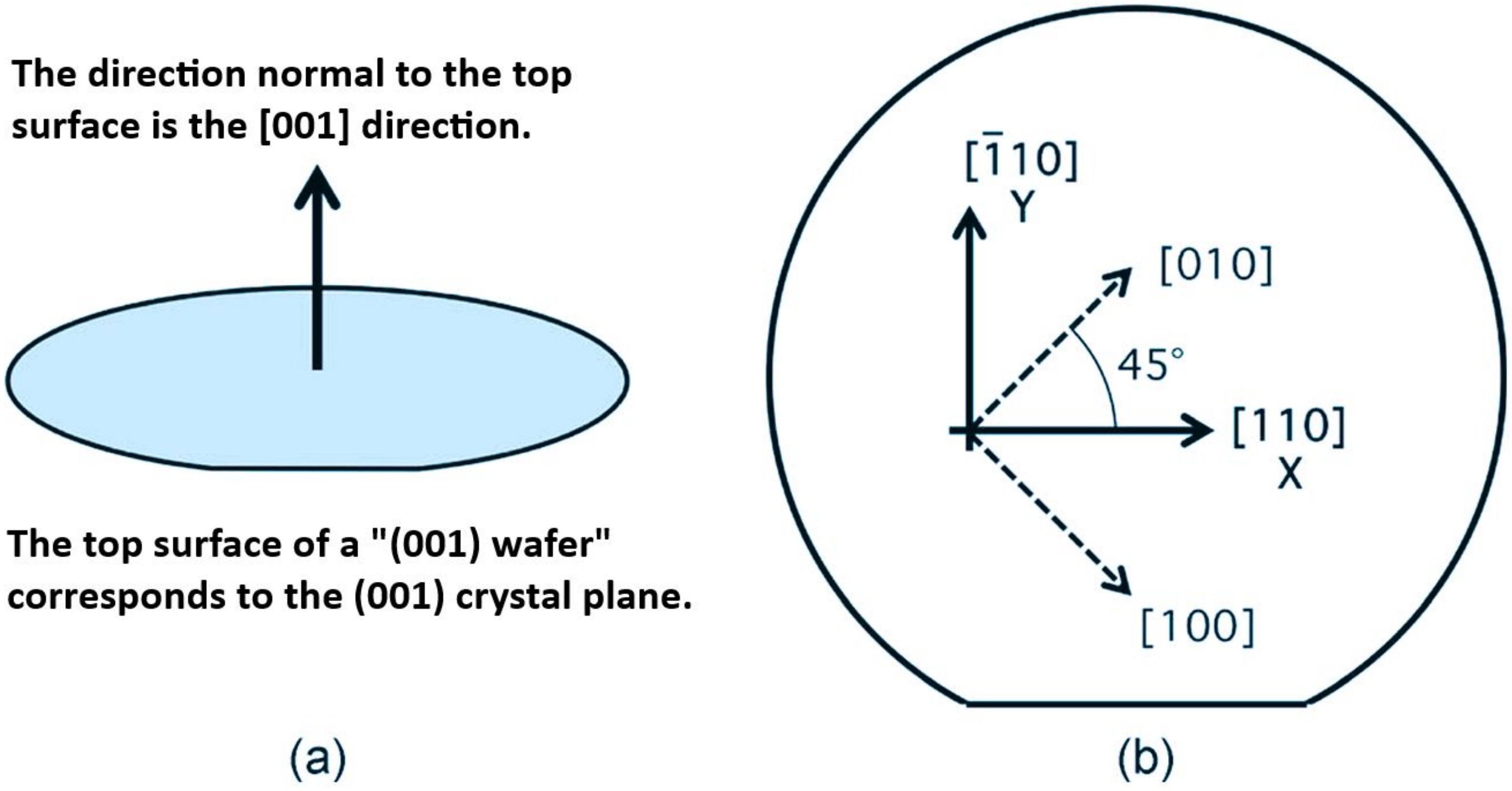
$$\langle 111 \rangle = [\bar{1}11], [1\bar{1}1], [11\bar{1}], [\overline{111}], [1\overline{11}], [\bar{1}1\bar{1}], [\overline{11}1]$$

- A bar on top is used to denote negative direction or intercepts
- Put a bar on top of all to reverse the direction
- Angle between two directions can be found by

$$\cos(\theta) = \frac{h_1 h_2 + k_1 k_2 + l_1 l_2}{[(h_1^2 + k_1^2 + l_1^2)(h_2^2 + k_2^2 + l_2^2)]^{1/2}}$$



Orientation of most commonly used Silicon wafer



Crystal orientation of a standard (001) silicon wafer. (a) The wafer surface is aligned with the (001) plane. (b) The in-plane X- and Y-axes are oriented along the [110] and [$\bar{1}$10] directions, respectively.

What is wavevector k ?

Wavenumber of a wave is a measure of oscillations (cycles) per meter like frequency is cycles per second

Wavenumber is the magnitude of the wavevector. The direction of wave vector is perpendicular to the wavefront. Angular wavenumber or wavevector is obtained by multiplying with 2π

Wavevector is the position gradient of phase of the wave. Its dot product with a unit vector gives the rate of change of the phase along that unit vector's direction (in radians/length). If the wave vector varies from place to place, it is a vector field k(r)

For a plane wave, k is constant and uniform throughout 3- dimensional space. The instantaneous loci of uniform phase (must be everywhere perpendicular to k) form a set of parallel planes

In an isotropic media, direction of wavevector is also the direction of wave propagation (energy flow, S ⊥ E & H) while it isn't in an anisotropic media (where D&E and B&H aren't always in same direction)*

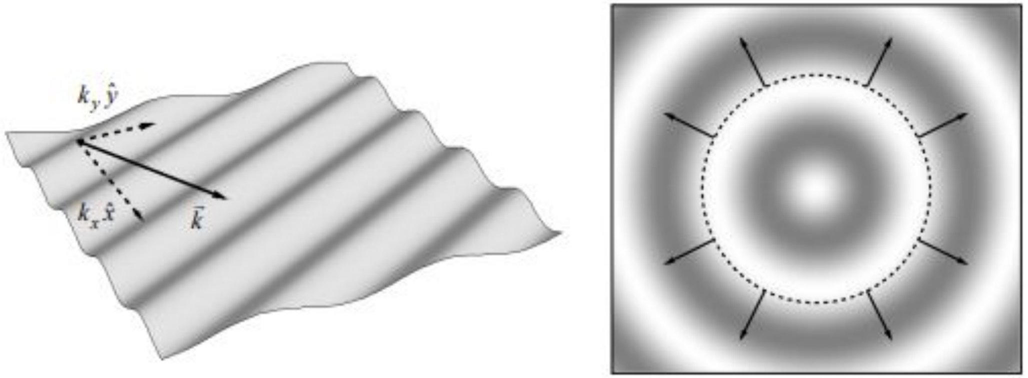
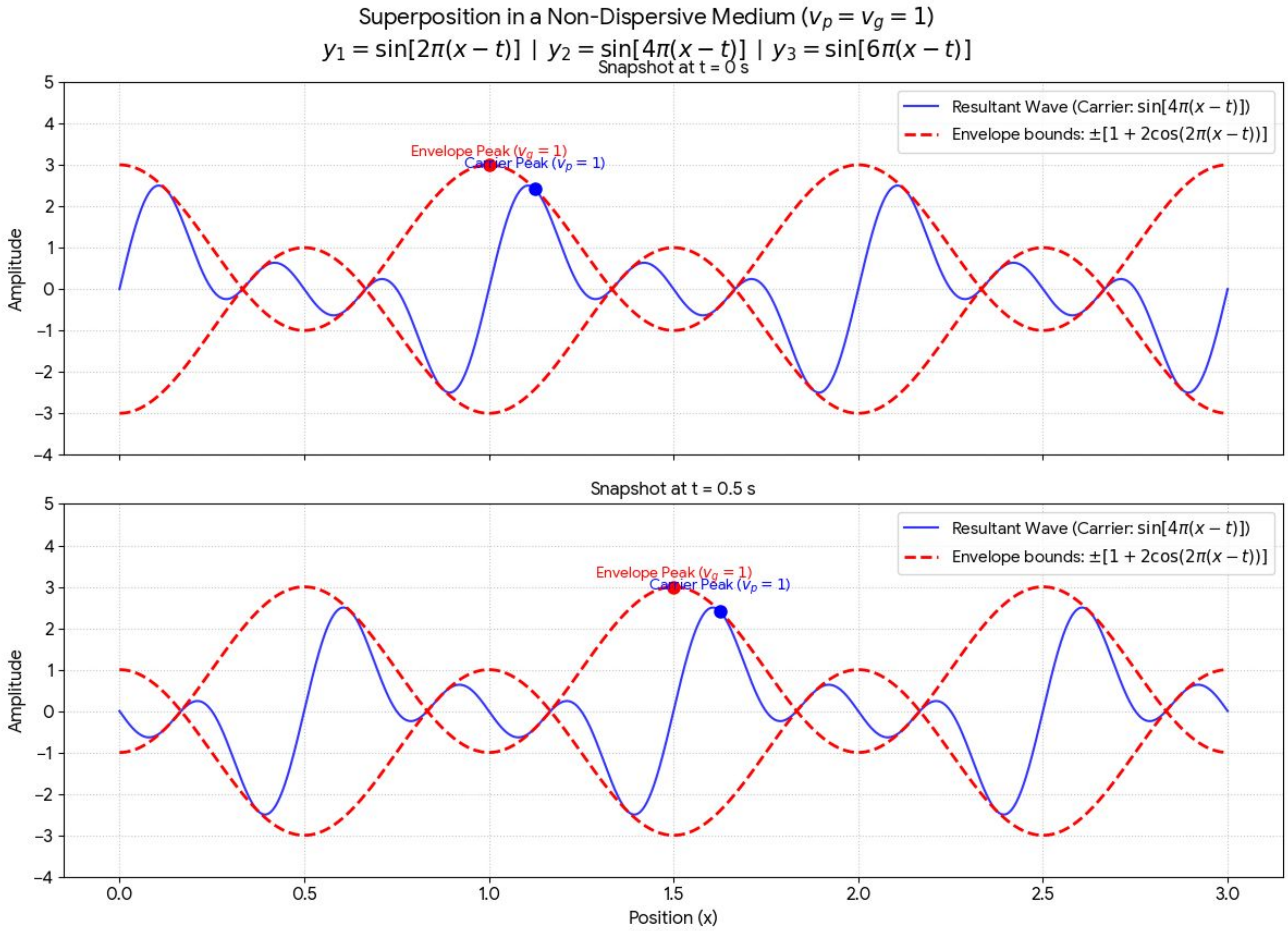


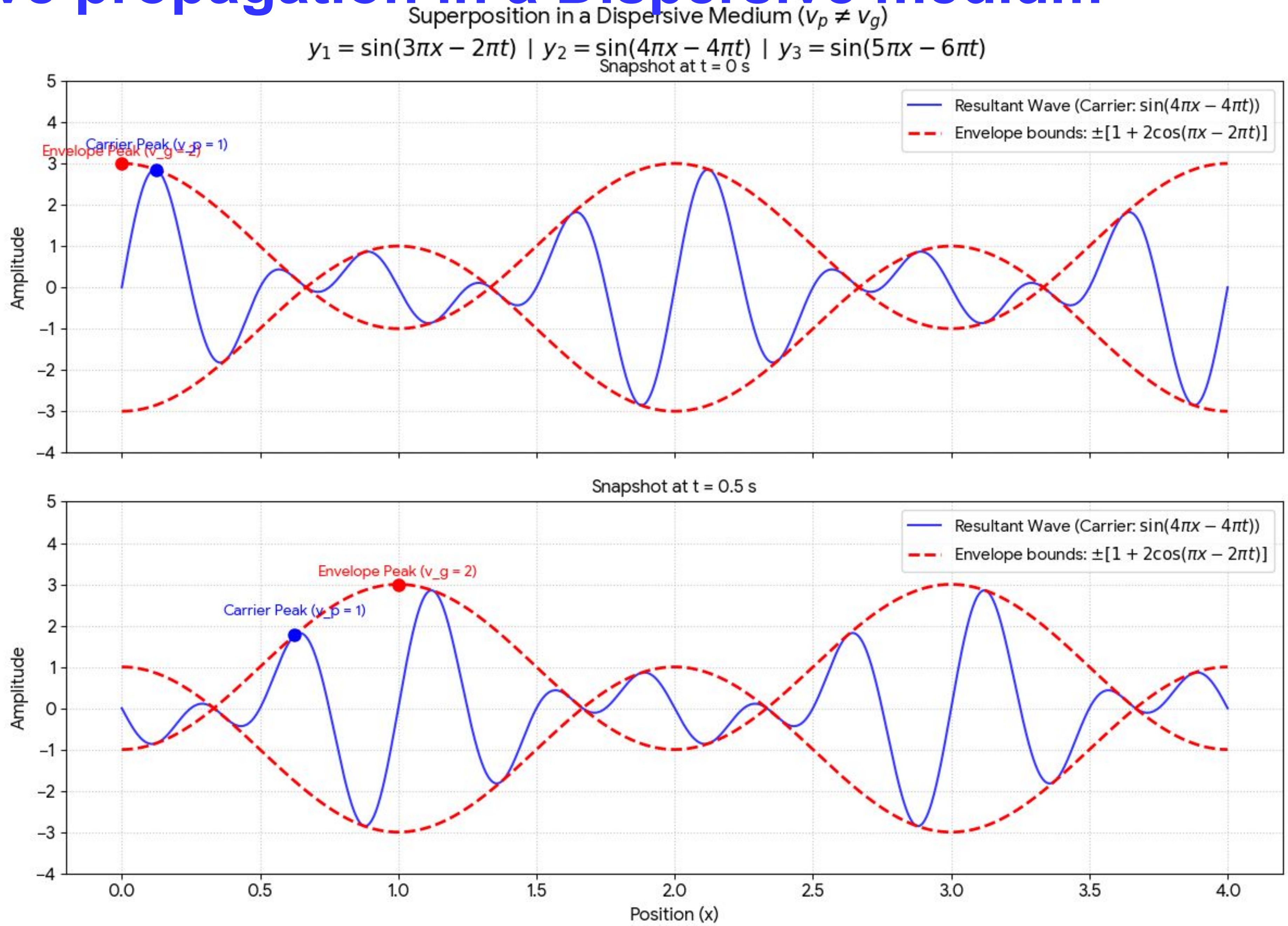
Figure D-1: (Left) a transverse plane wave propagating along a surface in the direction given by its wave vector $\vec{k}$. Note that the magnitudes of the wave vector's components give the rate of change of the wave's phase along their respective directions. (Right) a circular wave propagating outward. A selection of wave vectors is shown; each vector is perpendicular to the surface of constant phase at its location (in this case, a circle centered on the source).

http://www.sophphx.caltech.edu/Physics_6/Appendix_D_wave_vector.pdf
*Tailoring the Phase and Power Flow of Electromagnetic Fields

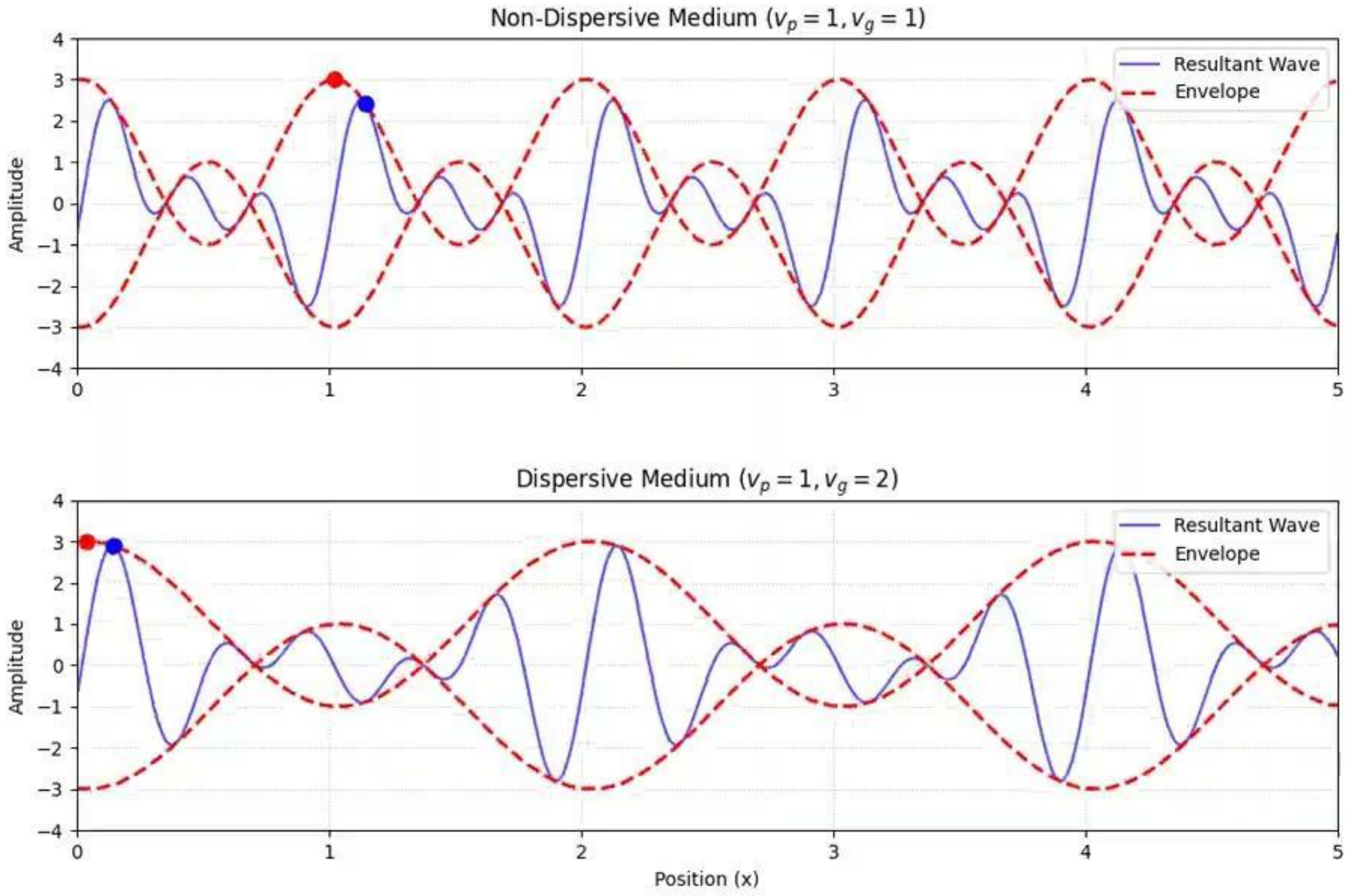
Wave propagation in a non-dispersive medium



Wave propagation in a Dispersive medium



Wave propagation (animation)



Reciprocal lattice

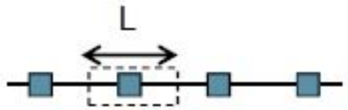
- Consider a set of points R constituting a Bravais Lattice and a plane wave e^{ikr}. For a general wave vector k, the plane wave will not have any periodicity of the Bravais lattice.
- But for certain wave vectors G, $e^{iG(r+R)}=e^{iGr}$ for all r and R provided $e^{iGR}=1$
- G is the set of all wave vectors that yield plane waves with the periodicity of a given Bravais Lattice
- Given a direct (in real space) lattice of points R, a point G is a point in the reciprocal lattice, iff $e^{iGR}=1$ for all points R
- A reciprocal lattice is defined wrf to a particular Bravais lattice

Reciprocal lattice (1D)

- Reciprocal lattice can be thought as being a Fourier transform of the direct lattice
- Direct lattice is given by R_n = nL. We can put a delta function at each lattice points and write the density of lattice points as

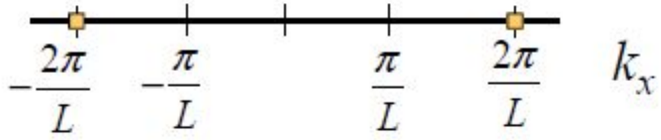
$$\rho(r) = \sum_n \delta(r - nL) \qquad (1)$$

Real-space



Fourier transforming this function gives

1st B-Z



$$\mathcal{F}[\rho(r)] = \int e^{ikr}\rho(r)\,dr = \sum_n \int e^{ikr}\delta(r - nL)\,dr = \sum_n e^{inkL} \qquad (2)$$

$$= \frac{2\pi}{|L|}\sum_m \delta\left(k - \frac{2\pi m}{L}\right)$$

- e^{inkL} is unity if k = 2πm/L, i.e., if k is a point on the reciprocal lattice.
- Points are periodic in k space with periodicity 2π/L

How to define reciprocal space?

- In real space, the primitive unit vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$
- We will define the unit vectors $\mathbf{b}_1$, $\mathbf{b}_2$, and $\mathbf{b}_3$ in reciprocal space, which relate to real space vectors $\mathbf{a}_1$, $\mathbf{a}_2$, and $\mathbf{a}_3$ as

$$\begin{aligned} \mathbf{b_1} &= \frac{2\pi\, \mathbf{a_2} \times \mathbf{a_3}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})} \\ \mathbf{b_2} &= \frac{2\pi\, \mathbf{a_3} \times \mathbf{a_1}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})} \\ \mathbf{b_3} &= \frac{2\pi\, \mathbf{a_1} \times \mathbf{a_2}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})} \end{aligned}$$

$$\mathbf{a_i} \cdot \mathbf{b_j} = 2\pi\delta_{ij}$$

where δ_{ij} is the Kronecker delta.[2]

- Does the above make reciprocal lattice a Bravais lattice?

Reciprocal lattice is a Bravais Lattice

- Let us write an arbitrary point in reciprocal space as $G = m_1b_1 + m_2b_2 + m_3b_3$.
- We can also write similarly for a point R in direct lattice, $R = n_1a_1 + n_2a_2 + n_3a_3$ where n_1, n_2 and n_3 are integers. But for certain wave vectors G, $e^{iG(r+R)} = e^{iGr}$ for all r and R provided $e^{iGR} = 1$
- Given a direct (in real space) lattice of points R, G is a point in the reciprocal lattice, iff $e^{iGR} = 1$ for all points in R

$$e^{i\mathbf{G}\cdot\mathbf{R}} = e^{i(m_1\mathbf{b_1}+m_2\mathbf{b_2}+m_3\mathbf{b_3})\cdot(n_1\mathbf{a_1}+n_2\mathbf{a_2}+n_3\mathbf{a_3})} = e^{2\pi i(n_1m_1+n_2m_2+n_3m_3)}$$

- It can be unity only if m_1, m_2 and m_3 are integers. This shows all points G constitutes a Bravais Lattice

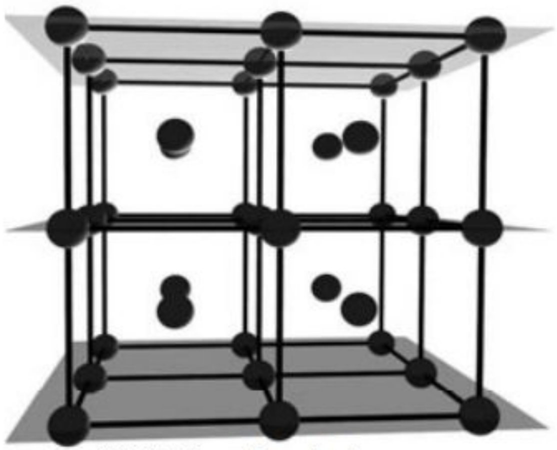
Vector normal to any lattice plane

- A lattice plane is a plane containing at least three non-collinear points of a lattice

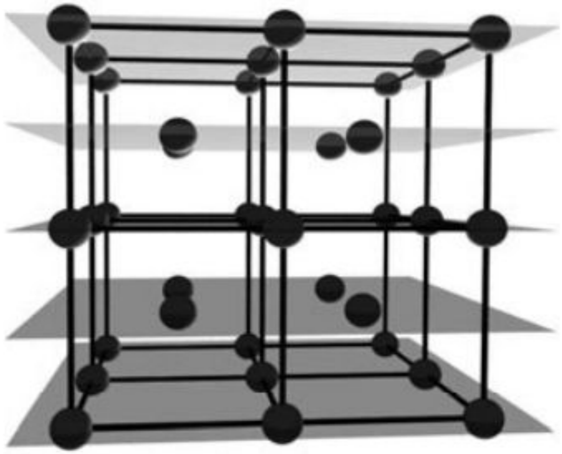
- A family of parallel lattice planes is an infinite set of equally separated parallel lattice planes which taken together contain all points of the lattice.

- Let (hkl) be any plane not passing through origin, with intercepts (x_i,y_i, and z_i,) on the respective axes with unit cell of dimensions (a,b,c)

$$h : k : l :: \frac{a}{x_i} : \frac{b}{y_i} : \frac{c}{z_i}$$



(010) family of planes
(not all lattice points included)

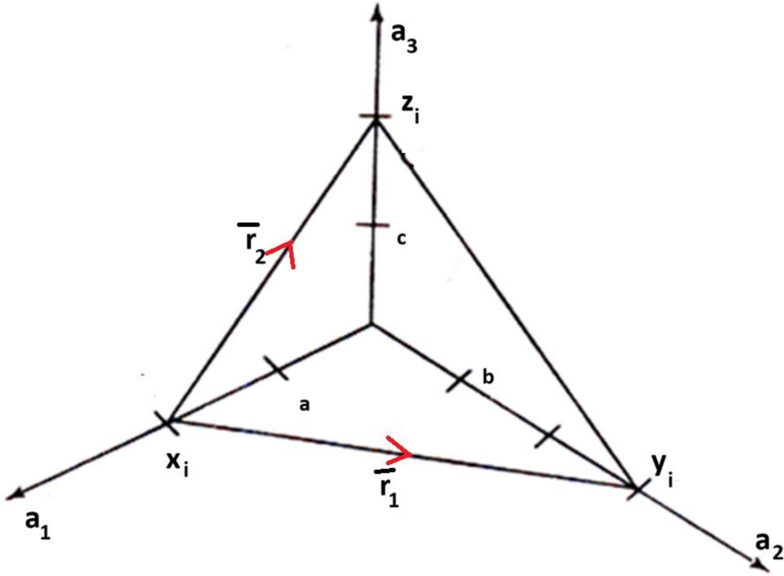


(020) family of lattice planes

- It can be proved that $G = hb_1 + kb_2 + lb_3$ the reciprocal lattice vector is normal to the above real lattice plane with miller index (hkl)

$$\vec{r_1} = \left(\frac{y_i}{b}\right)\vec{a_2} - \left(\frac{x_i}{a}\right)\vec{a_1}$$

$$\vec{r_2} = \left(\frac{z_i}{c}\right)\vec{a_3} - \left(\frac{x_i}{a}\right)\vec{a_1}$$



$$\vec{r_1} \times \vec{r_2} = \left(\frac{y_i z_i}{bc}\right)(\vec{a_2} \times \vec{a_3}) + \left(\frac{x_i z_i}{ac}\right)(\vec{a_3} \times \vec{a_1}) + \left(\frac{x_i y_i}{ab}\right)(\vec{a_1} \times \vec{a_2})$$

$$= \left(\frac{a y_i z_i}{abc}\right)(\vec{a_2} \times \vec{a_3}) + \left(\frac{b x_i z_i}{abc}\right)(\vec{a_3} \times \vec{a_1}) + \left(\frac{c x_i y_i}{abc}\right)(\vec{a_1} \times \vec{a_2})$$

$$= \left(\frac{x_i y_i z_i}{2\pi}\right)\left[\left(\frac{a}{x_i}\right)\vec{b_1} + \left(\frac{b}{y_i}\right)\vec{b_2} + \left(\frac{c}{z_i}\right)\vec{b_3}\right]$$

$$= A\left[h\vec{b_1} + k\vec{b_2} + l\vec{b_3}\right]$$

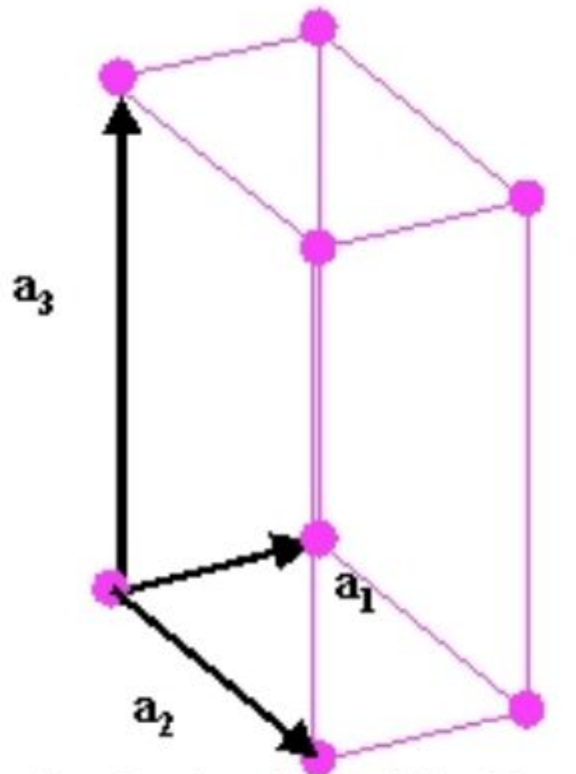
$$\mathbf{b_1} = \frac{2\pi\,\mathbf{a_2} \times \mathbf{a_3}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})}$$

$$\mathbf{b_2} = \frac{2\pi\,\mathbf{a_3} \times \mathbf{a_1}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})}$$

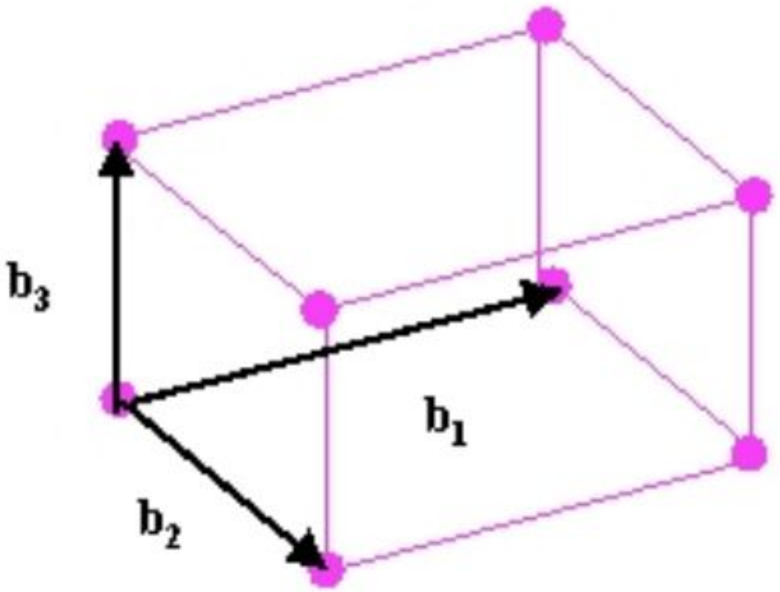
$$\mathbf{b_3} = \frac{2\pi\,\mathbf{a_1} \times \mathbf{a_2}}{\mathbf{a_1} \cdot (\mathbf{a_2} \times \mathbf{a_3})}$$

- $G = hb_1 + kb_2 + lb_3$ the shortest reciprocal lattice vector is normal to the above real lattice plane with miller index (hkl)

Constructing Reciprocal lattice



Simple Orthorhombic Bravais Lattice
with $a_3 > a_2 > a_1$

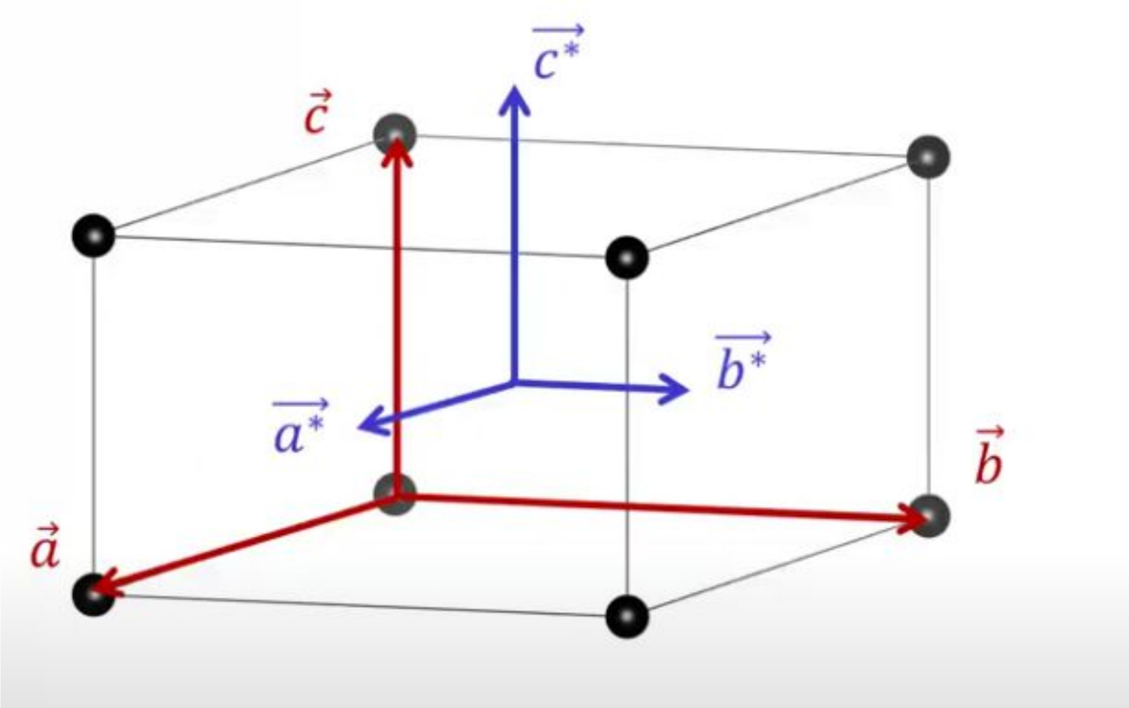


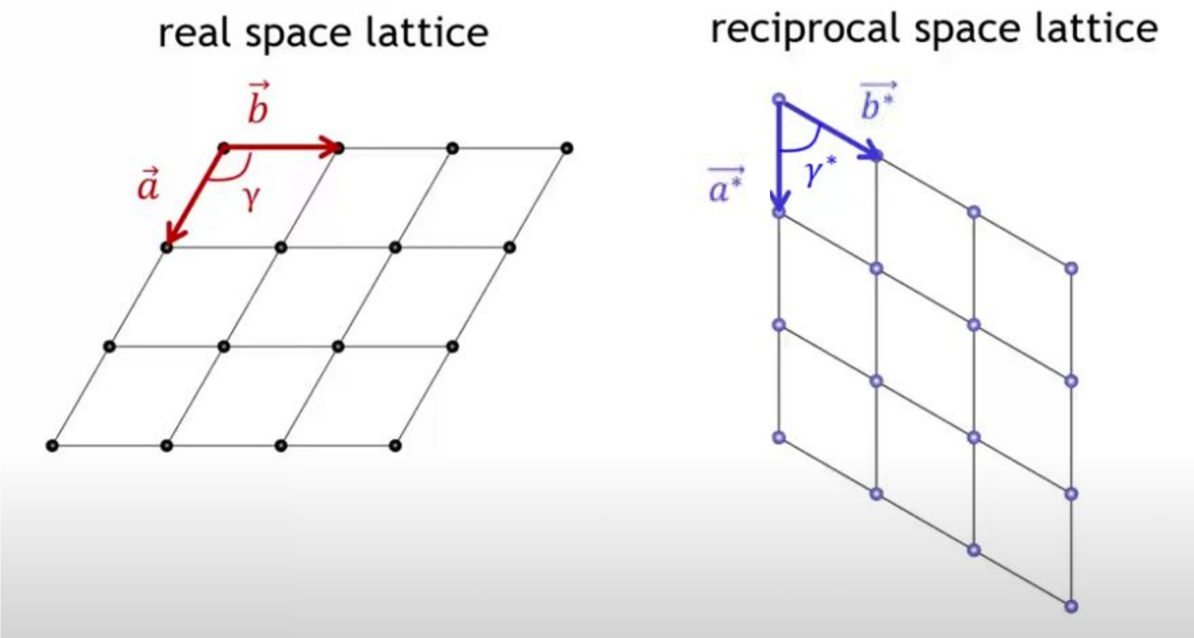
Reciprocal Lattice
Note: $b_1 > b_2 > b_3$

Long lengths in real space imply short lengths in reciprocal space and vice versa.

What can you say about their directions?

Direction of reciprocal lattice vector





$$\vec{b}\cdot\vec{a^*}=0 \qquad \vec{a}\cdot\vec{b^*}=0 \qquad \vec{a}\cdot\vec{c^*}=0$$

$$\vec{c}\cdot\vec{a^*}=0 \qquad \vec{c}\cdot\vec{b^*}=0 \qquad \vec{b}\cdot\vec{c^*}=0$$

b* is perpendicular to the ac plane/face of unit cell
a&a*/ b&b* / c&c* are parallel only if the real space lattice vectors are orthogonal

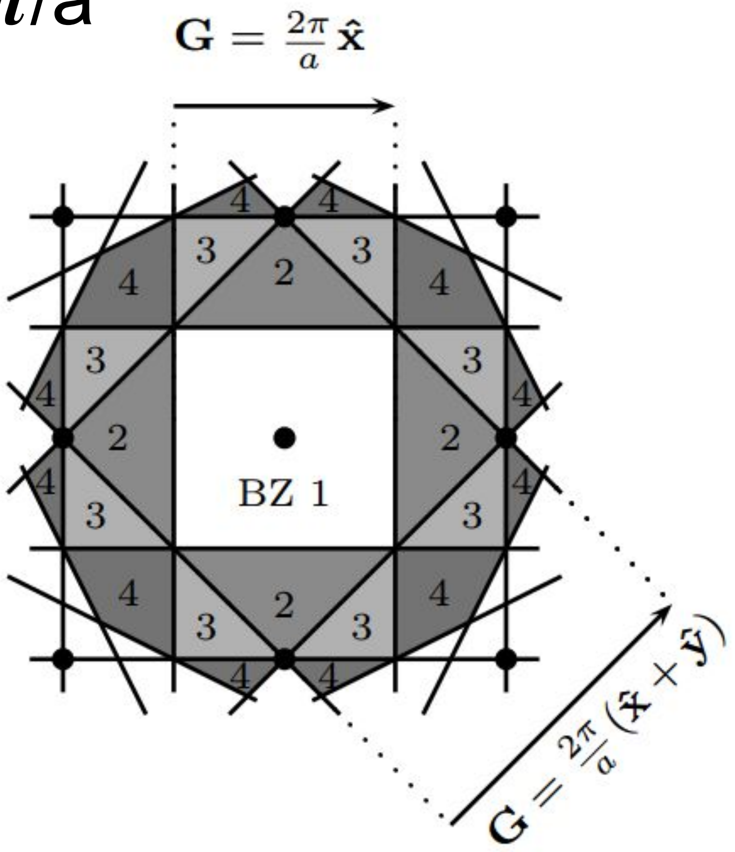
A reciprocal lattice vector G <hkl> is perpendicular to the plane (hkl) in real space

Brillouin zone

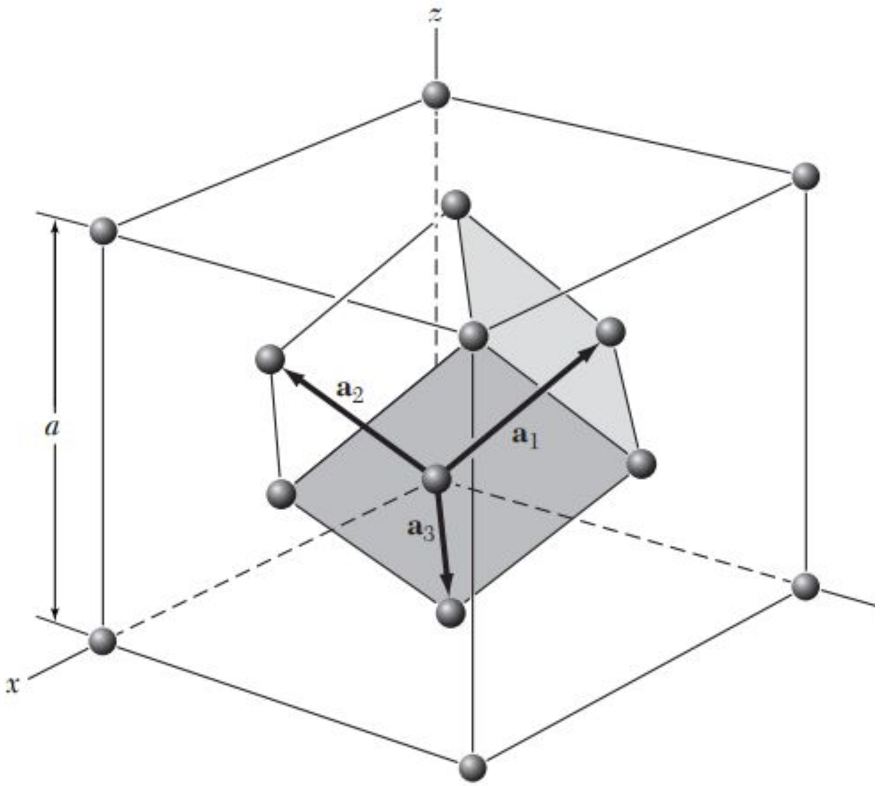
- 1st **Brillouin zone** is the Wigner-Seitz primitive cell of the reciprocal lattice

- Since waves are physically equivalent under shifts of the wavevector k by a reciprocal lattice vector $2\pi/a$ (G), we can do all representation within the first Brillouin zone (BZ)

- N^{th} Brillouin zone (BZ) is the region in reciprocal space between n^{th} and $(n-1)^{th}$ **Bragg planes** (perpendicular bisectors to the line joining the origin and any reciprocal lattice points)

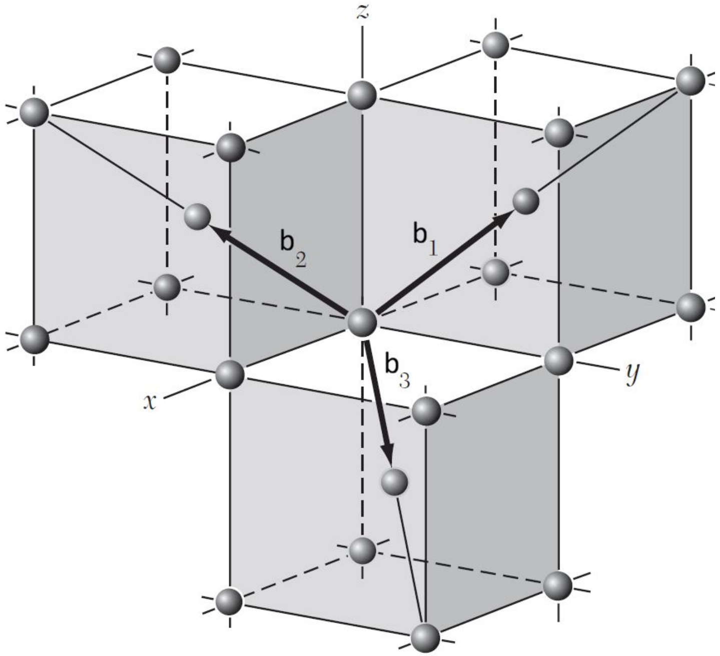


Reciprocal lattice of FCC

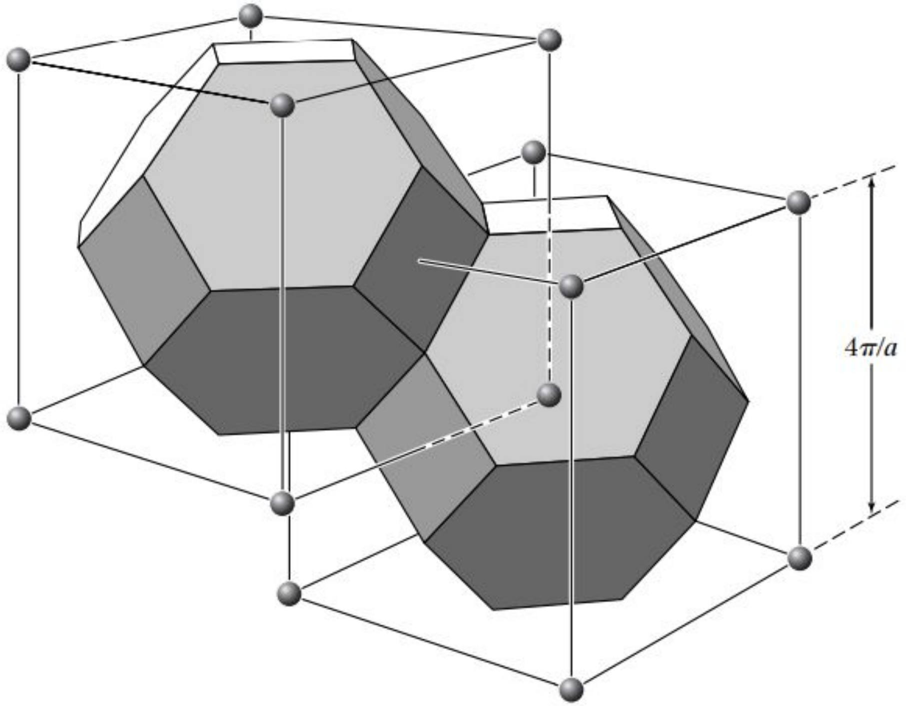


Direct lattice

$$\mathbf{a}_1 = \frac{1}{2}a(\hat{\mathbf{y}} + \hat{\mathbf{z}}) \ ; \quad \mathbf{a}_2 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{z}}) \ ; \quad \mathbf{a}_3 = \frac{1}{2}a(\hat{\mathbf{x}} + \hat{\mathbf{y}}) \ .$$



Reciprocal lattice



1st Brillouin Zone

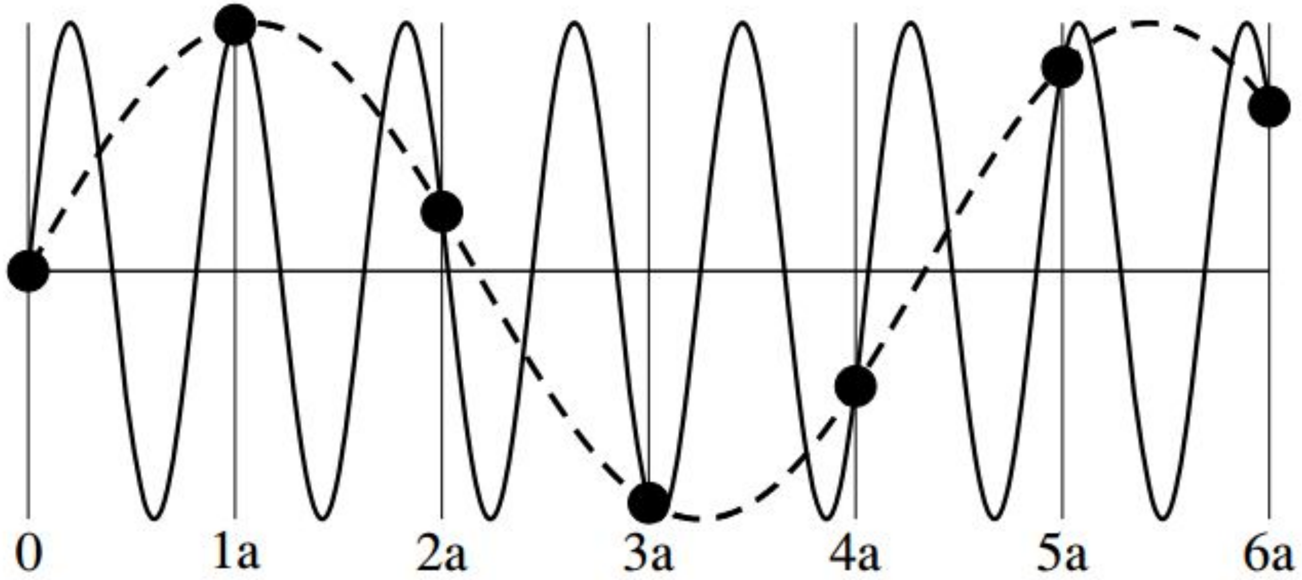
$$\mathbf{b}_1 = (2\pi/a)(-\hat{\mathbf{x}} + \hat{\mathbf{y}} + \hat{\mathbf{z}}) \ ; \qquad \mathbf{b}_2 = (2\pi/a)(\hat{\mathbf{x}} - \hat{\mathbf{y}} + \hat{\mathbf{z}}) \ ;$$

$$\mathbf{b}_3 = (2\pi/a)(\hat{\mathbf{x}} + \hat{\mathbf{y}} - \hat{\mathbf{z}}) \ .$$

Aliasing

A wavevector k describes the same wave as the wavevector k + 2π/a.

$e^{ir(k+2\pi/a)}=e^{ikr}$, only at r=na where $e^{i2\pi r/a}=1$



The dashed curve has wavevector k whereas the solid curve has wavevector k + 2π/a. These two waves have the same value (solid dots) at the location of the lattice points x_n = na, but disagree between lattice points. If the physical wave is ONLY defined at these lattice points the two waves are fully equivalent.

Similar aliasing happens when sampling rate in time domain is lower than Nyquist rate of signals

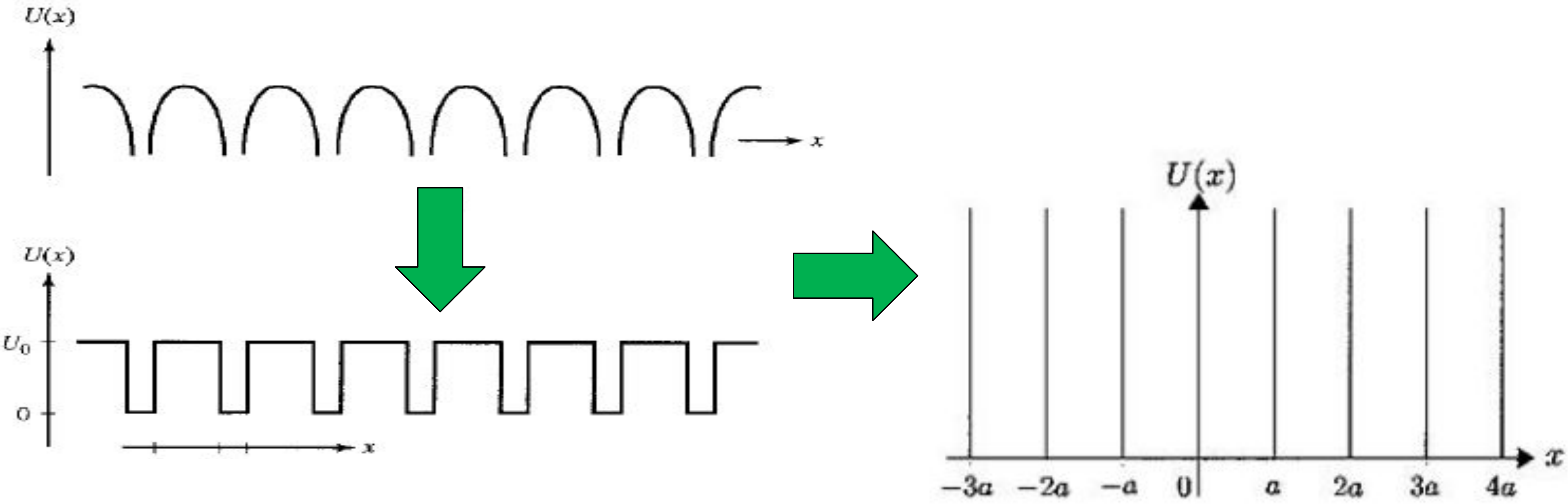
Try this problem

An electron is moving freely inside a one-dimensional infinite potential box with walls at $x = 0$ and $x = a$. If the electron is initially in the ground state ($n = 1$) of the box and if we *suddenly* quadruple the size of the box (i.e., the right-hand side wall is moved instantaneously from $x = a$ to $x = 4a$), calculate the probability of finding the electron in:

(a) the ground state of the new box and

(b) the first excited state of the new box.

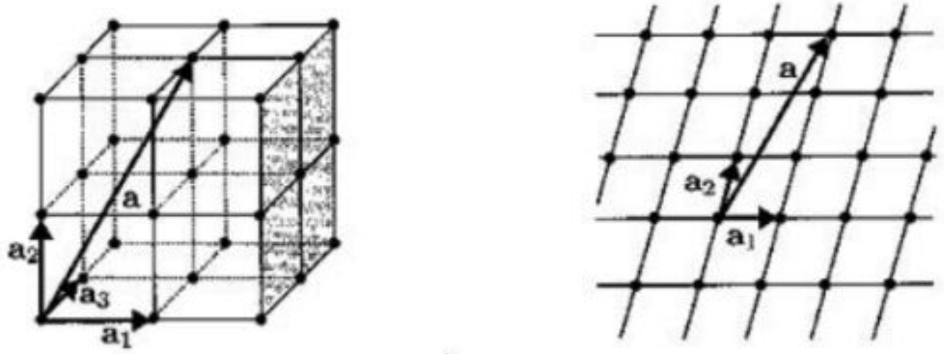
Particles in a periodic potential well
The Kronig-Penney model



Approximating the periodic Coulombic well to a periodic square well or an infinite Dirac comb

Bloch functions

Many solids have crystalline structures with atoms being arranged in regular, periodically repeated patterns which extend throughout the samples. These crystalline structures can be studied by using the concept of the crystal lattice. In the case of a crystal lattice, one deals with three- or two-dimensional arrays of periodically located points (sites) in space (see



(a) (b)

Fig. 5.1

Figure 5.1). Such arrays can be fully defined by the primitive basis vectors $\mathbf{a}_1$, $\mathbf{a}_2$ and $\mathbf{a}_3$ for the three-dimensional case and $\mathbf{a}_1$ and $\mathbf{a}_2$ for the two-dimensional case. These basis vectors are the shortest in length and independent vectors connecting lattice point sites. All vectors $\mathbf{a}$ connecting lattice sites can be represented as linear (integer) combinations of primitive vectors:

$$\mathbf{a} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 + n_3\mathbf{a}_3 \tag{5.1.1}$$

in the three-dimensional case, and

$$\mathbf{a} = n_1\mathbf{a}_1 + n_2\mathbf{a}_2 \tag{5.1.2}$$

in the two-dimensional case, with n_1, n_2 and n_3 being integers.

Next, we assume that **identical** (positive) charges (ions) exist at lattice points. Then, the electric field and its potential have **discrete translation symmetry**, which implies that for any lattice vector $\mathbf{a}$ we have

$$U(\mathbf{r}+\mathbf{a}) = U(\mathbf{r}). \tag{5.1.3}$$

This leads to the translational symmetry of the Hamiltonian operator

$$\hat{H}(\mathbf{r}) = \hat{H}(\mathbf{r}+\mathbf{a}), \tag{5.1.4}$$

where

$$\hat{H}(\mathbf{r}) = -\frac{\hbar^2}{2m}\nabla^2 + U(\mathbf{r}), \tag{5.1.5}$$

and

$$\hat{H}(\mathbf{r}+\mathbf{a}) = -\frac{\hbar^2}{2m}\nabla^2 + U(\mathbf{r}+\mathbf{a}). \tag{5.1.6}$$

The time-independent Schrödinger equations for the Hamiltonians $\hat{H}(\mathbf{r})$ and $\hat{H}(\mathbf{r}+\mathbf{a})$ can be written as follows, respectively,

$$\hat{H}(\mathbf{r})\Phi_n(\mathbf{r}) = \mathcal{E}_n\Phi_n(\mathbf{r}), \tag{5.1.7}$$

and

$$\hat{H}(\mathbf{r}+\mathbf{a})\Phi_n(\mathbf{r}+\mathbf{a}) = \mathcal{E}_n\Phi_n(\mathbf{r}+\mathbf{a}). \tag{5.1.8}$$

By using formula (5.1.4) in the last equation, we find

$$\hat{H}(\mathbf{r})\Phi_n(\mathbf{r}+\mathbf{a}) = \mathcal{E}_n\Phi_n(\mathbf{r}+\mathbf{a}). \tag{5.1.9}$$

By comparing equations (5.1.7) and (5.1.9), we conclude that

$$\Phi_n(\mathbf{r}+\mathbf{a}) = c(\mathbf{a})\Phi_n(\mathbf{r}), \tag{5.1.10}$$

which, in turn, leads to

$$|\Phi_n(\mathbf{r}+\mathbf{a})| = |c(\mathbf{a})||\Phi_n(\mathbf{r})|. \tag{5.1.11}$$

The last equation implies that

$$|\Phi_n(\mathbf{r}+2\mathbf{a})| = |c(\mathbf{a})||\Phi_n(\mathbf{r}+\mathbf{a})| = |c(\mathbf{a})|^2|\Phi_n(\mathbf{r})|. \tag{5.1.12}$$

Similarly, we find that

$$|\Phi_n(\mathbf{r}+m\mathbf{a})| = |c(\mathbf{a})|^m|\Phi_n(\mathbf{r})|. \tag{5.1.13}$$

It can be likewise proven that

$$|\Phi_n(\mathbf{r} - m\mathbf{a})| = \frac{1}{|c(\mathbf{a})|^m}|\Phi_n(\mathbf{r})|. \tag{5.1.14}$$

By using the last two formulas, it is easy to establish that

$$|c(\mathbf{a})| = 1. \tag{5.1.15}$$

Indeed, if

$$|c(\mathbf{a})| > 1, \tag{5.1.16}$$

then according to formula (5.1.13), we have

$$\lim_{m\to\infty} |\Phi_n(\mathbf{r} + m\mathbf{a})| = \infty, \tag{5.1.17}$$

which is not possible because $\Phi_n(\mathbf{r})$ must be finite.

On the other hand, if

$$|c(\mathbf{a})| < 1, \tag{5.1.18}$$

then, according to formula (5.1.14), it follows that

$$\lim_{m\to\infty} |\Phi_n(\mathbf{r} - m\mathbf{a})| = \infty, \tag{5.1.19}$$

which is not possible as well.

Thus, equality (5.1.15) is established. This equality implies that

$$c(\mathbf{a}) = e^{i\nu(\mathbf{a})}, \tag{5.1.20}$$

where $\nu(\mathbf{a})$ is some function of $\mathbf{a}$.

By combining formulas (5.1.10) and (5.1.20), we end up with

$$\Phi_n(\mathbf{r} + \mathbf{a}) = e^{i\nu(\mathbf{a})}\Phi_n(\mathbf{r}). \tag{5.1.21}$$

The last equation suggests that

$$\Phi_n(\mathbf{r} + \mathbf{a} + \mathbf{b}) = e^{i\nu(\mathbf{a}+\mathbf{b})}\Phi_n(\mathbf{r}). \tag{5.1.22}$$

On the other hand, we find

$$\Phi_n(\mathbf{r} + \mathbf{a} + \mathbf{b}) = e^{i[\nu(\mathbf{a})+\nu(\mathbf{b})]}\Phi_n(\mathbf{r}). \tag{5.1.23}$$

From the last two formulas, it follows that

$$\nu(\mathbf{a} + \mathbf{b}) = \nu(\mathbf{a}) + \nu(\mathbf{b}). \tag{5.1.24}$$

Thus, $\nu(\mathbf{a})$ is an additive function, and it can be represented as

$$\nu(\mathbf{a}) = \mathbf{k}\cdot\mathbf{a}, \tag{5.1.25}$$

where $\mathbf{k}$ is an arbitrary vector.

By substituting the last formula into equation (5.1.21), we obtain

$$\boxed{\Phi_n(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k}\cdot\mathbf{a}}\Phi_n(\mathbf{r}).} \tag{5.1.26}$$

The last formula admits the following interpretation. Consider the operator $\hat{T}_{\mathbf{a}}$ of translation by $\mathbf{a}$ defined as

$$\hat{T}_{\mathbf{a}}\Phi(\mathbf{r}) = \Phi(\mathbf{r} + \mathbf{a}). \tag{5.1.27}$$

It is clear from the last two formulas that

$$\hat{T}_{\mathbf{a}}\Phi_n(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{a}}\Phi_n(\mathbf{r}). \tag{5.1.28}$$

This means that the eigenvalues $\lambda(\mathbf{a})$ of the translation operator $\hat{T}_{\mathbf{a}}$ are

$$\lambda(\mathbf{a}) = e^{i\mathbf{k}\cdot\mathbf{a}}. \tag{5.1.29}$$

Furthermore, the operators $\hat{H}(\mathbf{r})$ and $\hat{T}_{\mathbf{a}}$ have the same eigenfunctions. This implies that these two operators commute

$$\hat{H}\hat{T}_{\mathbf{a}} - \hat{T}_{\mathbf{a}}\hat{H} = 0. \tag{5.1.30}$$

Now, by using formula (5.1.26), we shall establish the Bloch theorem, which is also known in mathematics as the Floquet theorem. This theorem states that the eigenfunctions $\Phi_n(\mathbf{r})$ are of the form

$$\boxed{\Phi_n(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r}),} \tag{5.1.31}$$

where functions $u_{n\mathbf{k}}(\mathbf{r})$ have the periodicity of the lattice

$$\boxed{u_{n\mathbf{k}}(\mathbf{r} + \mathbf{a}) = u_{n\mathbf{k}}(\mathbf{r}).} \tag{5.1.32}$$

Since any bounded function can be represented in the form (5.1.31), the only thing that must be proven is the validity of formula (5.1.32). The proof proceeds as follows. By using formulas (5.1.26) and (5.1.31), we find

$$\Phi_n(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k}\cdot\mathbf{a}}\Phi_n(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{a}}e^{i\mathbf{k}\cdot\mathbf{r}}u_{n\mathbf{k}}(\mathbf{r}), \tag{5.1.33}$$

or

$$\Phi_n(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{a})}u_{n\mathbf{k}}(\mathbf{r}). \tag{5.1.34}$$

On the other hand, it follows from formula (5.1.31) that

$$\Phi_n(\mathbf{r} + \mathbf{a}) = e^{i\mathbf{k}\cdot(\mathbf{r}+\mathbf{a})}u_{n\mathbf{k}}(\mathbf{r} + \mathbf{a}). \tag{5.1.35}$$

The validity of equality (5.1.32) is the immediate consequence of the last two formulas.

Functions $\Phi_n(\mathbf{r})$ defined by formulas (5.1.31) and (5.1.32) are called Bloch functions or Bloch waves. They can be viewed as periodically modulated plane waves. It is remarkable that electrons described by Bloch functions move through periodically varying electric fields without any scattering. The scattering appears only when the ideal periodicity of the lattice structure is perturbed by thermal vibrations of lattice sites or by introduction of impurities, that is, when not all lattice sites are occupied by identical ions.

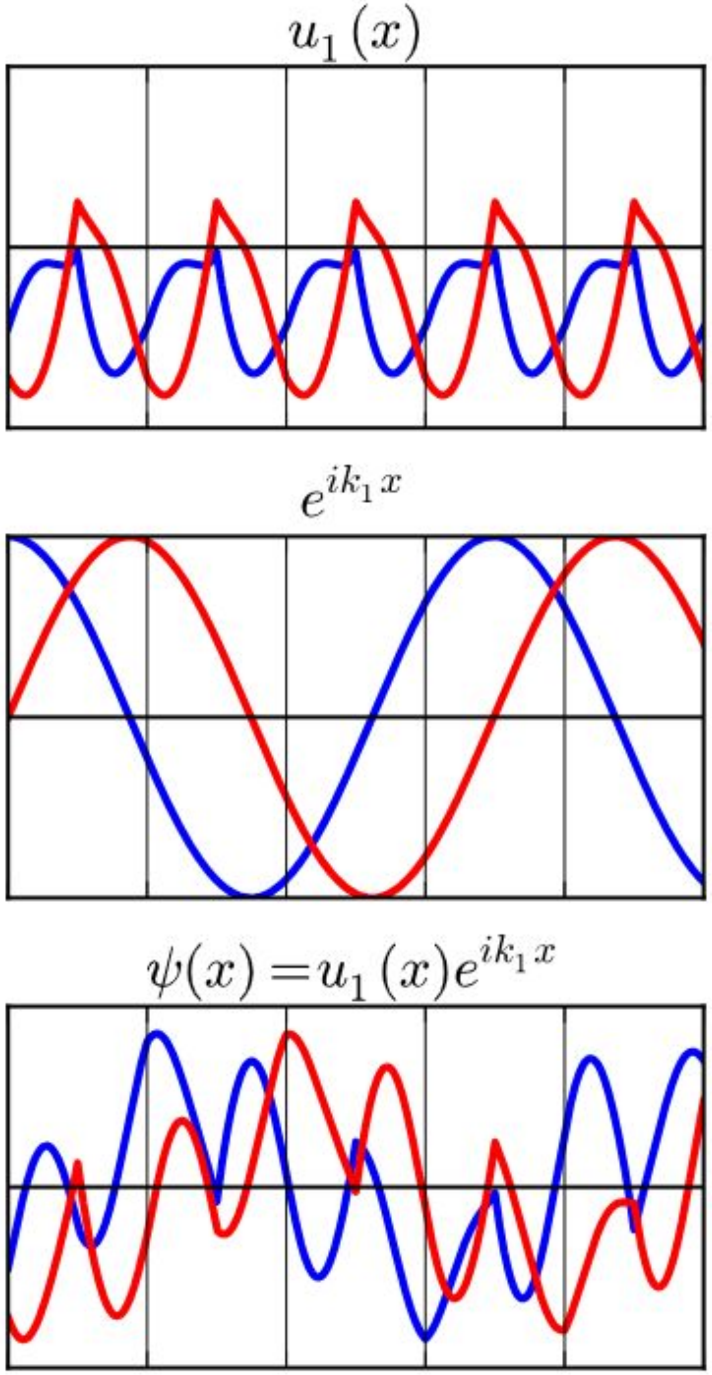
It is interesting to point out some similarity between the Bloch functions and the wave functions for free electrons with certain momentum $\mathbf{p}$. The latter functions can be written in the form (see formula (2.3.38) from Chapter 2):

$$\psi_{\mathbf{p}}(\mathbf{r}) = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}}, \tag{5.1.36}$$

where

$$\mathbf{p} = \hbar\mathbf{k}. \tag{5.1.37}$$

It is because of this similarity that the momentum $\mathbf{p}$ associated with the wave vector $\mathbf{k}$ in the Bloch function (5.1.31) is called **crystal momentum** or **quasimomentum**.



A Bloch wave function (bottom) can be broken up into the product of a periodic function (top) and a plane-wave (center). In all plots, blue is real part and red is imaginary part.

Dirac-Kronig-Penney model

It is discussed in the previous section that the energy band structure calculations require numerous solutions of the time-independent Schrödinger equations (5.1.41) for various wave vectors **k** within the first Brillouin zone. This is usually done numerically, and special techniques have been developed for this purpose. Most notable among them are the pseudo-potential method and the density functional method. It is apparent that band structure calculations are laborious and computer-intensive. It this section, we illustrate the energy band calculations by considering the simplest version of one-dimensional Kronig-Penney model in which the periodic potential $U(x)$ is modeled by periodically spaced Dirac delta functions (see Figure 5.3). For such a potential, the analytical treatment is possible, and the

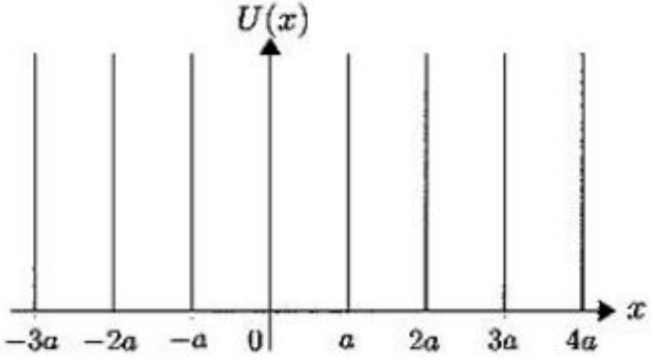


Fig. 5.3

main features of energy band structures discussed in the previous section can be vividly illustrated.

In our analysis, we shall use the following Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + U(x), \tag{5.2.1}$$

where

$$U(x) = \beta \sum_{\nu=-\infty}^{\infty} \delta(x - \nu a), \tag{5.2.2}$$

and β is some constant.

It is apparent that

$$U(x+a) = U(x). \tag{5.2.3}$$

The time-independent Schrödinger equation for the above potential can be written as follows

$$-\frac{\hbar^2}{2m}\frac{d^2\Phi(x)}{dx^2} + \beta\left(\sum_{\nu=-\infty}^{\infty}\delta(x-\nu a)\right)\Phi(x) = \mathcal{E}\Phi(x). \tag{5.2.4}$$

According to formula (5.1.26), any solution of this equation has the following property

$$\Phi(x+a) = e^{ika}\Phi(x), \tag{5.2.5}$$

which implies that if $\Phi(x)$ is known on the interval

$$0 < x < a, \tag{5.2.6}$$

then by using the last formula it can be found for any x, i.e., it can be extended to the entire x-axis. For this reason, it is sufficient to find $\Phi(x)$ on the above interval. It is clear that in the above interval equation (5.2.4) can be written as follows

$$\boxed{-\frac{\hbar^2}{2m}\frac{d^2\Phi(x)}{dx^2} = \mathcal{E}\Phi(x),} \quad (0 < x < a). \tag{5.2.7}$$

We shall next find the boundary conditions for $\Phi(x)$ at $x = 0_+$ and $x = a_-$, where subscripts "+" and "−" imply that the boundary points of the interval (5.2.6) are approached from "above" and "below," respectively, i.e., from within the interval.

Since $\Phi(x)$ is a continuous function, we have

$$\Phi(a_+) = \Phi(a_-). \tag{5.2.8}$$

According to formula (5.2.5), we have

$$\Phi(a_+) = e^{ika}\Phi(0_+). \tag{5.2.9}$$

By substituting the last formula into equation (5.2.8), we end up with the following boundary condition

$$\boxed{e^{ika}\Phi(0_+) = \Phi(a_-).} \tag{5.2.10}$$

These boundary conditions relate the values of $\Phi(x)$ at the ends of the interval (5.2.6).

Next, we shall derive the boundary condition that relates the values of $\frac{d\Phi}{dx}(x)$ at the ends of the same interval. The derivation proceeds as follows.

Consider the interval

$$0 < x < 2a. \tag{5.2.11}$$

On this interval, equation (5.2.4) has the form

$$-\frac{\hbar^2}{2m}\frac{d^2\Phi(x)}{dx^2} + \beta\delta(x-a)\Phi(x) = \mathcal{E}\Phi(x). \tag{5.2.12}$$

By integrating both sides of the last equation over the very small interval $a-\alpha < x < a+\alpha$, we obtain

$$-\frac{\hbar^2}{2m}\left[\frac{d\Phi}{dx}(a+\alpha) - \frac{d\Phi}{dx}(a-\alpha)\right] + \beta\Phi(a_+) = \mathcal{E}\int_{a-\alpha}^{a+\alpha}\Phi(x)dx. \tag{5.2.13}$$

In the limit when $\alpha \to 0$, from the last equation we find

$$\frac{d\Phi}{dx}(a_+) - \frac{d\Phi}{dx}(a_-) = \frac{2m\beta}{\hbar^2}\Phi(a_+). \tag{5.2.14}$$

From formula (5.2.5), it follows that

$$\frac{d\Phi}{dx}(a_+) = e^{ika}\frac{d\Phi}{dx}(0_+). \tag{5.2.15}$$

By substituting formulas (5.2.9) and (5.2.15) into equation (5.2.14), we end up with the boundary condition

$$\boxed{e^{ika}\frac{d\Phi}{dx}(0_+) - \frac{d\Phi}{dx}(a_-) = \frac{2m\beta}{\hbar^2}e^{ika}\Phi(0_+).} \tag{5.2.16}$$

Thus, in order to find $\Phi(x)$ in the interval (5.2.6) we have to find a solution of equation (5.2.7) subject to the boundary conditions (5.2.10) and (5.2.16).

A general solution of equation (5.2.7) can be written in the form

$$\Phi(x) = A\cos qx + B\sin qx, \quad (0 < x < a), \tag{5.2.17}$$

where

$$q = \frac{\sqrt{2m\mathcal{E}}}{\hbar}. \tag{5.2.18}$$

It is easy to derive the following equations for A and B from the boundary conditions (5.2.10) and (5.2.16), respectively,

$$e^{ika}A = A\cos qa + B\sin qa, \tag{5.2.19}$$

$$qa[e^{ika}B + A\sin qa - B\cos qa] = 2\gamma e^{ika}A, \tag{5.2.20}$$

where

$$\gamma = \frac{m\beta a}{\hbar^2}. \tag{5.2.21}$$

From equation (5.2.19), we find

$$\frac{A}{B} = \frac{\sin qa}{e^{ika} - \cos qa}. \tag{5.2.22}$$

Similarly, from equation (5.2.20) it follows that

$$\frac{A}{B} = \frac{e^{ika} - \cos qa}{\frac{2\gamma}{qa}e^{ika} - \sin qa}. \tag{5.2.23}$$

From the last two equations, we conclude that

$$\frac{\sin qa}{e^{ika} - \cos qa} = \frac{e^{ika} - \cos qa}{\frac{2\gamma}{qa}e^{ika} - \sin qa}, \tag{5.2.24}$$

which yields

$$(e^{ika} - \cos qa)^2 - \sin qa\left(\frac{2\gamma}{qa}e^{ika} - \sin qa\right) = 0. \tag{5.2.25}$$

Further algebraic transformations lead to

$$e^{i2ka} - 2e^{ika}\left(\cos qa + \gamma\frac{\sin qa}{qa}\right) + 1 = 0. \tag{5.2.26}$$

Dividing the last equation by $2e^{ika}$, we find

$$\cos qa + \gamma\frac{\sin qa}{qa} = \frac{e^{ika} + e^{-ika}}{2}, \tag{5.2.27}$$

which leads to the final equation

$$\boxed{\cos qa + \gamma\frac{\sin qa}{qa} = \cos ka.} \tag{5.2.28}$$

For any given k, the last equation can be solved numerically to find the corresponding values of q and, according to formula (5.2.18), the corresponding

Periodic Potentials 171

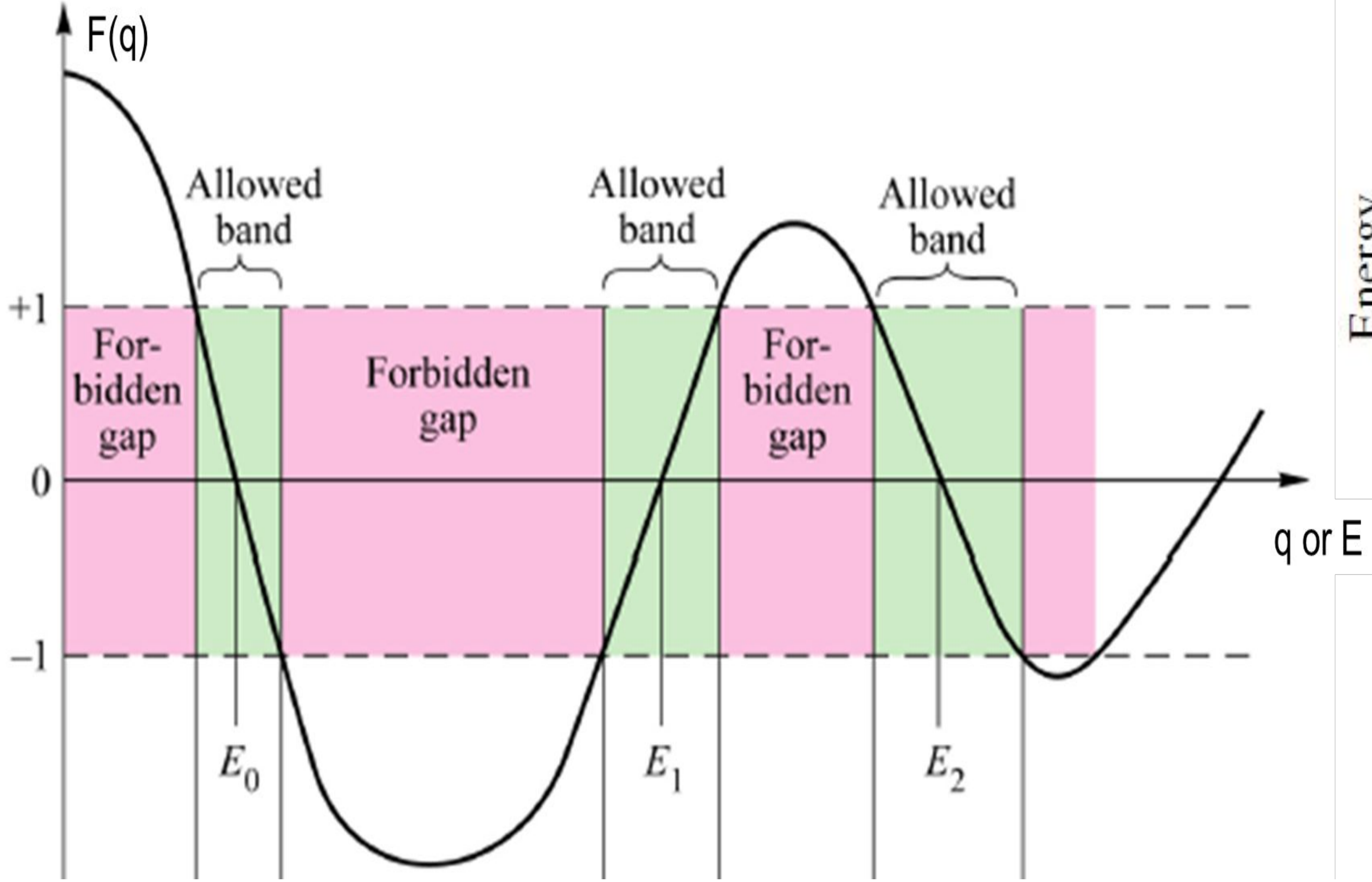
values of $\mathcal{E}$. In this way, the band structures $\mathcal{E}_n(k)$ can be computed. The described solution process can be illustrated graphically and the emergence of energy bands becomes clearly visible. To this end, we introduce the function

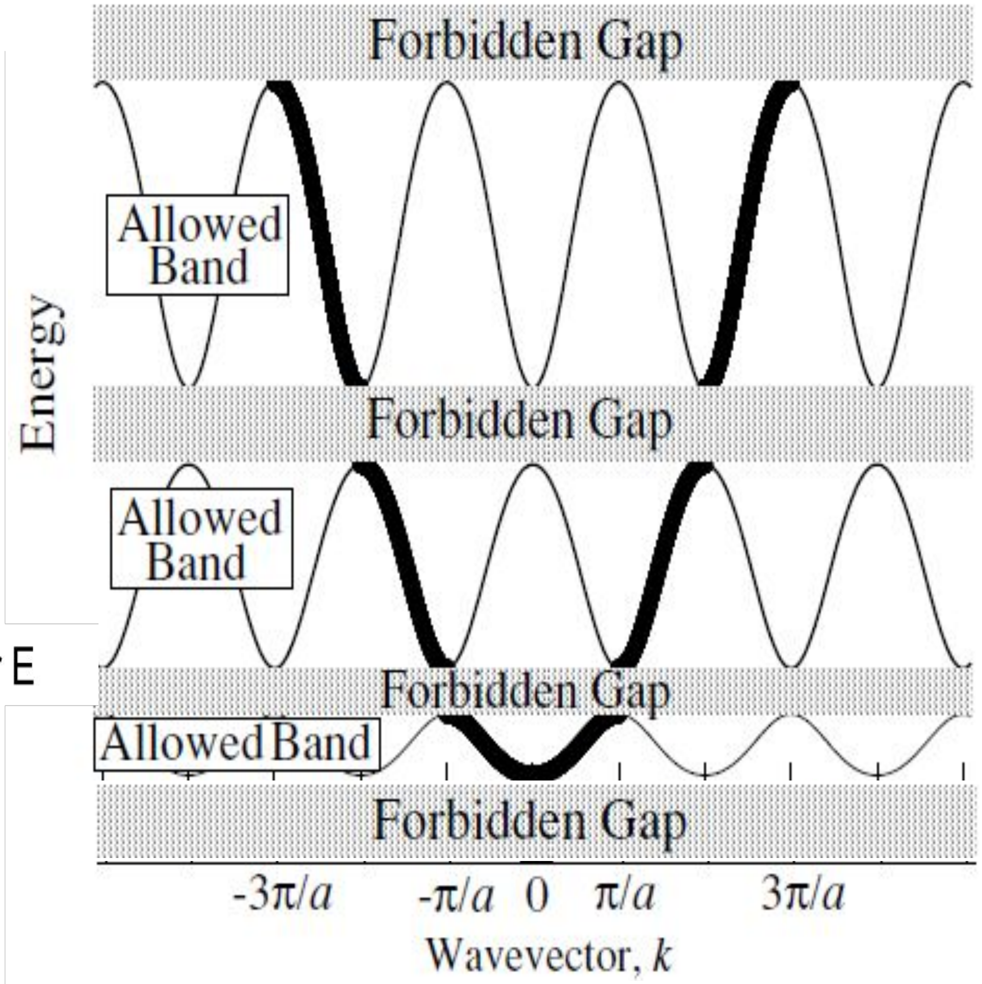
$$F(q) = \cos qa + \gamma \frac{\sin qa}{qa} \tag{5.2.29}$$

and write the equation (5.2.28) in the form

$$F(q) = \cos ka. \tag{5.2.30}$$

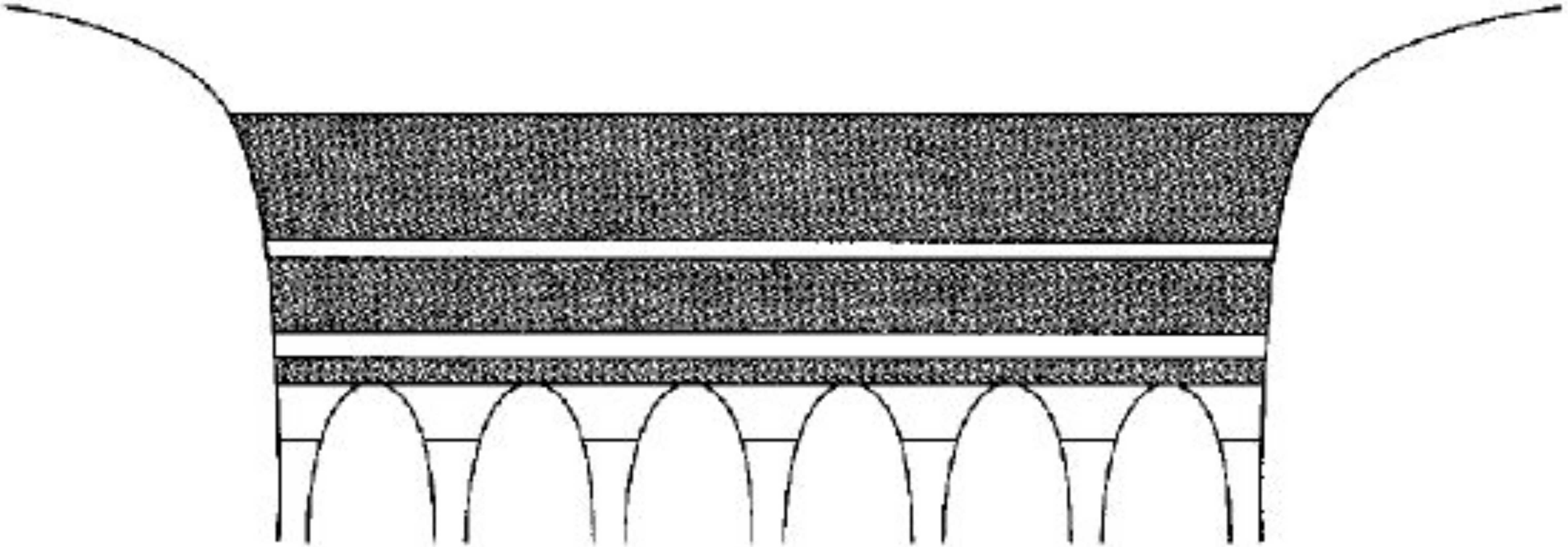
F(q) has a solution, only if $|F(q)| \leq 1$ because the cos(ka) on the right-hand side of the equation is limited to values ≤ 1.





F.E Schubert, Physical foundations of solid state devices

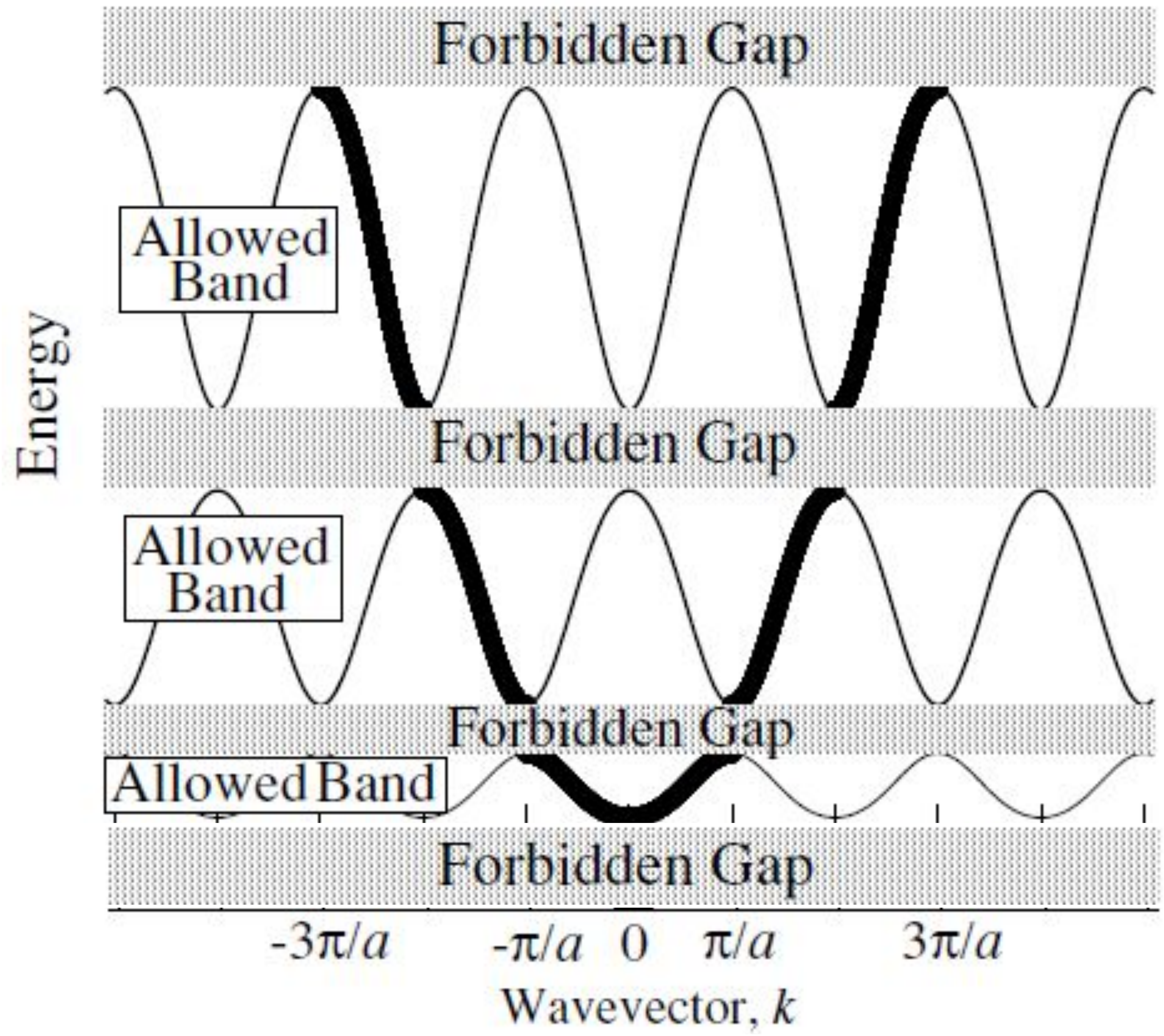
Emergence of energy bands and gaps



No discrete energy levels as in isolated atoms, but consists of allowed bands and forbidden gaps.

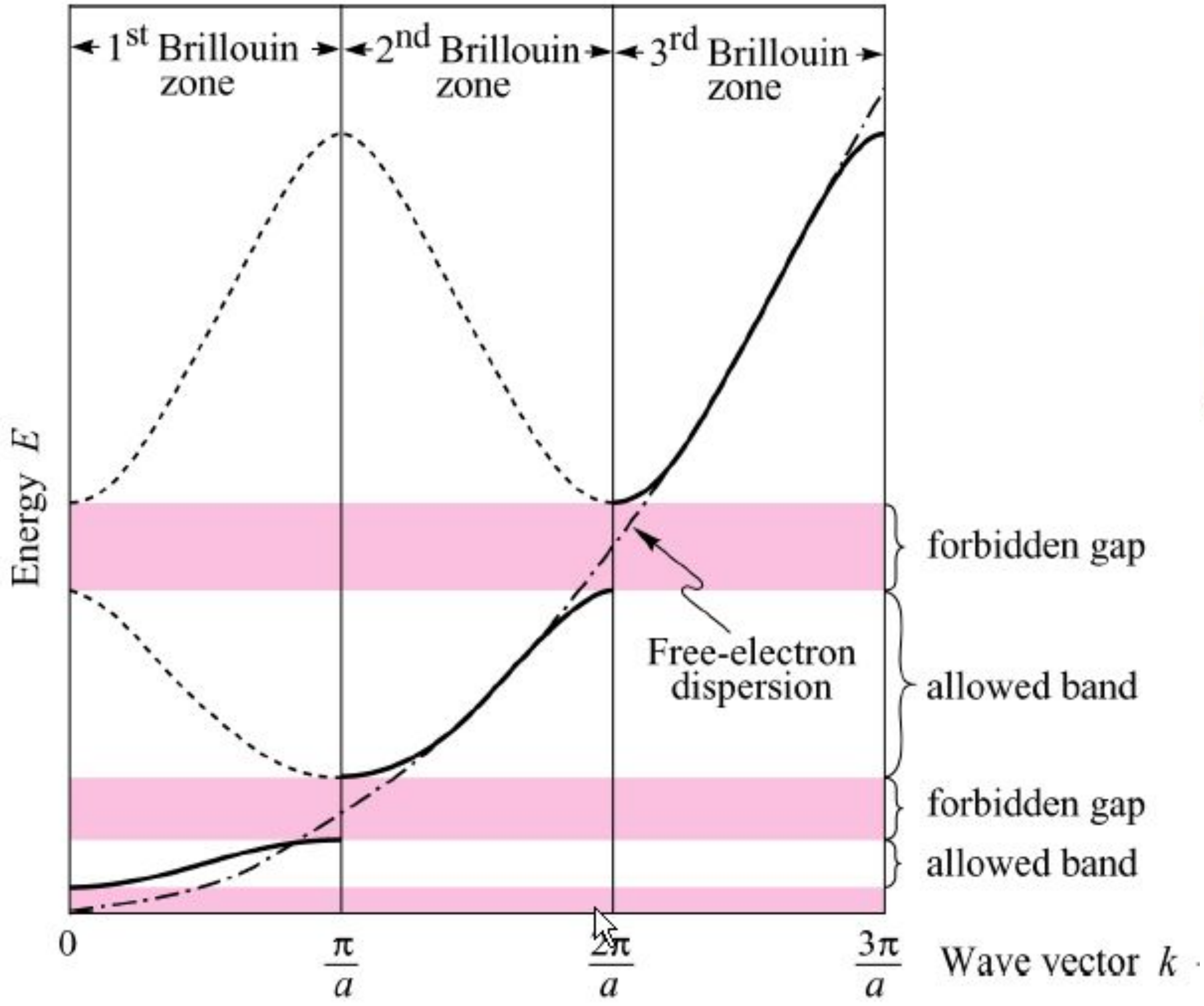
Energy bands naturally arise from the solution of Schrodinger equation in a periodic potential well

Periodic E-k diagram

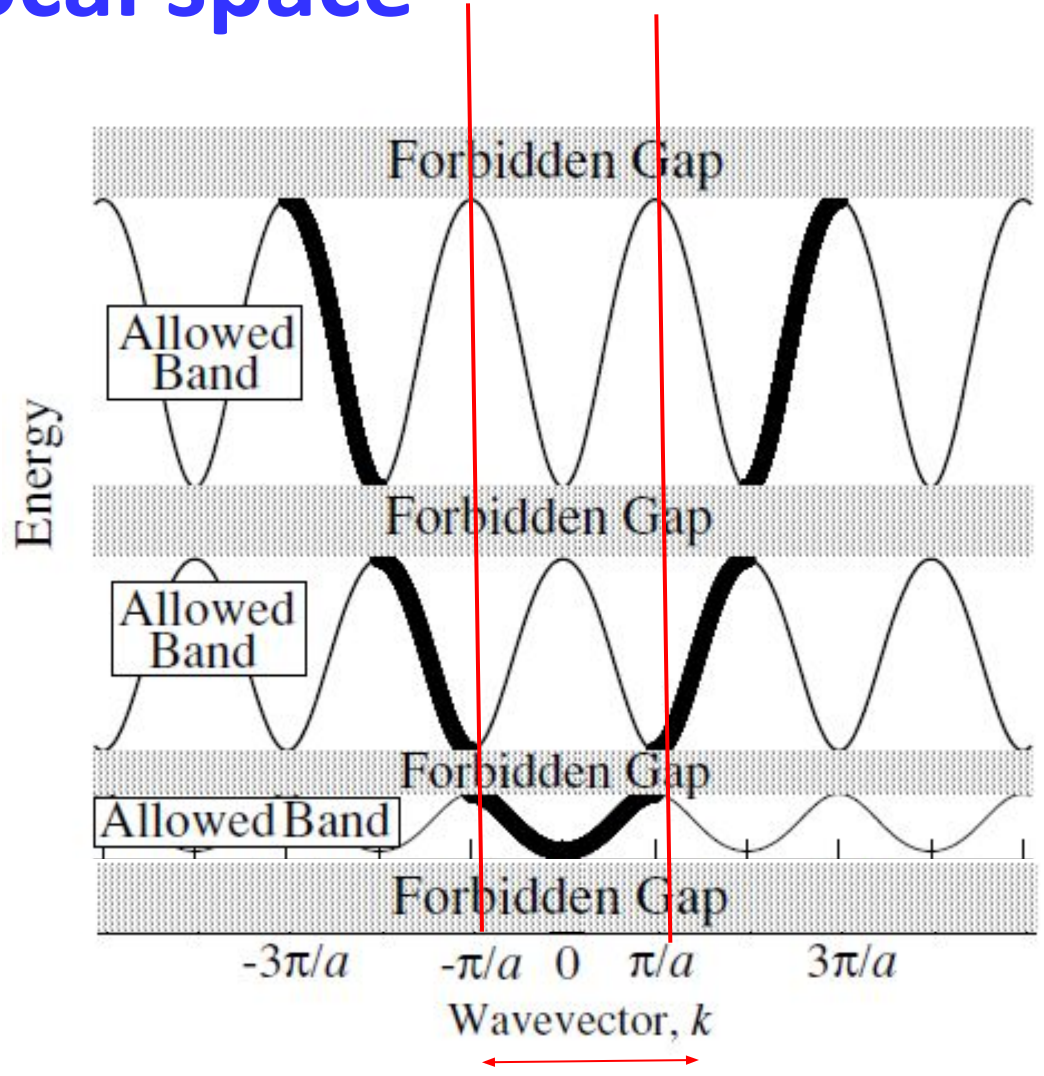


Please note that these aren't sinusoidal functions in general. It is periodic with a maxima and a minima

Energy bands and gaps in reciprocal space



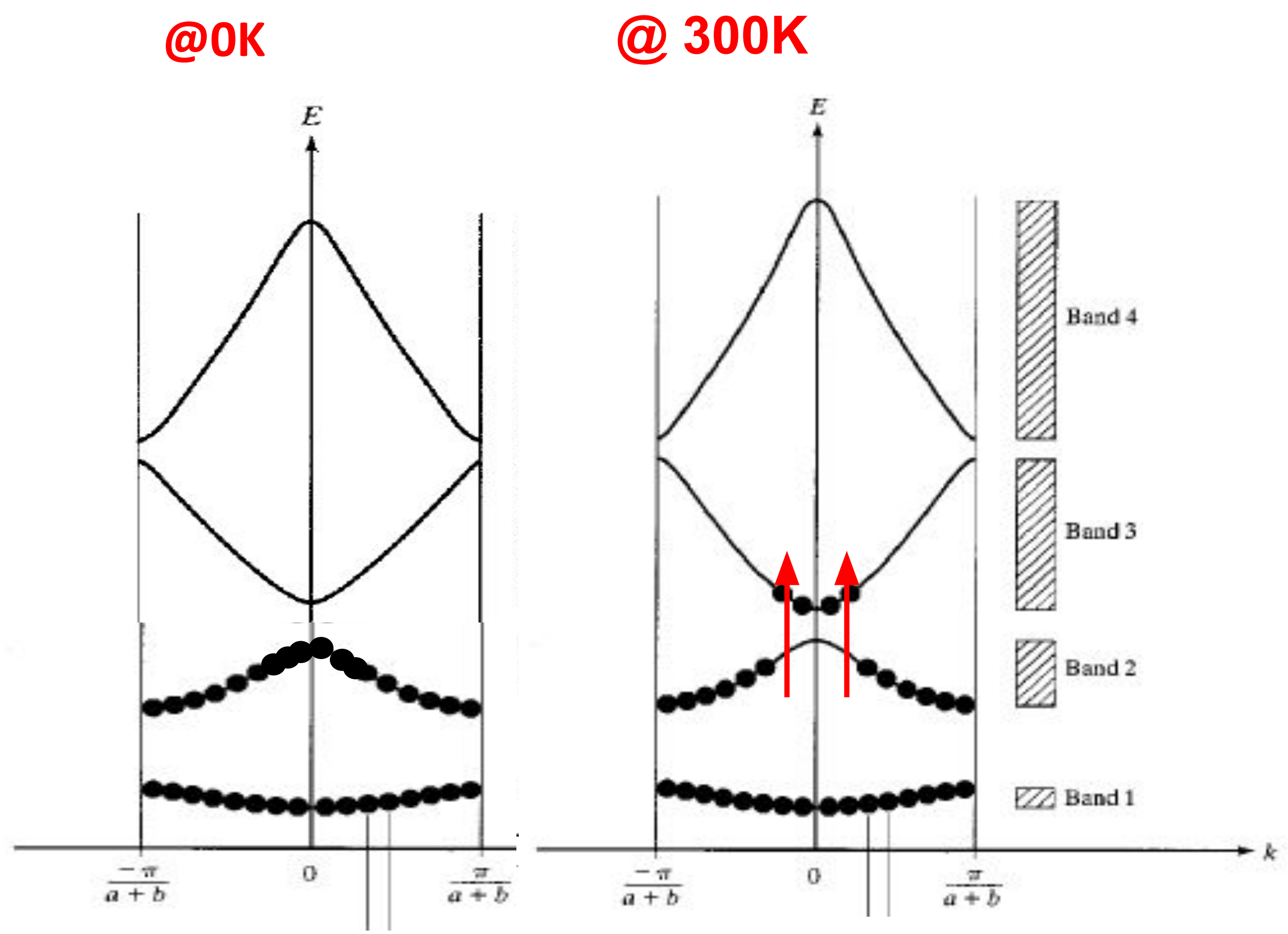
Extended Zone



Reduced Zone

Please note that these aren't sinusoidal functions in general.

How bands are occupied by electrons?



Top most completely filled band @ 0K – valence band (VB)

Band above VB – conduction band (CB)

At higher temp, electrons can gain energy and move to higher bands, leaving behind vacancies.

Completely filled bands don't contribute to current due to symmetry of E-k

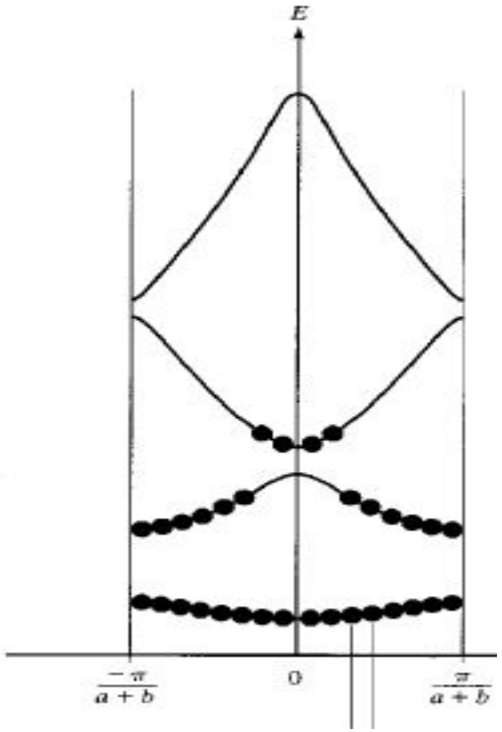
$$v(k) = \frac{1}{\hbar}\frac{dE}{dk}$$

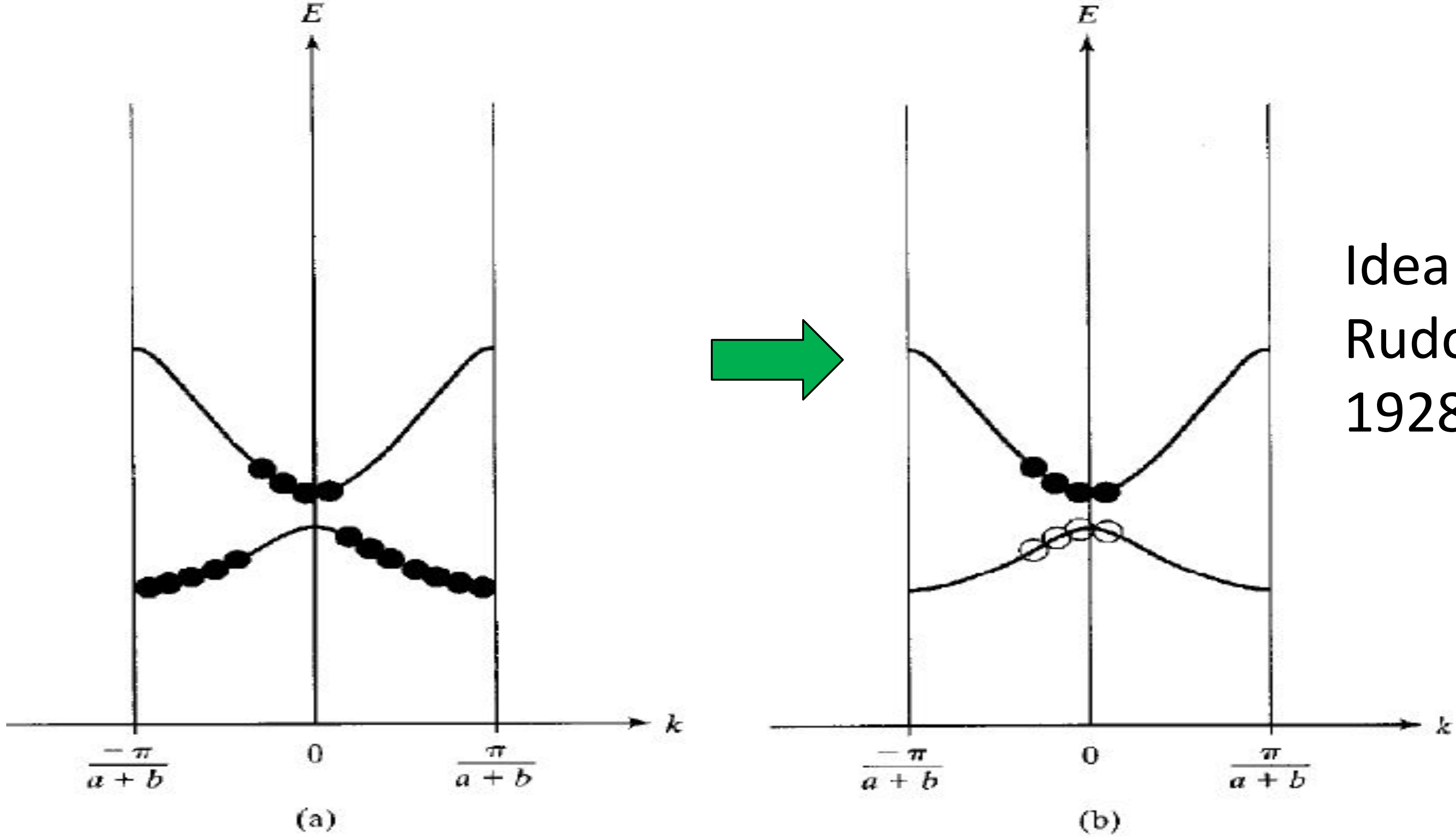
$$v(-k) = - v(k)$$

Partially occupied band & Concept of hole

Electric field in +ve x direction

Zero field @300K



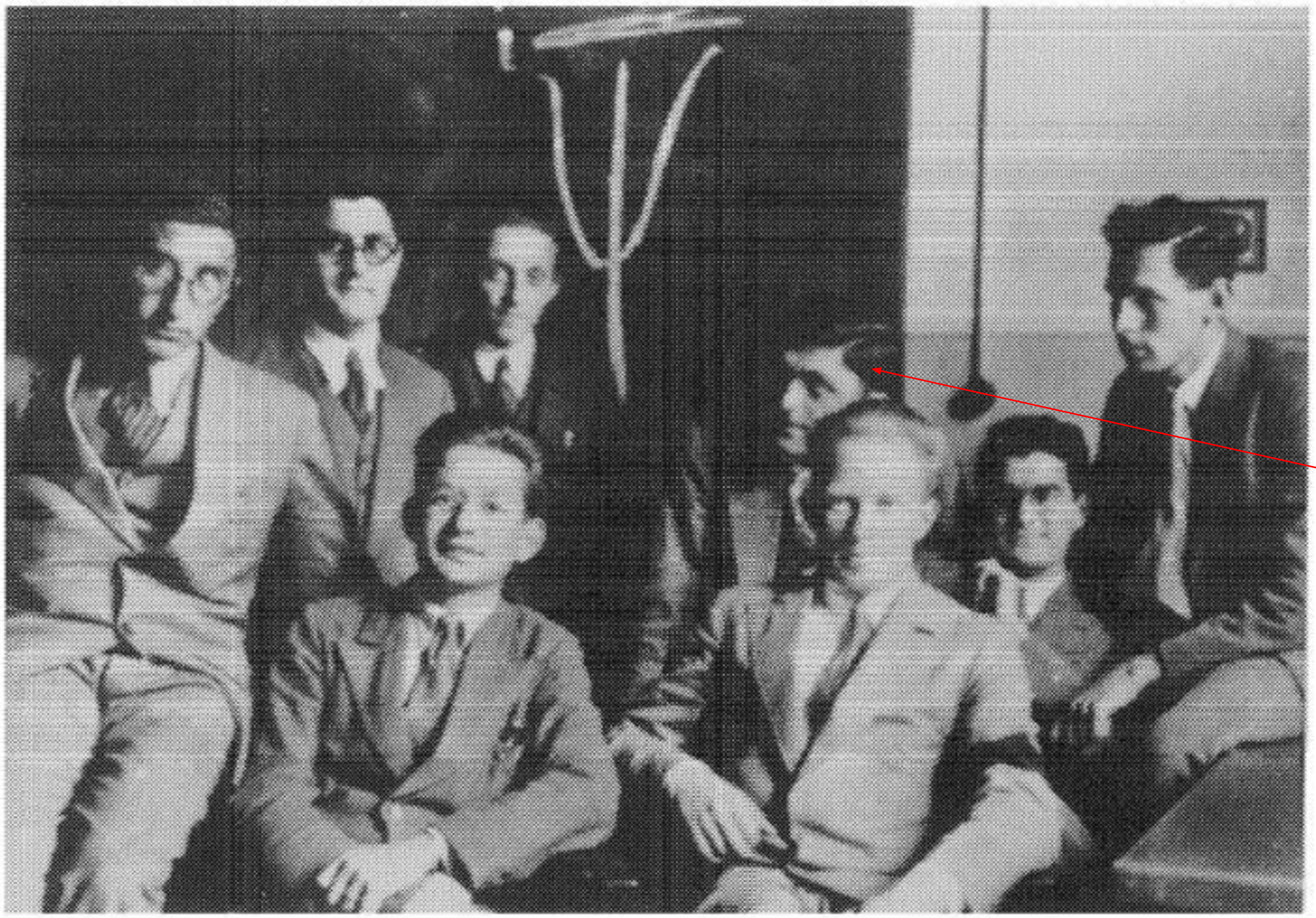


Idea of holes - Rudolf Peierls in 1928

$$J_{vb} = \frac{1}{V}\Big(\sum_{\text{all states}}(-q)v_i - \sum_{\text{empty states}}(-q)v_i \Big)$$

$$J_{vb} = \frac{1}{V} \sum_{\text{empty states}}(+q)v_i$$

Dynamics of large number of VB electrons with negative charge ≡ small number of vacant states with positive charge



Felix Bloch

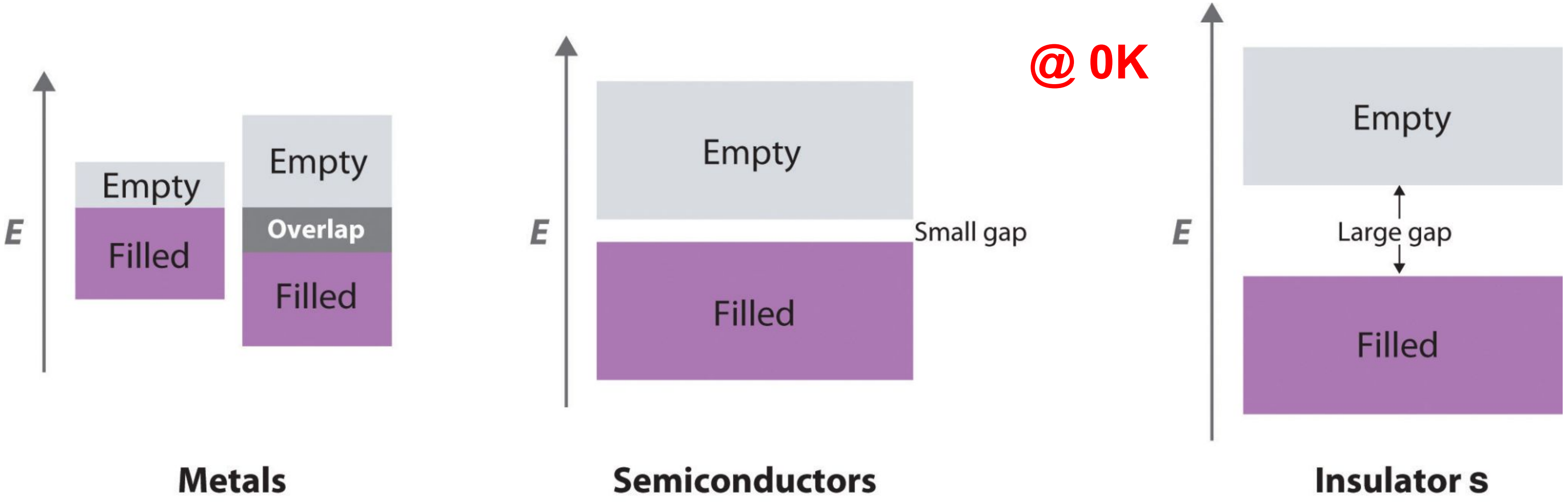
Heisenberg's Institute in Leipzig, 1931

Rudolf Peierls Werner Heisenberg

https://repository.aip.org/front-row-l-r-rudolf-peierls-w-heisenberg-back-row-l-r-g-placzek-g-gentile-g-wick-f-bloch-v

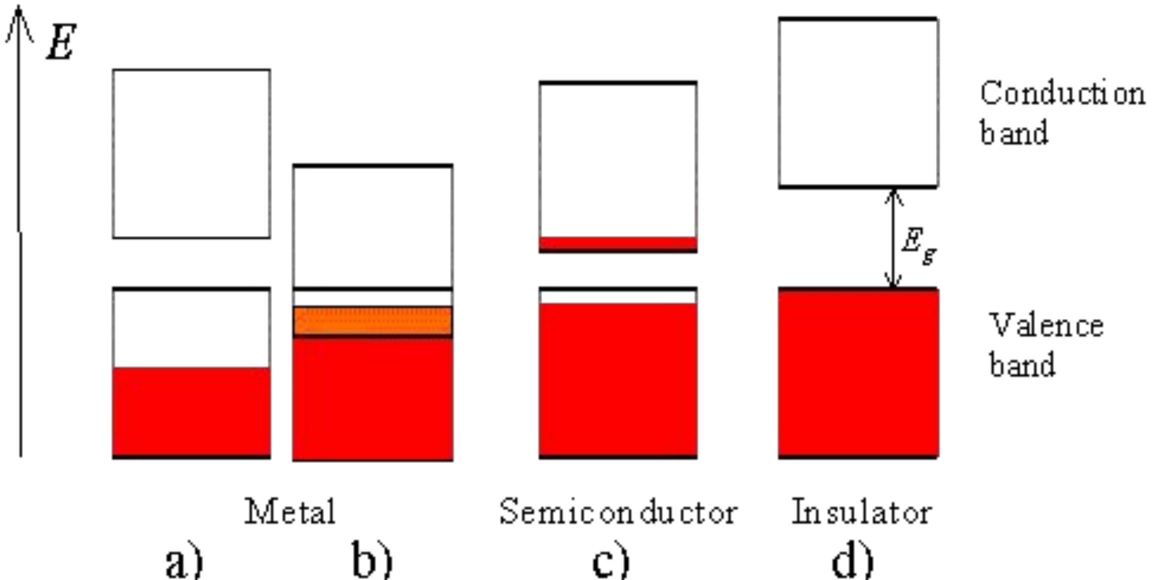
Materials classification

@ 0K



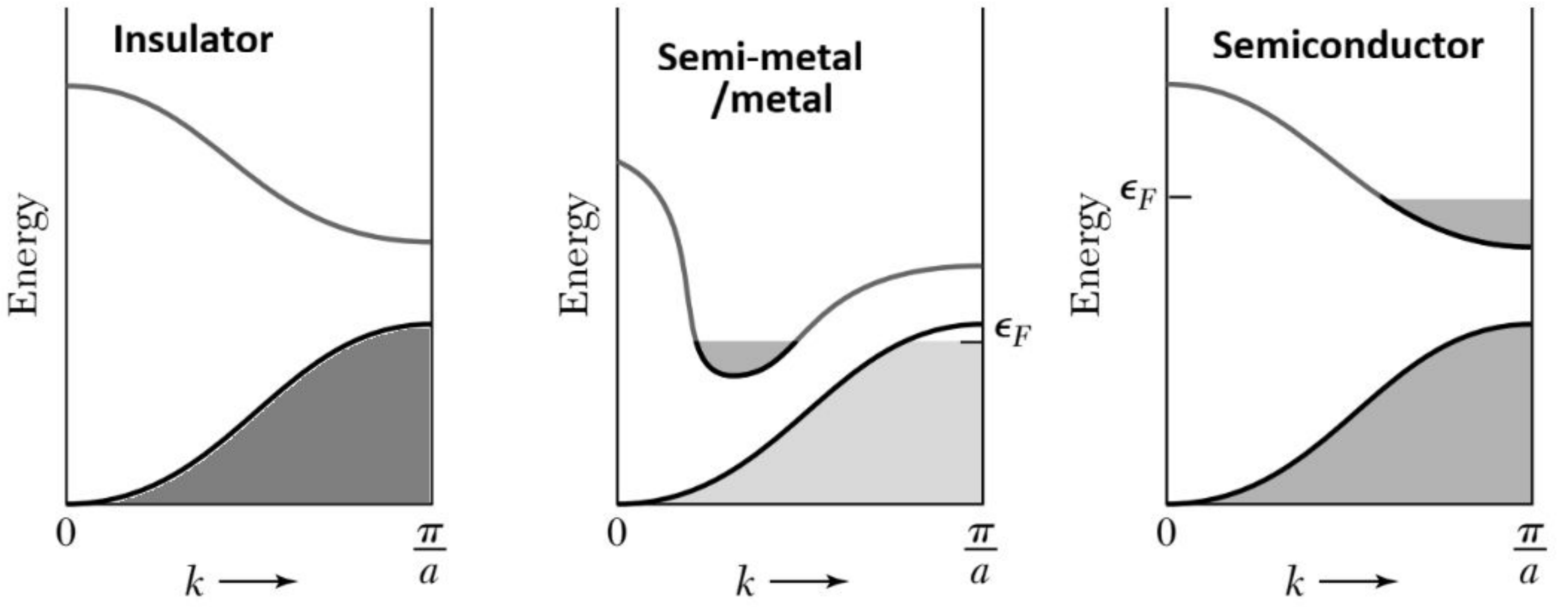
Alan Wilson in 1931

@ 300K

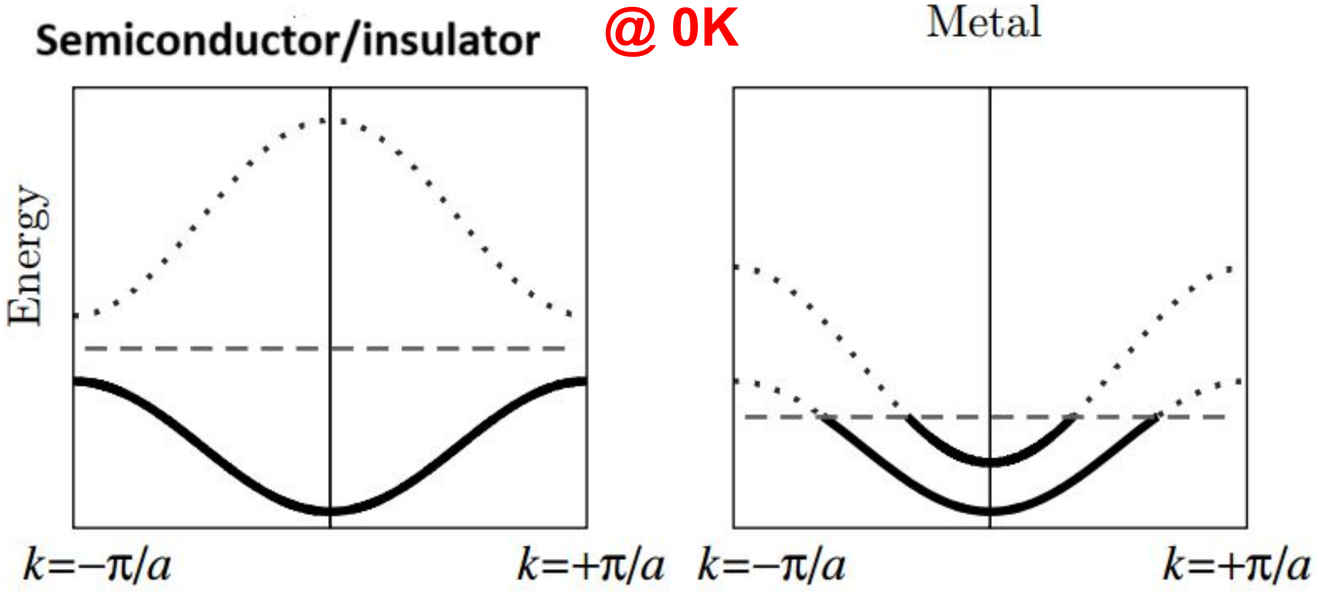


Material	E_g (eV)
Germanium	0.66
Silicon	1.12
Gallium arsenide	1.42
Silicon carbide (cubic)	2.4
Gallium nitride	3.4
Silicon nitride (Si_3N_4)	5
Diamond	5.5
Silicon dioxide (SiO_2)	9

Materials classification (bandstructure)



@ 300K

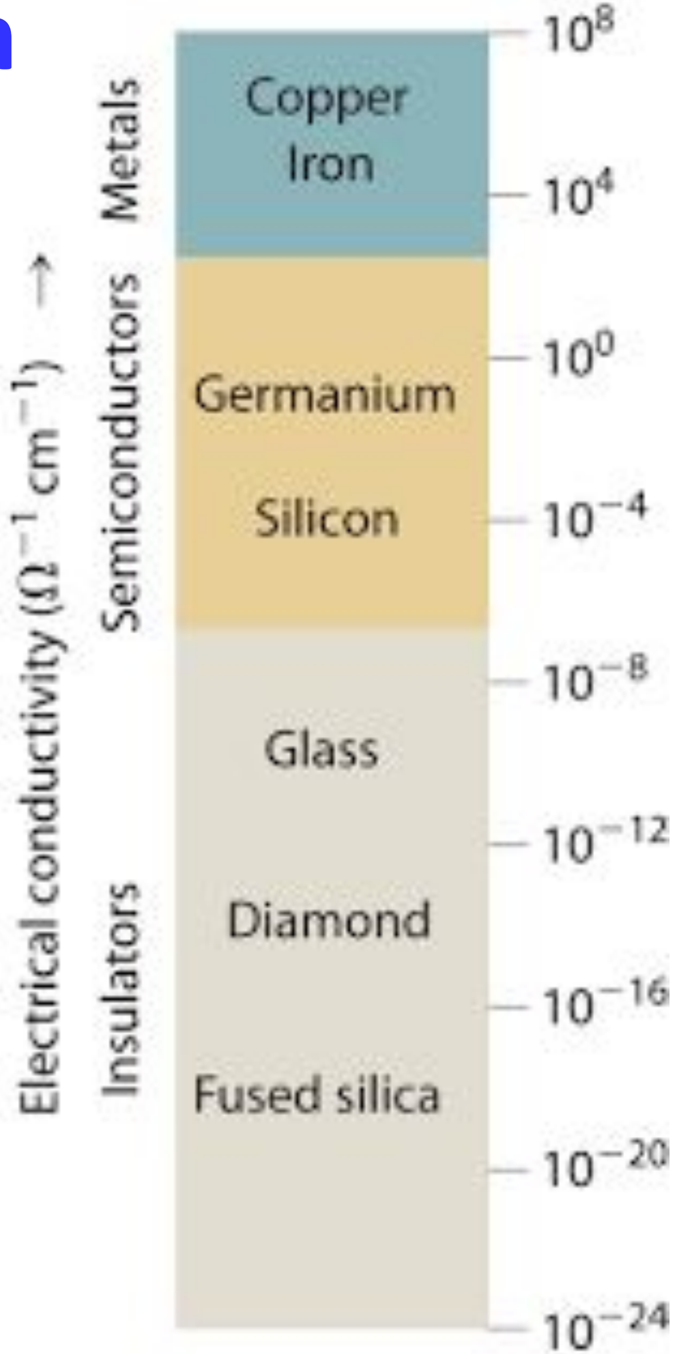


@ 0K

Clarity on the conductivity conundrum

TABLE 2.1 Energy-Gap Values for Different Materials at 300 K

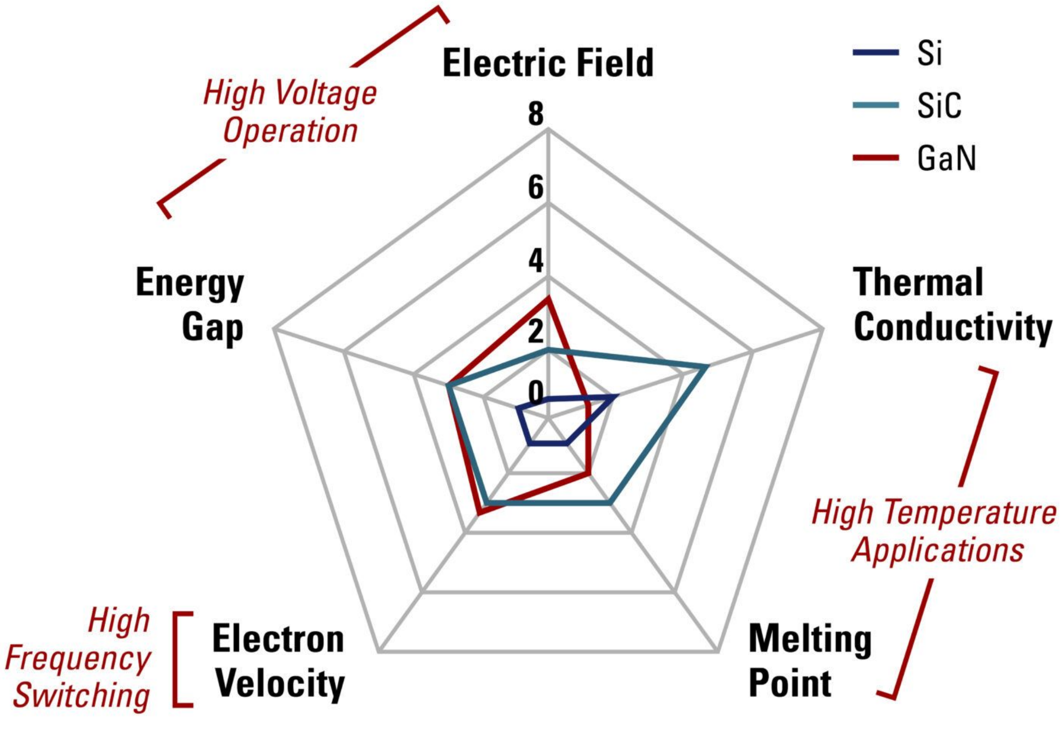
Material	E_g (eV)
Germanium	0.66
Silicon	1.12
Gallium arsenide	1.42
Silicon carbide (cubic)	2.4
Gallium nitride	3.4
Silicon nitride (Si_3N_4)	5
Diamond	5.5
Silicon dioxide (SiO_2)	9

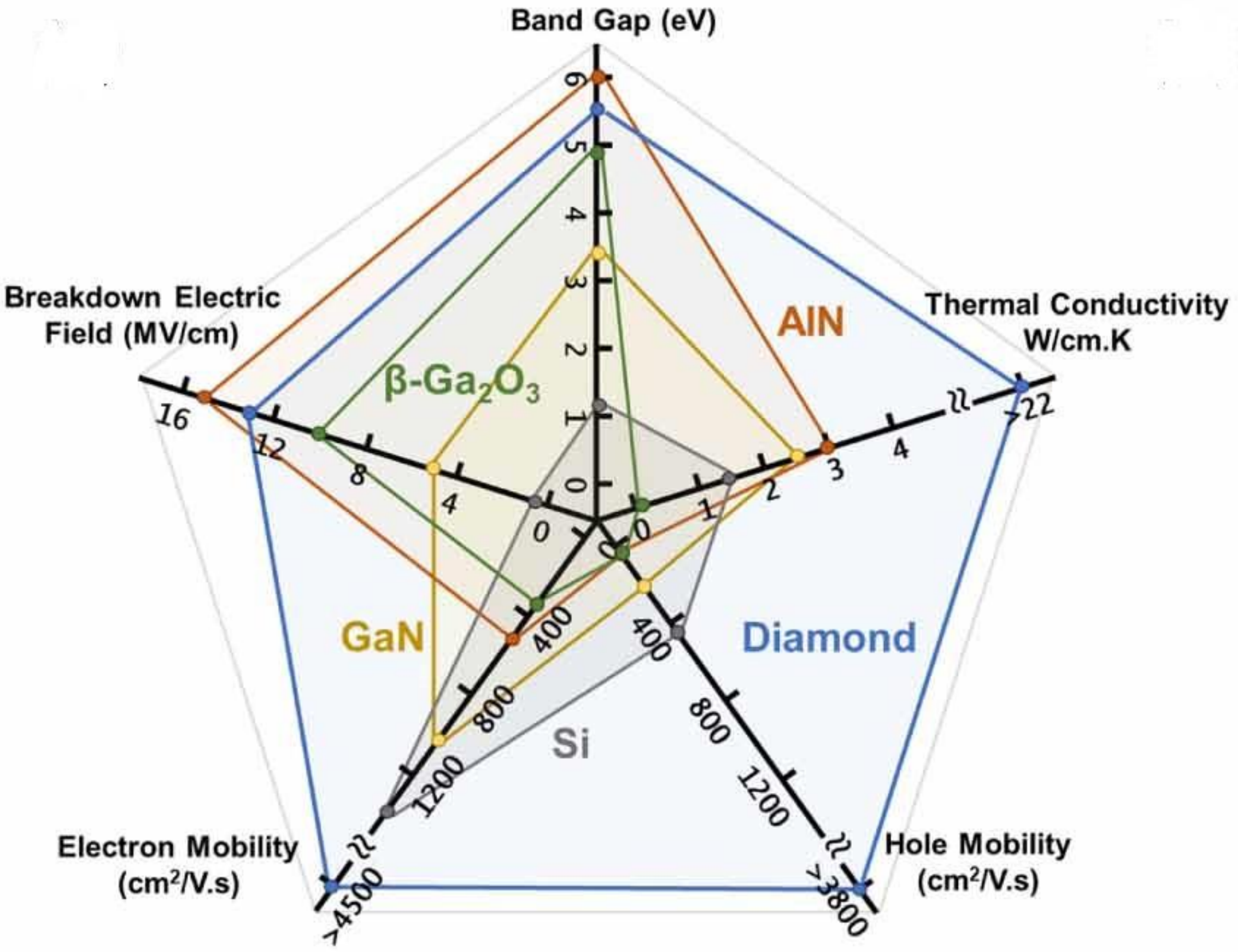


- Why does a material display a specific value for a specific property from the range of values possible for that property?

Wide bandgap and Ultra-wide bandgap semiconductors

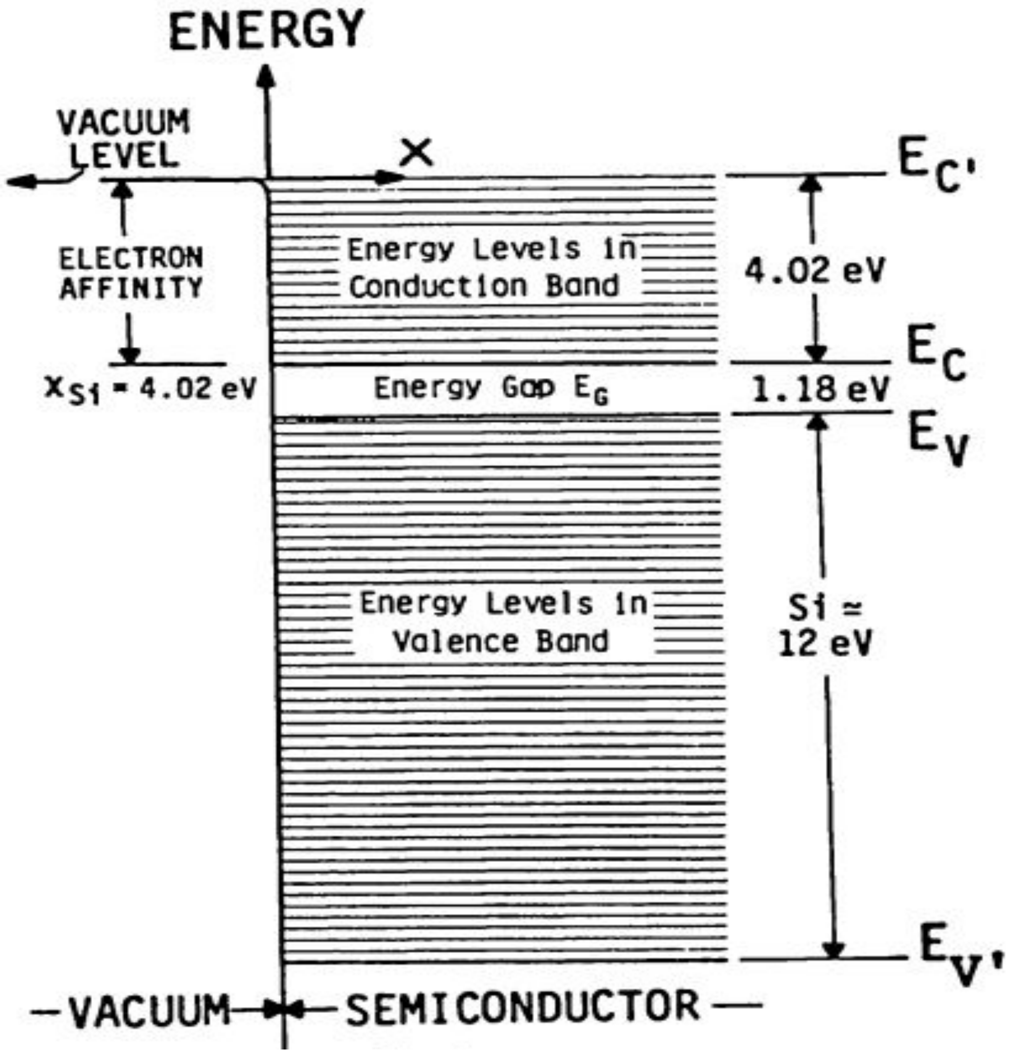
Motivation - light emission in the UV/visible range for optoelectronics and higher breakdown voltages for power electronics

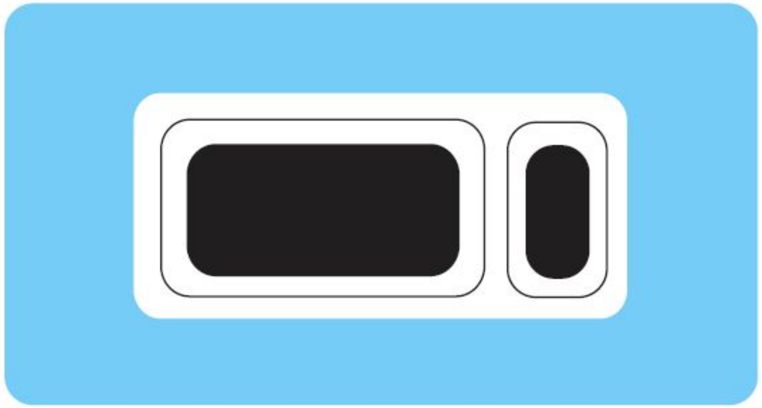




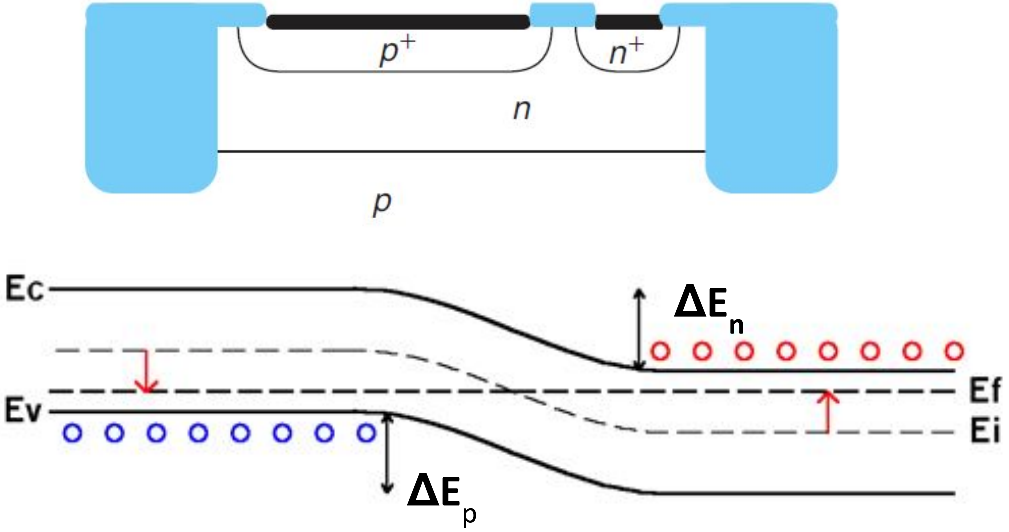
Energy band diagram (E-x)

E-x diagram becomes useful when there is an irregularity of material composition and/or electric field in real space (happens in devices)



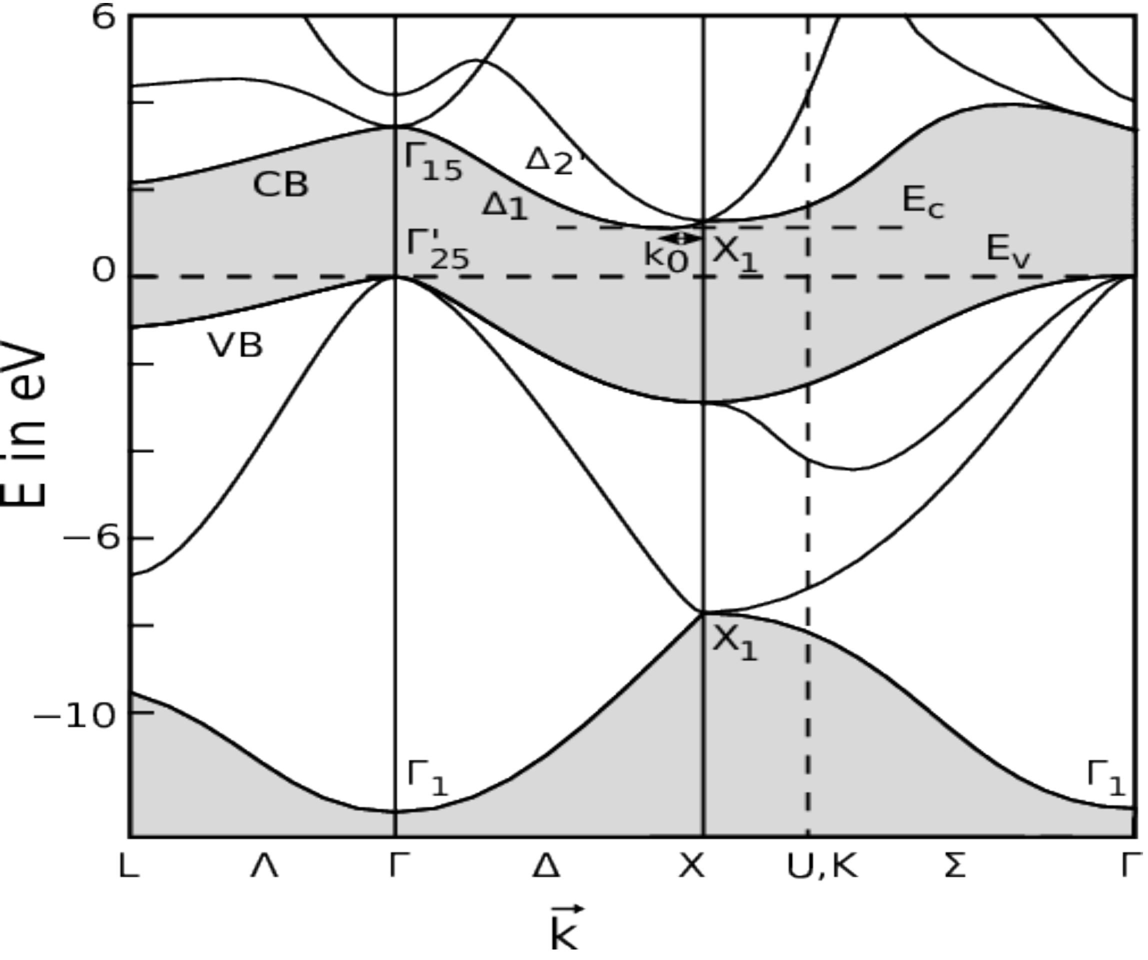


Top view and cross-sectional view of integrated planar PN junction diode

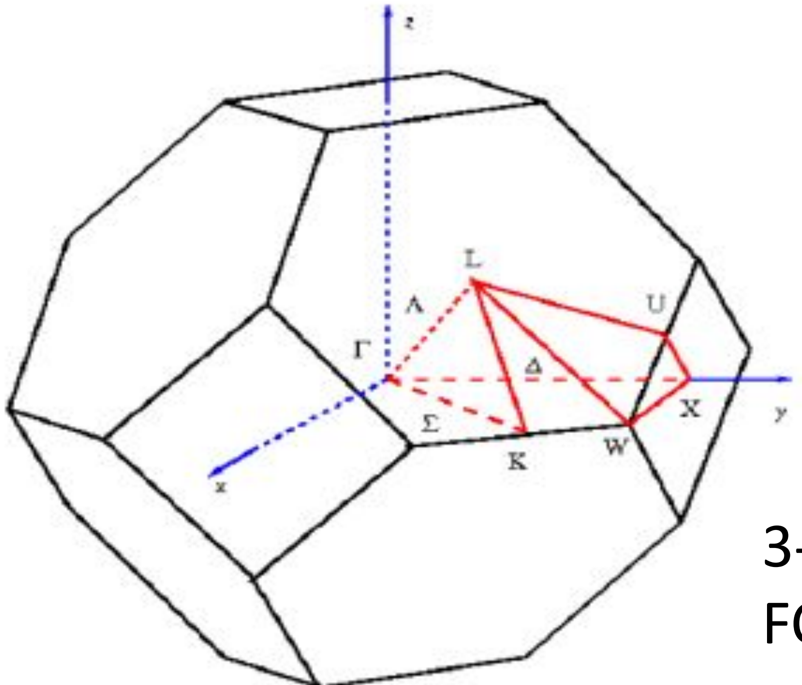


Energy band structure (E-k)

E-k diagram gives a richer and more compact representation in the case of an infinite homogeneous crystal in real space (eg: Optoelectronics)

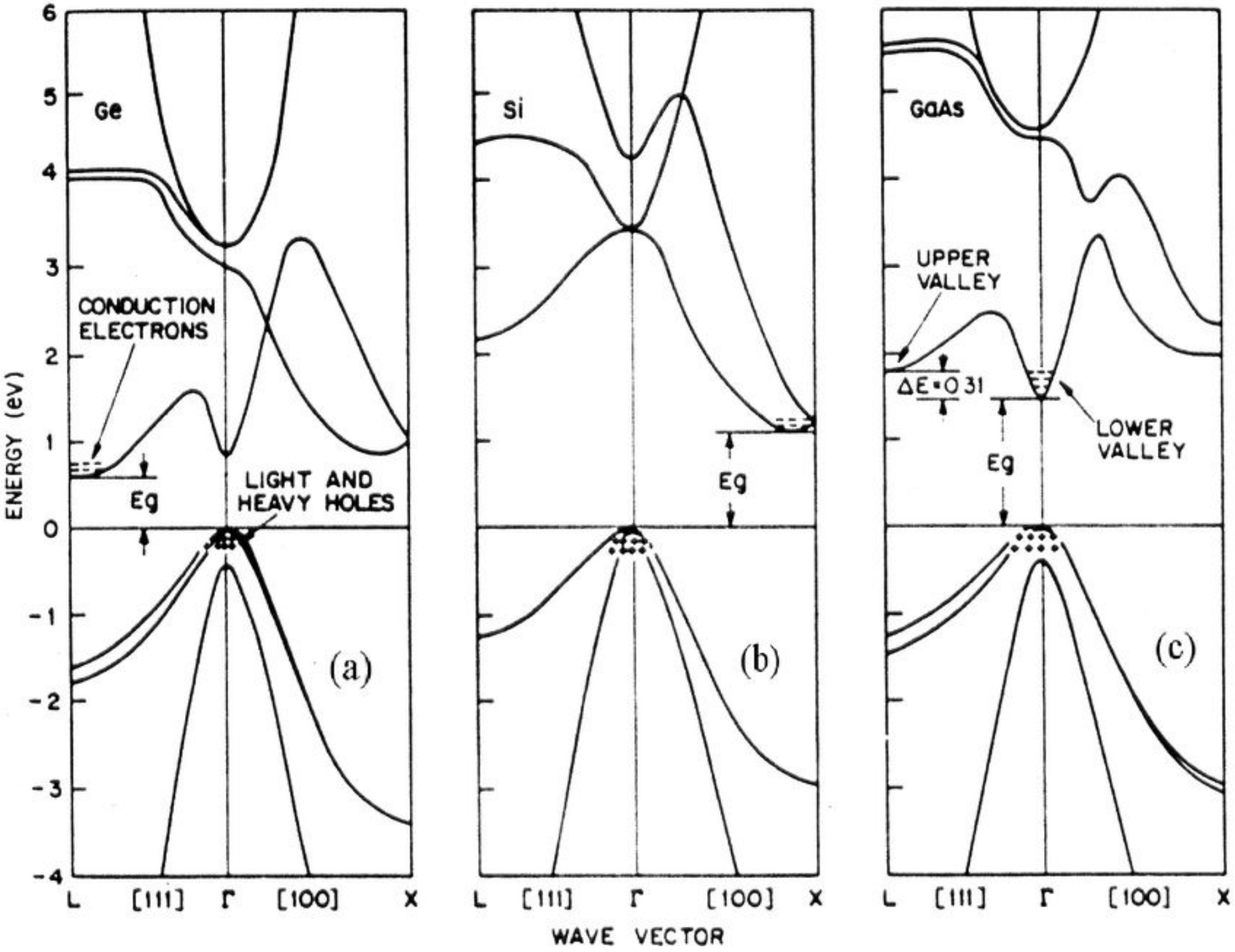


Symmetric Points/ Directions	Coordinate	Remark
Γ	$(0,0,0)$	Origin of **k** space
X	$(1,0,0)$	Middle of square faces
L	$(\frac{1}{2},\frac{1}{2},\frac{1}{2})$	Middle of hexagonal faces
K	$(\frac{3}{4},\frac{3}{4},\frac{3}{4})$	Middle of edge shared by two hexagons
U		Middle of edge shared by a hexagons and a square
W		Middle of edge shared by two hexagons and a square
Δ	$\langle 1,0,0\rangle$	Directed from Γ to X
Λ	$\langle 1,1,1\rangle$	Directed from Γ to L
Σ	$\langle 1,1,0\rangle$	Directed from Γ to K



3-D Brillouin zone of FCC lattice

Direct and indirect semiconductors

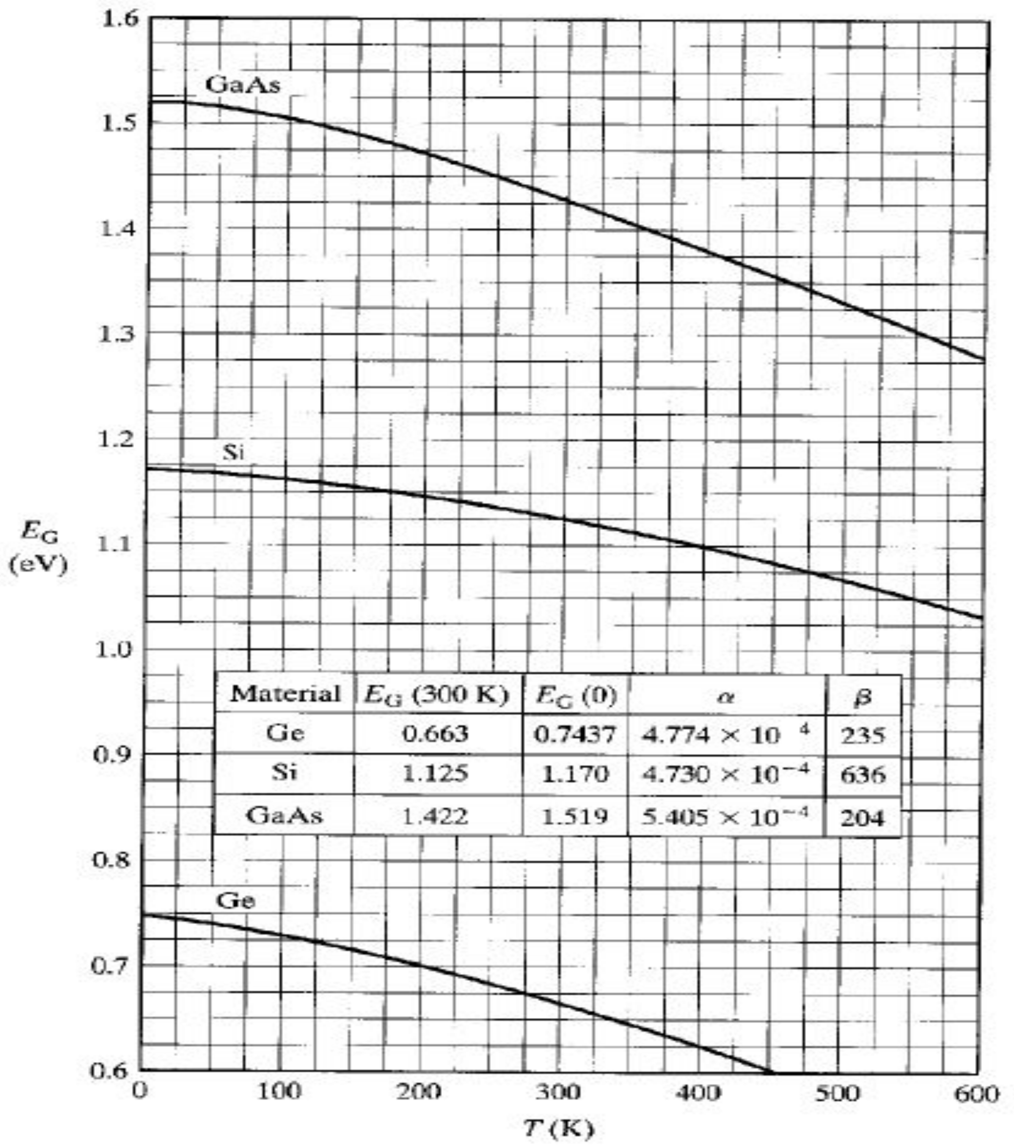


CB min and VB max lies

at same k – Direct (GaAs)

at diff k – Indirect (Si, Ge)

Energy band gap – Temperature dependence



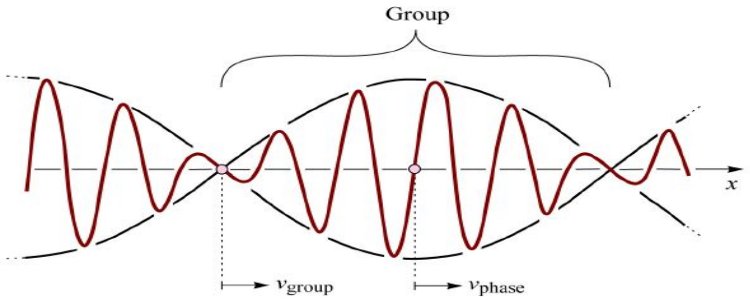
Material	E_G (300 K)	E_G (0)	α	β
Ge	0.663	0.7437	4.774×10^{-4}	235
Si	1.125	1.170	4.730×10^{-4}	636
GaAs	1.422	1.519	5.405×10^{-4}	204

$$E_G(T) = E_G(0) - \frac{\alpha T^2}{(T + \beta)}$$

The bandgap of a semiconductor decreases with increase in temperature due to the increase of lattice constant

Effective mass of a particle

Group velocity



$$F = m^* \frac{dv_g}{dt} \text{ ------(1)}$$

$$F dt = dp = \hbar dk$$

$$v_g = \frac{d\omega}{dk} = \frac{1}{\hbar}\frac{dE}{dk}$$

$$F = \hbar \frac{dk}{dt} \quad \text{-------(2)}$$

$$\frac{dv_g}{dt} = \frac{1}{\hbar}\frac{d}{dt}\left(\frac{dE}{dk}\right) = \frac{1}{\hbar}\frac{d}{dk}\left(\frac{dE}{dk}\right)\frac{dk}{dt}$$

Substituting in (1) and equating (1) and (2), we have

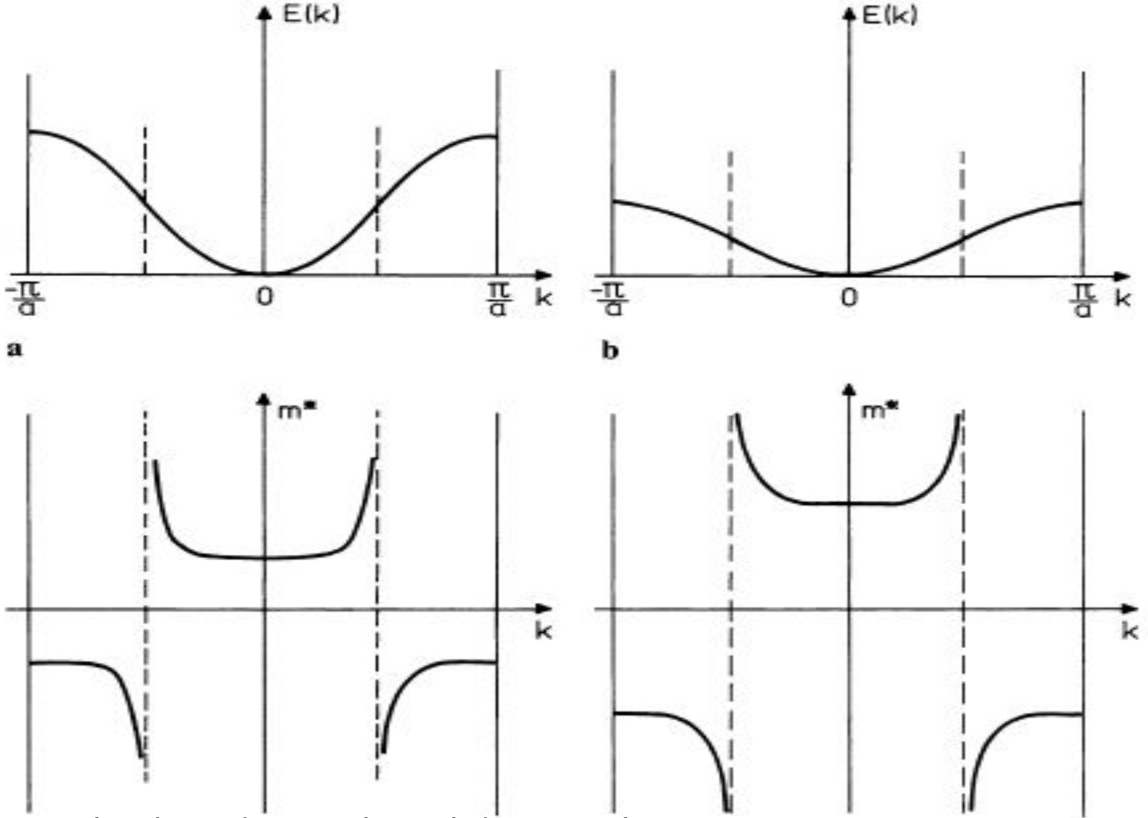
$$\hbar \frac{dk}{dt} = m^* \frac{1}{\hbar}\frac{d}{dk}\left(\frac{dE}{dk}\right)\frac{dk}{dt}$$

$$\frac{1}{m^*} = \frac{1}{\hbar^2}\left(\frac{d^2E}{dk^2}\right)$$

Rate of change of crystal momentum ($\hbar$k) = externally applied force

Electrons in a periodic potential behaves as a free electron with a different mass!

Band structure effective mass of a particle



H. Ibach and H. Luth, *Solid State Physics*, 2009

$$\frac{1}{m_{ij}} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_i \partial k_j}$$ **= Curvature of E-k**

m^* is positive near the bottoms of all bands

m^* is negative near the tops of all bands.

Band structure effective mass depends on

- **material**
- **energy**
- **direction of propagation.**

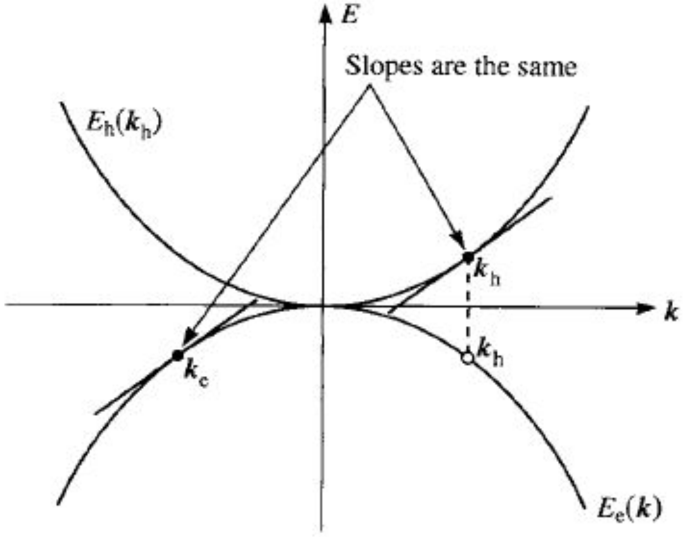
Effective mass is really a tensor

$$\frac{d\mathbf{v}}{dt} = \frac{1}{\mathbf{m}^*}\cdot\mathbf{F} \qquad \frac{1}{\mathbf{m}^*} = \begin{pmatrix} m_{xx}^{-1} & m_{xy}^{-1} & m_{xz}^{-1} \\ m_{yx}^{-1} & m_{yy}^{-1} & m_{yz}^{-1} \\ m_{zx}^{-1} & m_{zy}^{-1} & m_{zz}^{-1} \end{pmatrix}$$

- Electron/hole in a crystal can be treated semi-classically as a free particle but with free space mass replaced by effective mass
- Helps to perform device analysis with minimal use of quantum mechanics
- Use appropriate "average" of band structure effective masses to compute density of states (DoS), conductivity/mobility etc

Comparison of a missing electron in VB & a hole

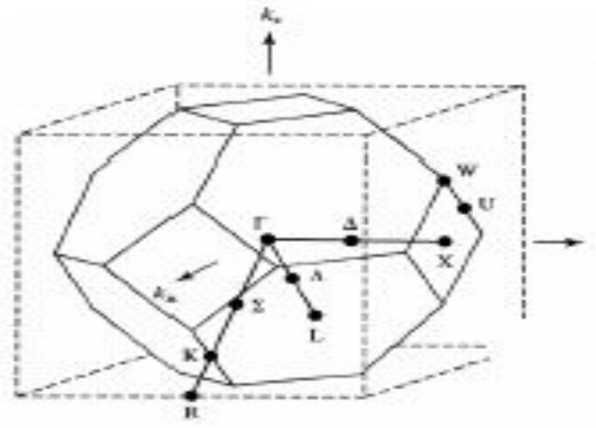
Missing electron (near top of VB)	Assignment for Hole
-ve charge	+ve charge
-ve energy (top of VB is taken as 0 eV)	+ve energy (energy of hole increases downwards)
Wavenumber, k_e	$k_h = - k_e$
Group velocity, v_e	$v_h = v_e$
-ve effective mass	+ve effective mass
Valence band electrons are "real". They can be emitted to vacuum*	Holes are Quasi particles. Holes cannot be emitted to vacuum. Their effect can only be observed while they are inside VB of a solid



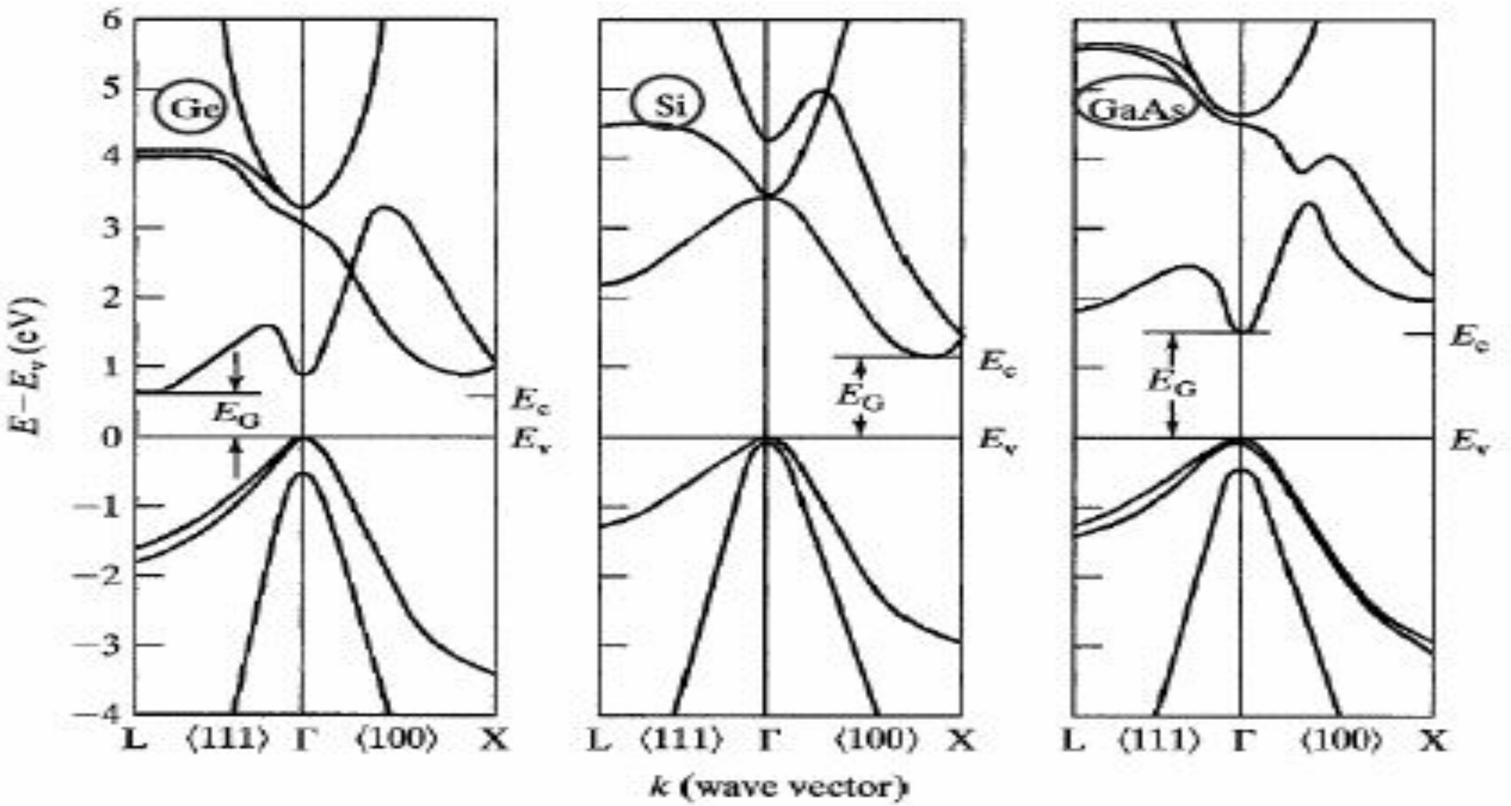
$$\frac{h}{2\pi}\frac{dk_h}{dt} = F_h = +|e|(E + v_h \times B).$$

*M. Ding, H. Kim, A.I. Akinwande, "**Observation of valence band electron emission from n-type silicon field emitter arrays**" *Appl. Phys. Lett., 75 (1999), pp. 823-825*

4D nature of dispersion (E-k) relation



- 3 valence bands (light hole, heavy hole, split-off) valence bands near k=0 is essentially E ~ k^2
- Minima may not be at zone center (Ge: 8 L valley, Si: 6 X valley, and GaAs: Γ valley). May transfer from Γ to L to X.

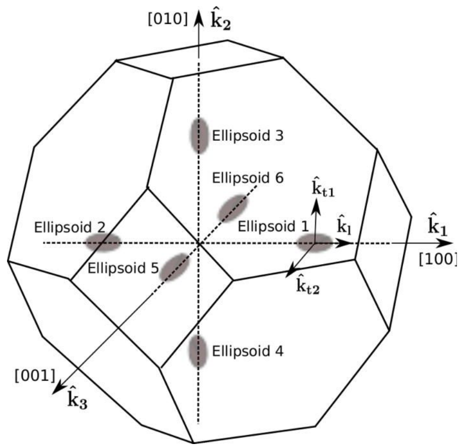


CB minimum and VB maximum lies at same k – Direct (GaAs)

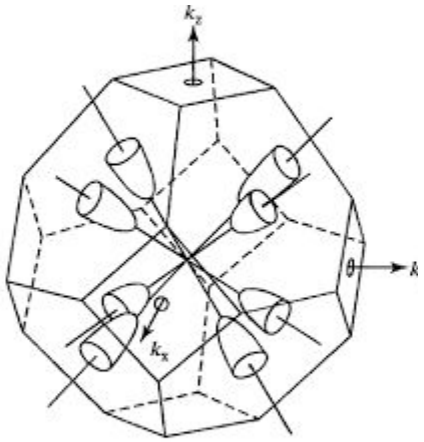
at diff k – Indirect (Si, Ge)

R.F. Pierret, Advanced Semiconductor Fundamentals

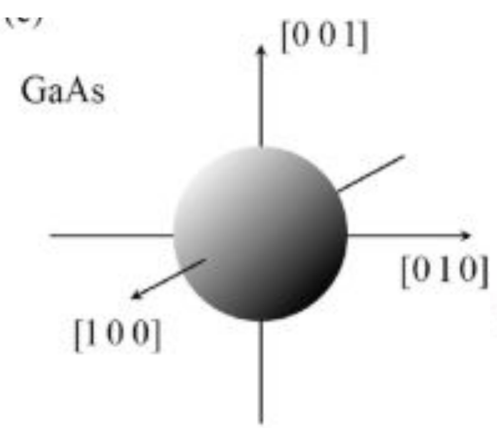
Constant energy surfaces of electrons



Si



Ge



Electrons – Ellipsoids or Sphere

Electron effective mass ($/m_0$) m_t: 0.19 m_l: 0.916

$$E_{c100} = E_g + \frac{\hbar^2}{2m_l}\left(k_x - \frac{1.7\pi}{a}\right)^2 + \frac{\hbar^2}{2m_t}k_y^2 + \frac{\hbar^2}{2m_t}k_z^2,$$

$$E_{c\bar{1}00} = E_g + \frac{\hbar^2}{2m_l}\left(k_x + \frac{1.7\pi}{a}\right)^2 + \frac{\hbar^2}{2m_t}k_y^2 + \frac{\hbar^2}{2m_t}k_z^2,$$

$$E_{c010} = E_g + \frac{\hbar^2}{2m_t}k_x^2 + \frac{\hbar^2}{2m_l}\left(k_y - \frac{1.7\pi}{a}\right)^2 + \frac{\hbar^2}{2m_t}k_z^2,$$

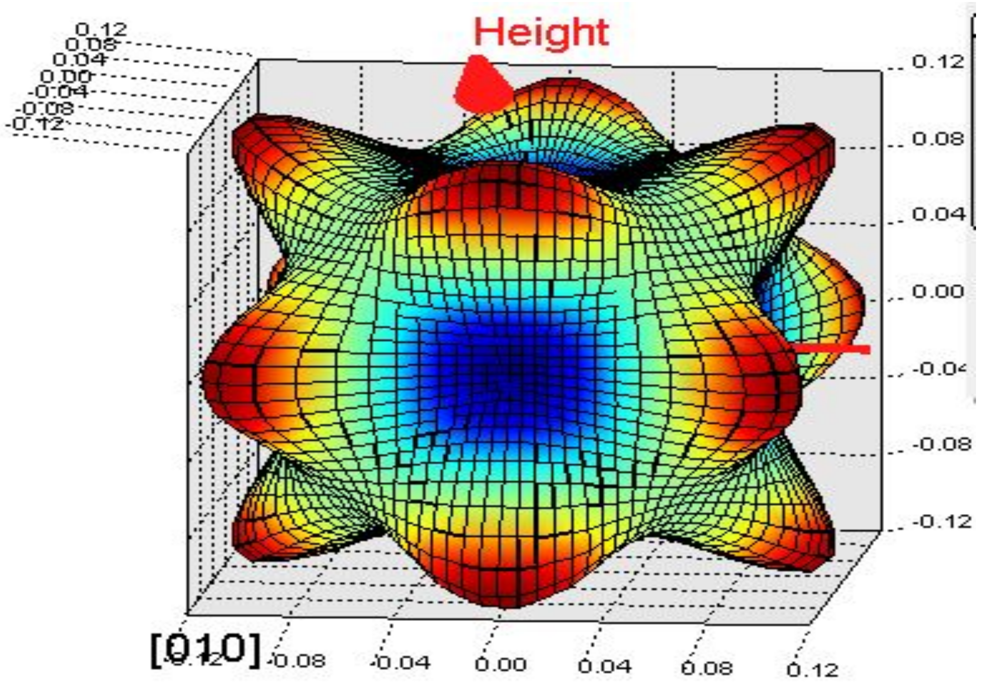
$$E_{c0\bar{1}0} = E_g + \frac{\hbar^2}{2m_t}k_x^2 + \frac{\hbar^2}{2m_l}\left(k_y + \frac{1.7\pi}{a}\right)^2 + \frac{\hbar^2}{2m_t}k_z^2,$$

$$E_{c001} = E_g + \frac{\hbar^2}{2m_t}k_x^2 + \frac{\hbar^2}{2m_t}k_y^2 + \frac{\hbar^2}{2m_l}\left(k_z - \frac{1.7\pi}{a}\right)^2,$$

$$E_{c00\bar{1}} = E_g + \frac{\hbar^2}{2m_t}k_x^2 + \frac{\hbar^2}{2m_t}k_y^2 + \frac{\hbar^2}{2m_l}\left(k_z + \frac{1.7\pi}{a}\right)^2.$$

In Ge we have 8 conduction band minima (along 111 direction), in the case of Si we have 6 conduction band equivalent valleys (along 100 direction) and in the case of GaAs we have only one constant energy surface at the center of the Brillouin zone.

Constant energy surfaces of holes

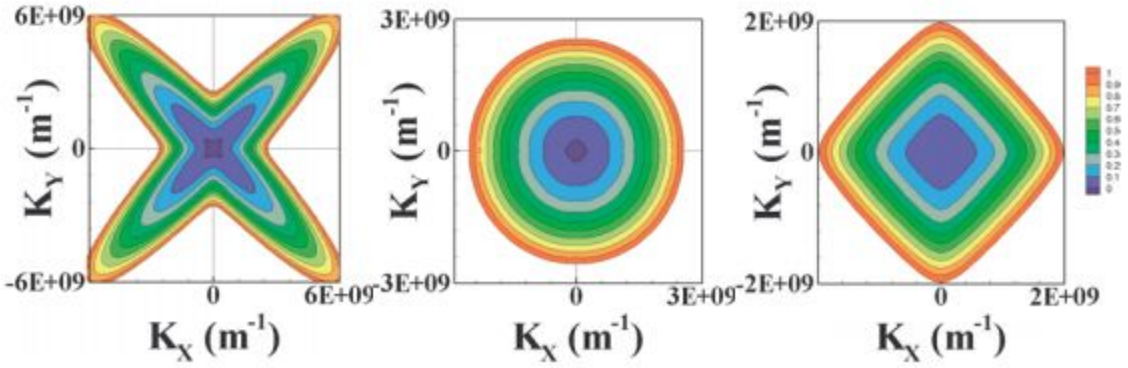


Holes – Warped sphere

$$E_{hh} = -Ak^2 - [B^2k^4 + C^2(k_x^2k_y^2 + k_y^2k_z^2 + k_z^2k_x^2)]^{\frac{1}{2}},$$
$$E_{lh} = -Ak^2 + [B^2k^4 + C^2(k_x^2k_y^2 + k_y^2k_z^2 + k_z^2k_x^2)]^{\frac{1}{2}}.$$

Hole effective mass ($/m_0$) m_{HH}: 0.49

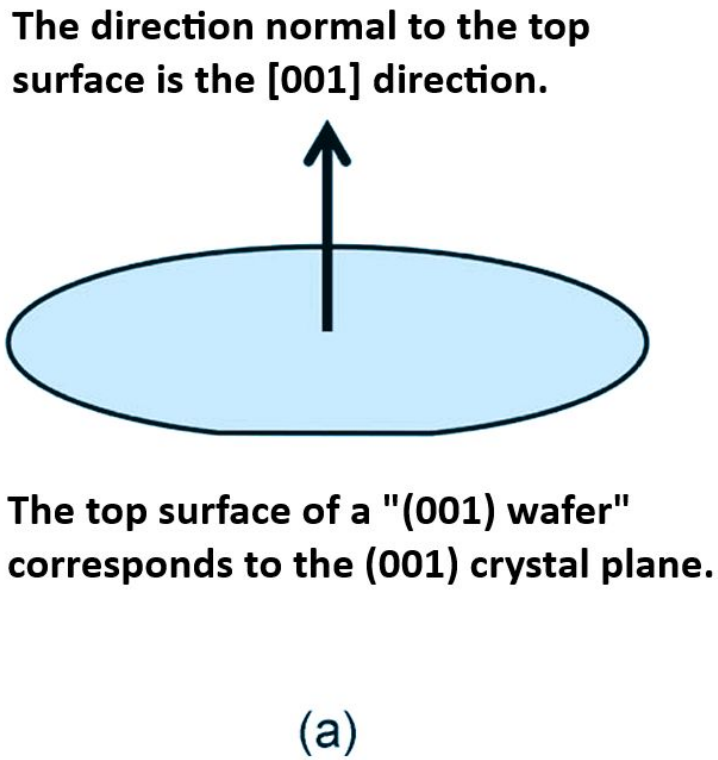
m_{LH}: 0.16

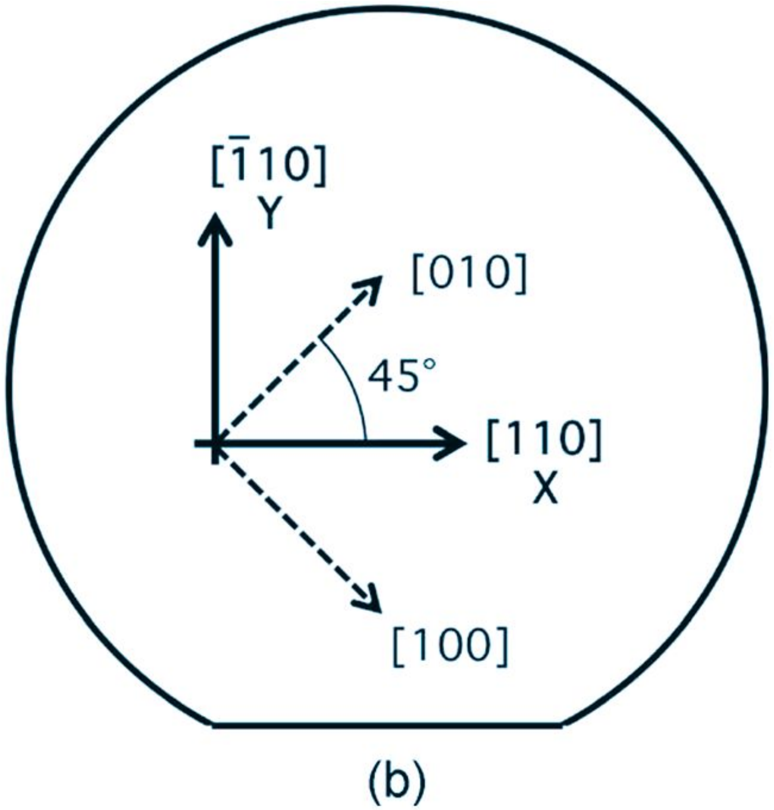


3D equienergy surfaces of heavy hole, light hole and split off band in Si for $k_z = 0$.

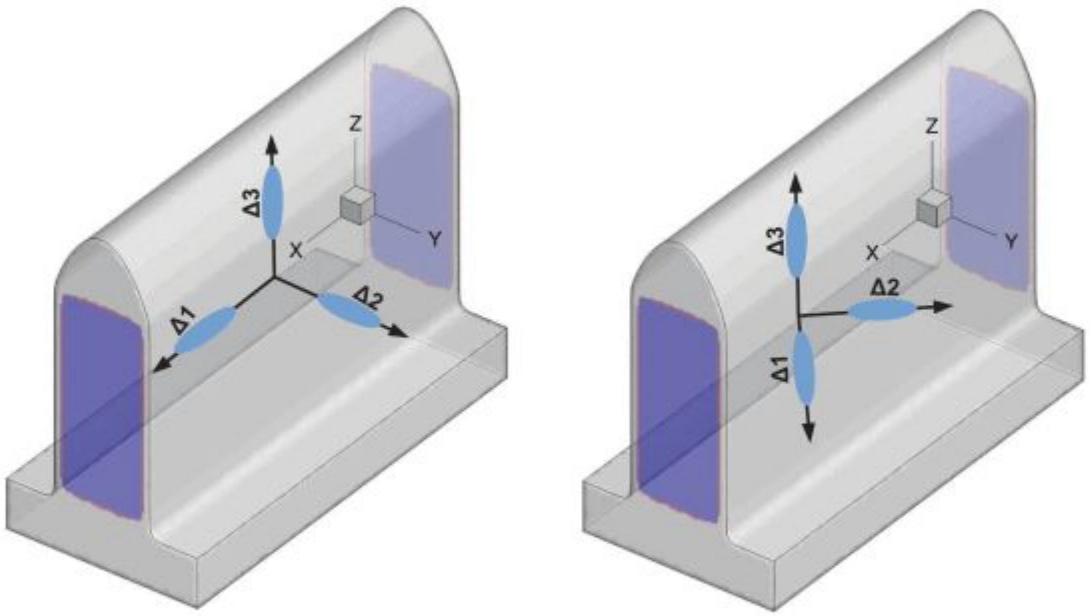
Courtesy: Mastar

Orientation of most commonly used Silicon wafer

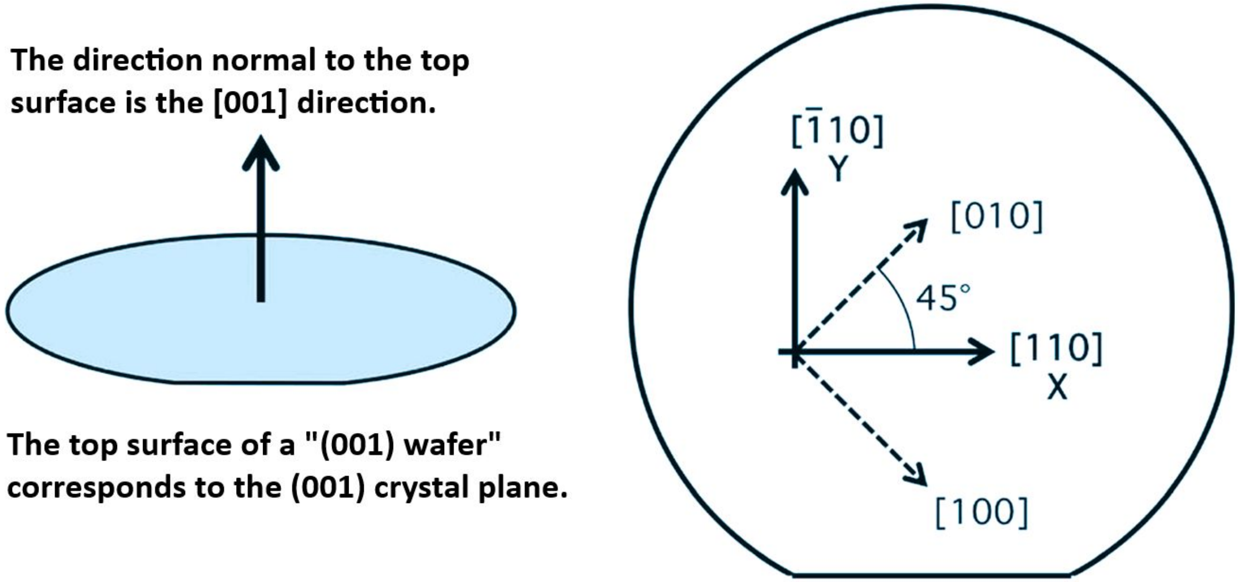




Crystal orientation of a standard (001) silicon wafer. (a) The wafer surface is aligned with the (001) plane. (b) The in-plane X- and Y-axes are oriented along the [110] and [$\bar{1}$10] directions, respectively.



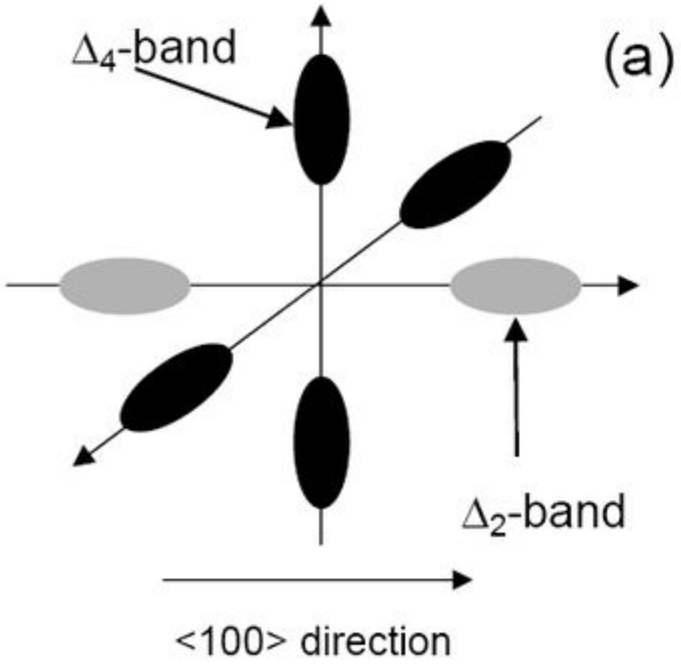
(a) (b)



(a) (b)

a) Channel along <100> direction
b) Channel along <110> direction

Orientation	Valley	$1/m^*_{yy}$	$1/m^*_{zz}$	m^*_{Tr}
$\langle 100 \rangle$	$\Delta 1$	$1/m_t$	$1/m_t$	m_l
$\langle 100 \rangle$	$\Delta 2$	$1/m_l$	$1/m_t$	m_t
$\langle 100 \rangle$	$\Delta 3$	$1/m_t$	$1/m_l$	m_t
$\langle 110 \rangle$	$\Delta 1$	$(m_t+m_l)/(2m_tm_l)$	$1/m_t$	$(m_t+m_l)/2$
$\langle 110 \rangle$	$\Delta 2$	$(m_t+m_l)/(2m_tm_l)$	$1/m_t$	$(m_t+m_l)/2$
$\langle 110 \rangle$	$\Delta 3$	$1/m_t$	$1/m_l$	m_t

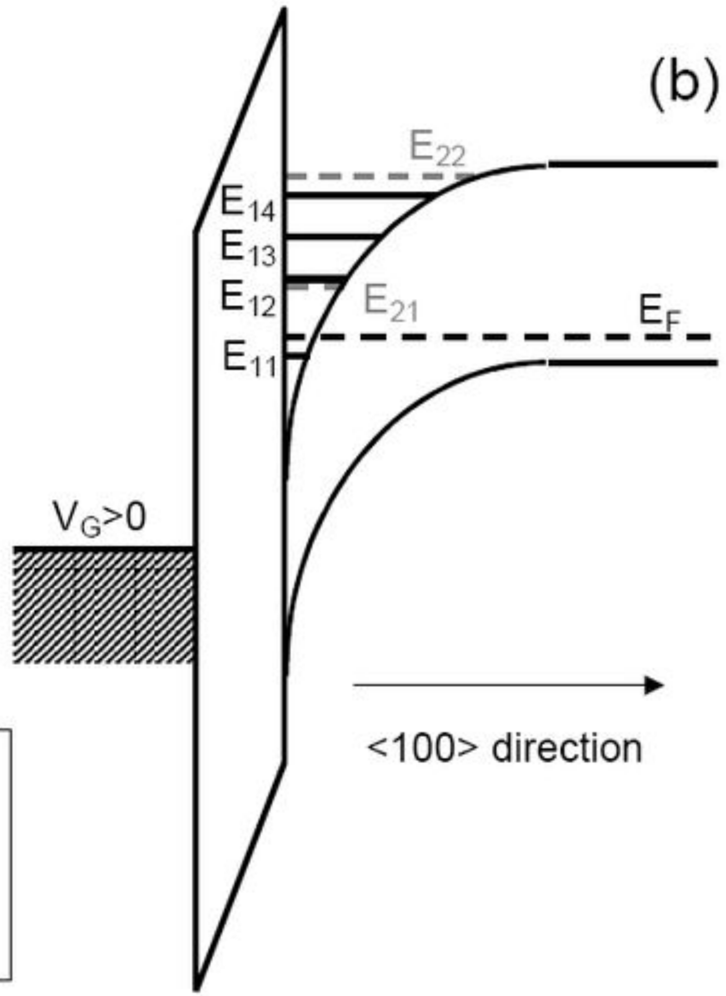


Δ_2-band:

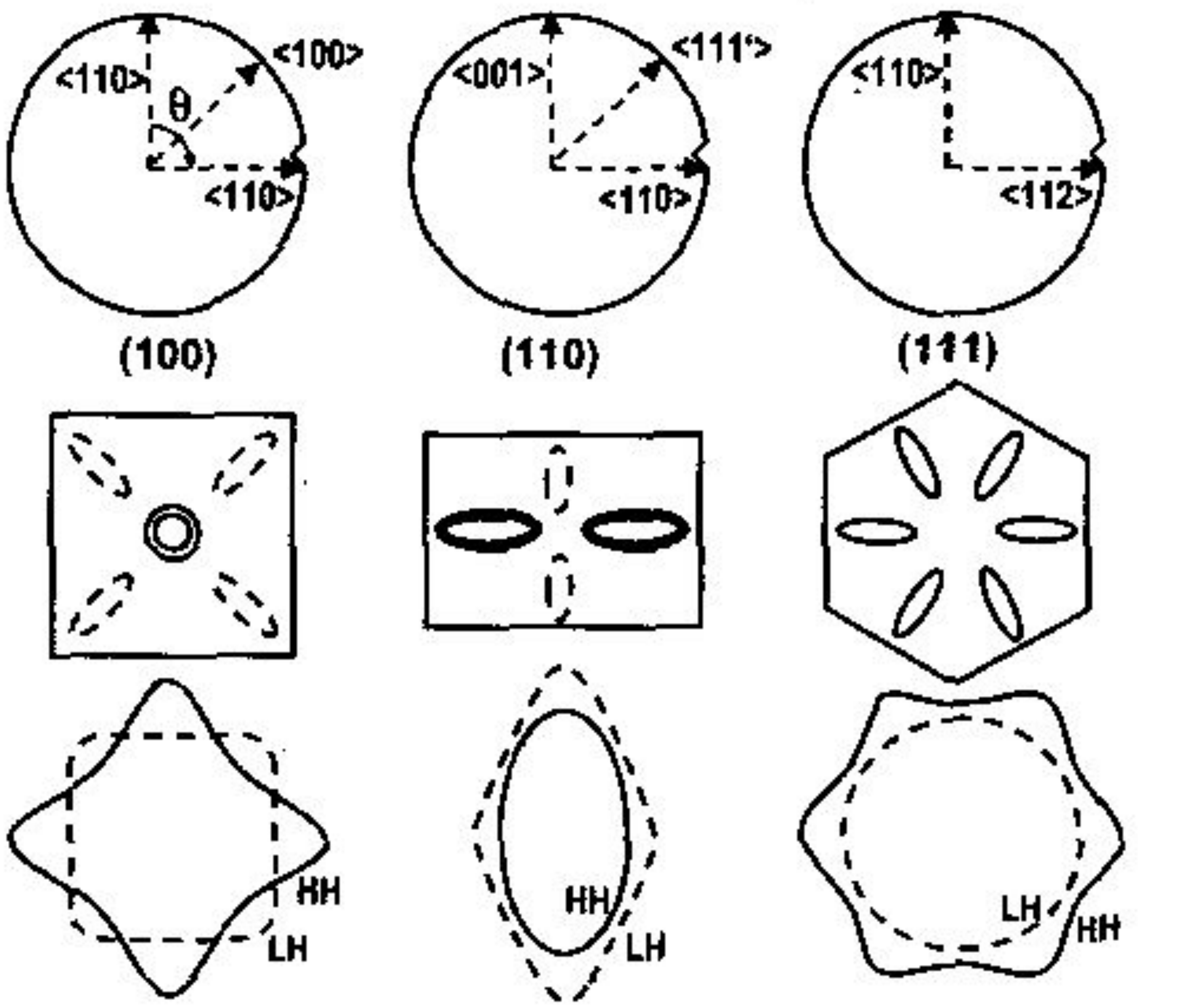
$m_{\perp}=m_l=0.916m_0$, $m_{||}=m_t=0.196m_0$

Δ_4-band:

$m_{\perp}=m_t=0.196m_0$, $m_{||}=(m_l\,m_t)^{1/2}=0.42m_0$



Constant energy surfaces with different wafer orientation.



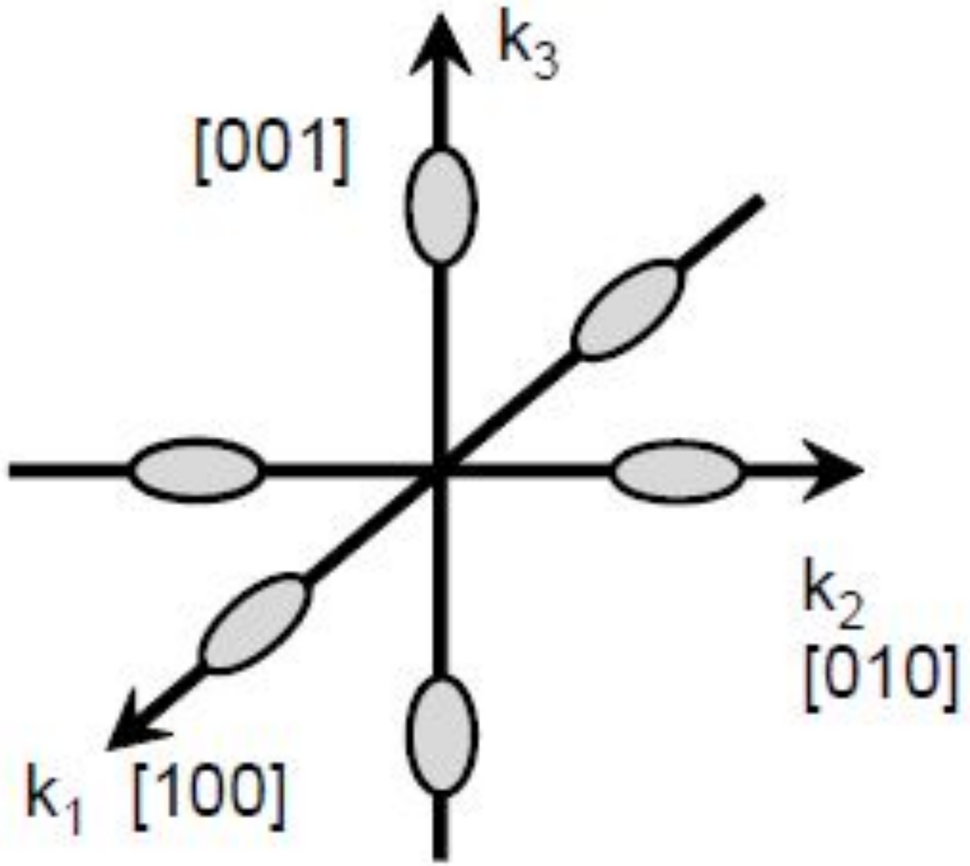


TABLE I. Transport, width, and confinement effective masses and subband degeneracies for three different technologically important semiconductor wafer orientations.

(Wafer)/[transport]/[width]	Valley	m_x	m_y	m_z	Degeneracy
(001)/[100]/[010]	Δ	m_t	m_t	m_l	2
		m_l	m_t	m_t	2
		m_t	m_l		2
	Λ	$m_t[(2m_l+m_t)/(m_l+2m_t)]$	$(m_l+2m_t)/3$	$3m_l m_t/(2m_l+m_t)$	4
(111)/$[\bar{2}11]$/$[0\bar{1}1]$	Δ	$(2m_l+m_t)/3$	m_t	$3m_l m_t/(2m_l+m_t)$	2
		$(2/3)m_t[(2m_l+m_t)/(m_l+m_t)]$	$(m_l+m_t)/2$	$3m_l m_t/(2m_l+m_t)$	4
	Λ	m_t	m_t	m_l	1
		$(8m_l+m_t)/9$	m_t	$9m_l m_t/(8m_l+m_t)$	1
		$(m_t/3)[(8m_l+m_t)/(2m_l+m_t)]$	$(2m_l+m_t)/3$		2
(110)/[001]/$[0\bar{1}0]$	Δ	m_t	$(m_l+m_t)/2$	$2m_l m_t/(m_l+m_t)$	4
		m_l	m_t	m_t	2
	Λ	$(m_l+2m_t)/3$	m_t	$3m_l m_t/(m_l+2m_t)$	2
		$3m_l m_t/(2m_l+m_t)$	$(2m_l+m_t)/3$	m_t	2

Anisur Rahman, Mark S. Lundstrom, Avik W. Ghosh; Generalized effective-mass approach for n-type metal-oxide-semiconductor field-effect transistors on arbitrarily oriented wafers. *J. Appl. Phys.* 1 March 2005; 97 (5): 053702

Try this problem

Background & Setup

Consider a highly scaled Silicon NMOSFET fabricated on a (001) substrate, with the source-to-drain channel oriented along the $\langle 110 \rangle$ crystallographic direction. The channel length is $L = 50$ nm. A voltage of $V_{DS} = 1$ V is applied across the channel, producing a uniform, constant electric field. Assume an electron is released from rest at the top of the source-channel barrier ($x = 0$, $t = 0$, $k = 0$) and travels toward the drain. To isolate the effects of the band structure, assume the transport is purely **ballistic** (the electron experiences zero scattering events during its transit).

Given Parameters $m_0 = 9.11 \times 10^{-31}$ kg, $\hbar = 1.055 \times 10^{-34}$ J · s, $a = 5.431$ Å, $m^* = 0.19 m_0$, $a_{110} = a/\sqrt{2}$

Band Structure Approximations

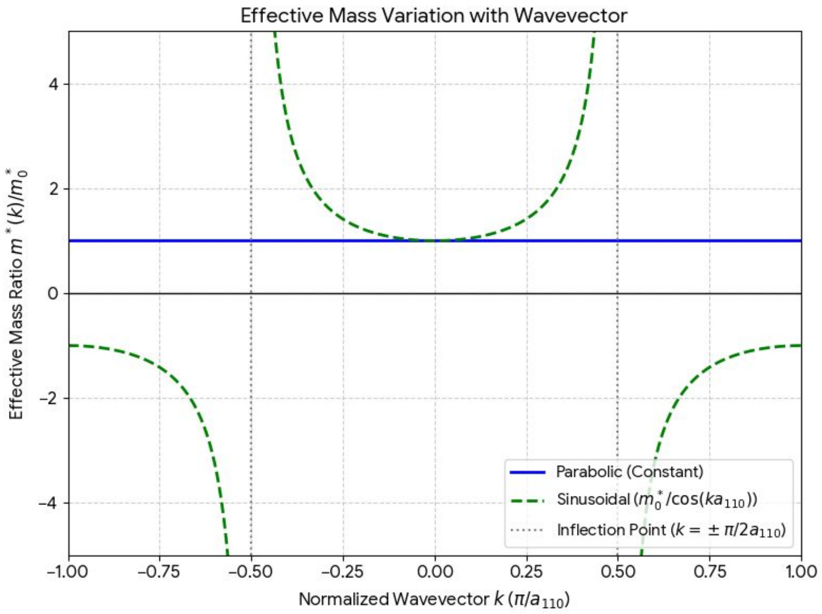
Evaluate the electron's trajectory using two different approximations for the lowest conduction band (Δ_2 valley) dispersion relation:

1. **The Parabolic Approximation**

$$E(k) = \frac{\hbar^2 k^2}{2m^*}$$

2. **The Sinusoidal Approximation**

$$E(k) = \frac{\hbar^2}{m^* a_{110}^2} [1 - \cos(k a_{110})]$$



- **Part A: Equations of Motion**
  Using the semiclassical equation of motion for a wave-packet in an electric field, determine the expression for the electron's wavevector $k(t)$ as a function of time. Does this expression depend on which band structure approximation is used?

- **Part B: Velocity and Trajectory**
  Derive analytical expressions for the electron's group velocity $v(t)$ and real-space position $x(t)$ as a function of time for both:
  1. The Parabolic Approximation
  2. The Sinusoidal Approximation

- **Part C: Transit Time Calculation**
  Calculate the exact numerical value of the transit time (t_{tr}) required for the electron to cross the 50 nm channel for both approximations under the 1 V bias. Express your answers in picoseconds (ps).

- **Part D: Drain Velocity**
  Calculate the numerical value of the electron's final velocity upon reaching the drain ($x = L$) for both approximations under the 1 V bias.

- **Part E: High Bias**
  Repeat A-D for when the applied voltage is increased significantly to $V_{DS} = 10$ V.

Density of states

The density of states (DOS) is essentially the number of different states at a particular energy level that electrons are allowed to occupy, i.e. the number of electron states per unit volume per unit energy.

$$D(E) = \frac{V}{(2\pi)^3} \int_S \frac{\mathrm{d}S}{|\nabla_k E|}$$

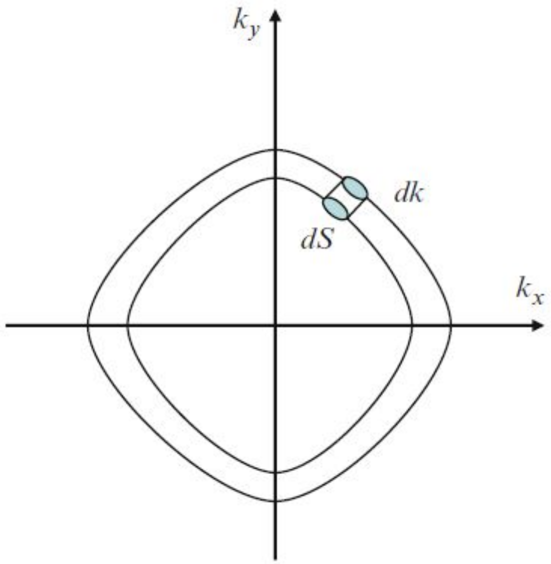
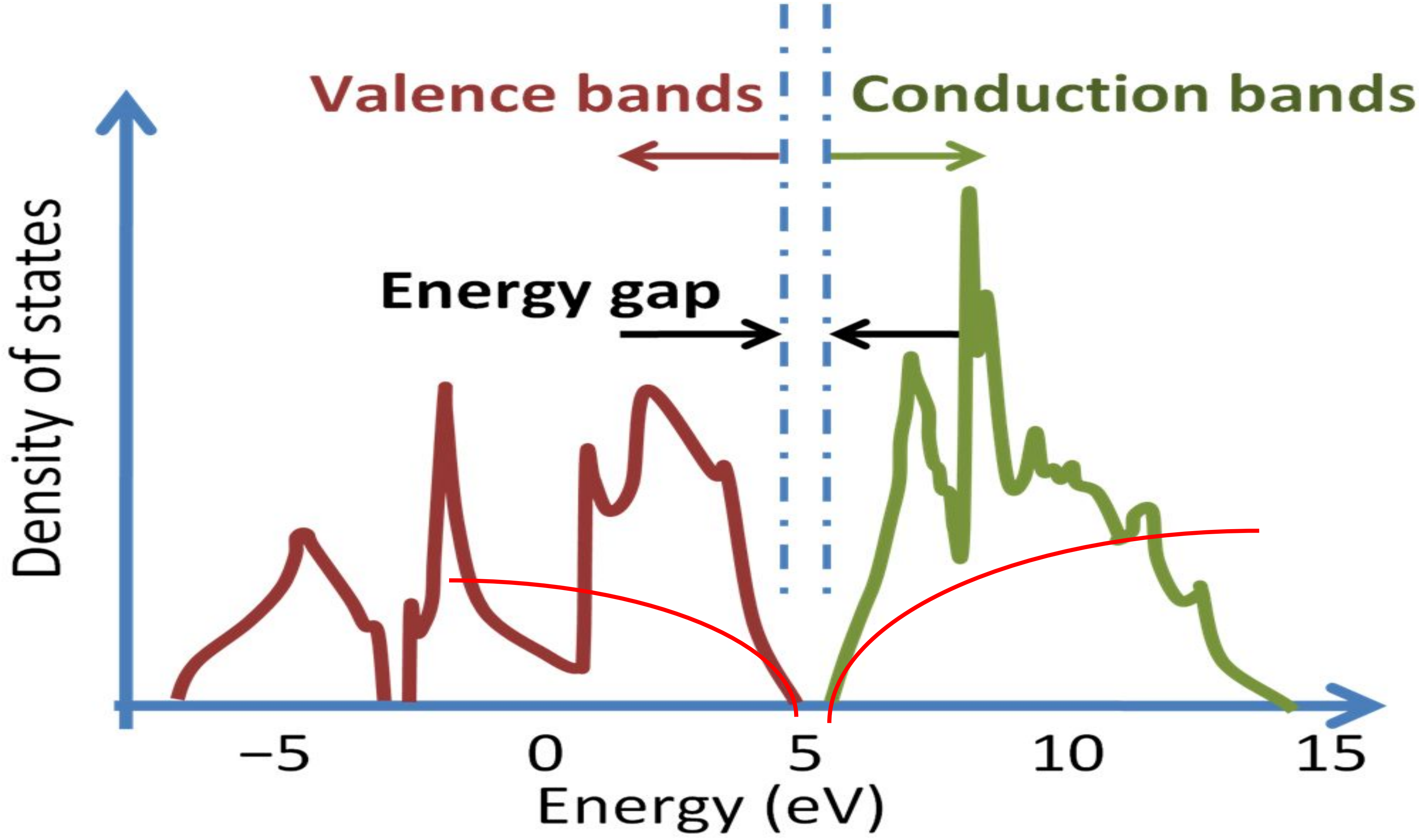


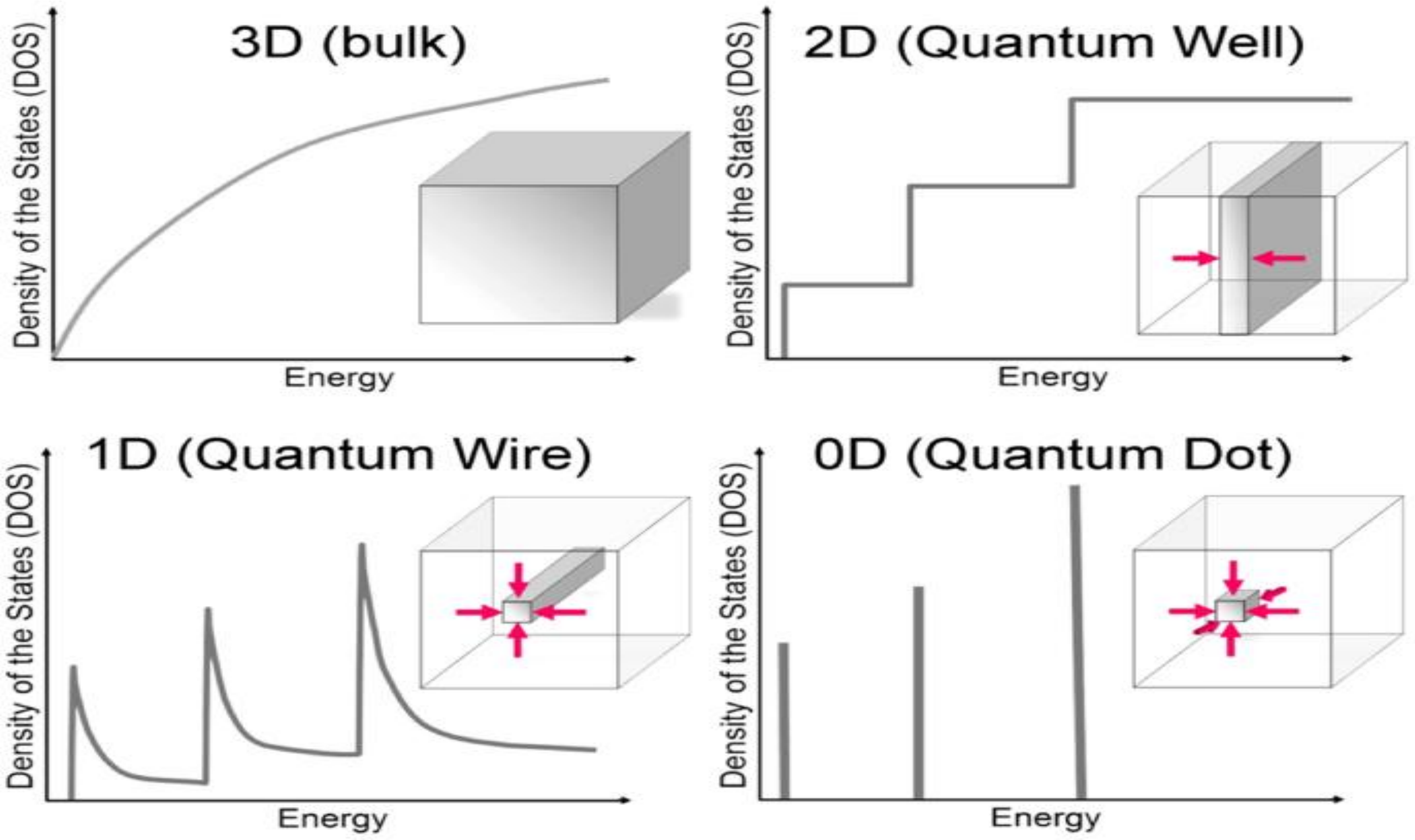
Fig. 4.22. Illustration of a unit volume enclosed by equi-energy shift in the k-space. DOS is the number of states enclosed by the equi-energy surface shift by a unit energy in the k-space

Density of states (seats) –Center court analogy

Density of states of Si–parabolic approximation



Density of states (free particle)



DOS in 3D

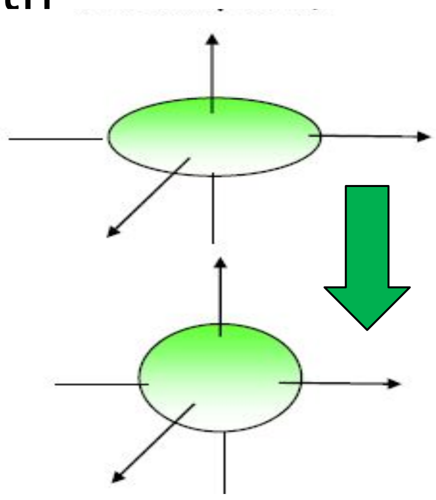
$$D_c(E) = \frac{m_n^* \sqrt{2m_n^*(E - E_c)}}{\pi^2 \hbar^3} \qquad D_v(E) = \frac{m_p^* \sqrt{2m_p^*(E_v - E)}}{\pi^2 \hbar^3}$$

DOS effective mass for various solids

$$m_n^* = N^{2/3}(m_l m_t^2)^{1/3}$$

$$m_p^* = (m_{hh}^{3/2} + m_{lh}^{3/2})^{2/3}$$

Find total k space volume of constant energy surfaces and find an eqvt sphere with same volume



	m_l	m_t	N	m_{hh}	m_{lh}	m_n^*	m_p^*
GaAs	0.067	0.067	1	0.45	0.082	0.067	0.473
Si	0.98	0.19	6	0.49	0.16	1.084	0.549
Ge	1.64	0.082	8	0.28	0.042	0.89	0.29

Fermi level

- Given a system of free electrons with chemical potential μ, the probability of state with energy E is occupied, $= \dfrac{1}{1+\exp(E-\mu)/kT}$
- Chemical potential is the change in Gibbs free energy when a particle is added/removed to the system
- Fermi energy/level, E_F is the chemical potential at temperature T = 0 K
- Fermi energy is the energy of the top most occupied electron state in metals where the CB is partially filled at T=0 K.
- This definition isn't true for semiconductors or quantum wells with discrete energy states. In such cases, E_F lies halfway between the highest occupied energy level, and the lowest unoccupied energy level. Recall for an intrinsic semiconductor, E_F (0 K) = $(E_C+E_V)/2$. **Find position of E_F in n/p type semiconductor at 0 K?**
- **In semiconductor textbooks, the symbol E_F is used instead of the μ of electron for all T>=0 K**
- For a device with many terminals, chemical potential μ at any terminal can be raised or lowered by the application of bias voltage.
- When a voltmeter is connected between two terminals (A&B), it measures the difference between levels of chemical potential.